Compte rendu de travaux pratiques

Audit et sécurité des accès
avec Oracle Database

Étude de cas : CHU Al Majd de Tanger

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Environnement

Le TP est réalisé sur **Oracle Database 26ai Free (23.26.3.0.0)** dans un conteneur Docker. La PDB utilisée est `FREEPDB1` (l'énoncé mentionne ORCLPDB). Toutes les connexions se font dans la PDB.

1 Jeu de données (Chapitre 2)

1.1 2.1 Charger et contrôler le jeu de données

Objectif. Charger le schéma CHU avec le script fourni et vérifier l'intégrité du chargement.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:21:16 2026
Version 23.26.3.0.0

Copyright (c) 1982, 2026, Oracle.  All rights reserved.

SQL> -- chu_2_1.sql : Chapitre 2 -- Charger et contr^oler le jeu de donn´ees CHU
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL> SHOW CON_NAME

CON_NAME
------------------------------
FREEPDB1
SQL>
SQL> -- Exécution du script de chargement du sch´ema CHU + donn´ees (déjà copié dans le conteneur)
SQL> PROMPT === Chargement du sch´ema CHU et des donn´ees ===
=== Chargement du sch´ema CHU et des donn´ees ===
SQL> @/tmp/chu_01_schema_dataset.sql
SQL> -- ========================================================================
SQL> -- TP Audit – CHU Al Majd de Tanger (établissement fictif)
SQL> -- 01_schema_dataset.sql : schéma CHU + jeu de données (données fictives)
SQL> -- À exécuter connecté en SYS dans la PDB (ex. ORCLPDB)
SQL> -- ========================================================================
SQL> CREATE USER chu NO AUTHENTICATION DEFAULT TABLESPACE users QUOTA UNLIMITED ON users;

User created.

SQL>
SQL> CREATE TABLE chu.services (
  2    service_id NUMBER PRIMARY KEY,
  3    nom    VARCHAR2(40) NOT NULL,
  4    batiment   VARCHAR2(10));

Table created.

SQL>
SQL> CREATE TABLE chu.personnel (
  2    pers_id       NUMBER PRIMARY KEY,
  3    nom       VARCHAR2(40) NOT NULL,
```

Figure 1 – Chapitre 2 – Jeu de donnees CHU (1/46)

```
  4    prenom        VARCHAR2(40),
  5    fonction VARCHAR2(12) CHECK (fonction IN
('MEDECIN','INFIRMIER','SECRETAIRE','PHARMACIEN','COMPTABLE','DIRECTEUR','DPO')),
  6    service_id   NUMBER REFERENCES chu.services,
  7    login         VARCHAR2(30) UNIQUE,
  8    date_embauche DATE);

Table created.

SQL>
SQL> CREATE TABLE chu.patients (
  2    patient_id    NUMBER PRIMARY KEY,
  3    cin        VARCHAR2(10) UNIQUE,
  4    nom        VARCHAR2(40) NOT NULL,
  5    prenom         VARCHAR2(40),
  6    date_naissance DATE,
  7    sexe       CHAR(1) CHECK (sexe IN ('F','M')),
  8    telephone     VARCHAR2(15),
  9    ville          VARCHAR2(30),
 10    mutuelle  VARCHAR2(10) CHECK (mutuelle IN ('CNSS','CNOPS','AMO','AUCUNE')),
 11    vip        CHAR(1) DEFAULT 'N' CHECK (vip IN ('O','N')));

Table created.

SQL>
SQL> CREATE TABLE chu.sejours (
  2    sejour_id   NUMBER PRIMARY KEY,
  3    patient_id  NUMBER NOT NULL REFERENCES chu.patients,
  4    service_id  NUMBER NOT NULL REFERENCES chu.services,
  5    medecin_id  NUMBER NOT NULL REFERENCES chu.personnel,
  6    date_entree DATE NOT NULL,
  7    date_sortie DATE,
  8    motif       VARCHAR2(60));

Table created.

SQL>
SQL> CREATE TABLE chu.dossiers_medicaux (
  2    dossier_id   NUMBER PRIMARY KEY,
  3    sejour_id    NUMBER NOT NULL UNIQUE REFERENCES chu.sejours,
```

FIGURE 2 – Chapitre 2 – Jeu de donnees CHU (2/46)

```
  4    diagnostic   VARCHAR2(80),
  5    antecedents  VARCHAR2(80),
  6    traitement   VARCHAR2(80),
  7    confidentiel CHAR(1) DEFAULT 'N' CHECK (confidentiel IN ('O','N')),
  8    maj_le       DATE);

Table created.

SQL>
SQL> CREATE TABLE chu.prescriptions (
  2    presc_id   NUMBER PRIMARY KEY,
  3    sejour_id  NUMBER NOT NULL REFERENCES chu.sejours,
  4    medecin_id NUMBER NOT NULL REFERENCES chu.personnel,
  5    medicament VARCHAR2(40) NOT NULL,
  6    dose   VARCHAR2(30),
  7    date_presc DATE,
  8    statut     VARCHAR2(10) CHECK (statut IN ('PRESCRITE','DELIVREE','ANNULEE')));

Table created.

SQL>
SQL> CREATE TABLE chu.factures (
  2    facture_id   NUMBER PRIMARY KEY,
  3    sejour_id    NUMBER NOT NULL REFERENCES chu.sejours,
  4    montant       NUMBER(10,2),
  5    part_mutuelle NUMBER(10,2),
  6    statut        VARCHAR2(8) CHECK (statut IN ('EMISE','PAYEE')),
  7    date_emission DATE);

Table created.

SQL>
SQL> -- Services
SQL> INSERT INTO chu.services VALUES (10, 'Urgences', 'A');

1 row created.

SQL> INSERT INTO chu.services VALUES (20, 'Cardiologie', 'B');

1 row created.
```

FIGURE 3 – Chapitre 2 – Jeu de donnees CHU (3/46)

```
SQL> INSERT INTO chu.services VALUES (30, 'Pédiatrie', 'C');

1 row created.

SQL> INSERT INTO chu.services VALUES (40, 'Oncologie', 'D');

1 row created.

SQL> INSERT INTO chu.services VALUES (50, 'Pharmacie', 'E');

1 row created.

SQL> INSERT INTO chu.services VALUES (60, 'Administration', 'F');

1 row created.

SQL> -- Personnel (le login est le nom du compte Oracle de la personne)
SQL> INSERT INTO chu.personnel VALUES (1, 'Alami', 'Youssef', 'MEDECIN', 10, 'DR_ALAMI', DATE '2012-01-01');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (2, 'Benali', 'Salma', 'MEDECIN', 20, 'DR_BENALI', DATE '2013-02-04');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (3, 'Chraibi', 'Karim', 'MEDECIN', 30, 'DR_CHRAIBI', DATE '2014-03-07');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (4, 'Daoudi', 'Nadia', 'MEDECIN', 40, 'DR_DAOUDI', DATE '2015-04-10');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (5, 'El Idrissi', 'Omar', 'MEDECIN', 10, 'DR_ELIDRISSI', DATE '2016-05-13');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (6, 'Fassi', 'Rim', 'MEDECIN', 20, 'DR_FASSI', DATE '2017-06-16');
```

FIGURE 4 – Chapitre 2 – Jeu de donnees CHU (4/46)

```
1 row created.

SQL> INSERT INTO chu.personnel VALUES (7, 'Ghazi', 'Hamza', 'INFIRMIER', 10, 'INF_GHAZI', DATE '2018-07-19');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (8, 'Haddad', 'Imane', 'INFIRMIER', 20, 'INF_HADDAD', DATE '2019-08-22');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (9, 'Idrissi', 'Anas', 'INFIRMIER', 30, 'INF_IDRISSI', DATE '2020-09-25');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (10, 'Jabri', 'Leila', 'INFIRMIER', 40, 'INF_JABRI', DATE '2021-10-01');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (11, 'Kettani', 'Sara', 'SECRETAIRE', 10, 'SEC_KETTANI', DATE '2022-11-04');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (12, 'Lahlou', 'Mehdi', 'SECRETAIRE', 20, 'SEC_LAHLOU', DATE '2012-12-07');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (13, 'Mansouri', 'Hind', 'PHARMACIEN', 50, 'PHA_MANSOURI', DATE '2013-01-10');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (14, 'Naciri', 'Adil', 'PHARMACIEN', 50, 'PHA_NACIRI', DATE '2014-02-13');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (15, 'Ouazzani', 'Yasmine', 'COMPTABLE', 60, 'CPT_OUAZZANI', DATE '2015-03-16');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (16, 'Rami', 'Khalid', 'DIRECTEUR', 60, 'DIR_RAMI', DATE '2016-04-19');
```

FIGURE 5 – Chapitre 2 – Jeu de donnees CHU (5/46)

```
1 row created.

SQL> INSERT INTO chu.personnel VALUES (17, 'Sbai', 'Meriem', 'DPO', 60, 'DPO_SBAI', DATE '2017-05-22');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (18, 'Tazi', 'Ilyas', 'INFIRMIER', 10, 'INF_TAZI', DATE '2018-06-25');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (19, 'Zerouali', 'Kawtar', 'MEDECIN', 30, 'DR_ZEROUALI', DATE '2019-07-01');

1 row created.

SQL> INSERT INTO chu.personnel VALUES (20, 'Amrani', 'Fouad', 'SECRETAIRE', 30, 'SEC_AMRANI', DATE '2020-08-04');

1 row created.

SQL> -- Patients (données fictives)
SQL> INSERT INTO chu.patients VALUES (1, 'K107919', 'Amrani', 'Salma', DATE '1947-06-12', 'F', '0661001234', 'Tétouan',
'CNOPS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (2, 'K115838', 'Bakkali', 'Yassine', DATE '1954-11-23', 'M', '0661002468',
'Larache', 'AMO', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (3, 'L123757', 'Belhaj', 'Kenza', DATE '1961-04-06', 'F', '0661003702', 'Asilah',
'AUCUNE', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (4, 'K131676', 'Benjelloun', 'Amine', DATE '1968-09-17', 'M', '0661004936',
'Fnideq', 'CNSS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (5, 'K139595', 'Berrada', 'Nour', DATE '1975-02-28', 'F', '0661006170', 'Ksar El
```

FIGURE 6 – Chapitre 2 – Jeu de donnees CHU (6/46)

```
Kebir', 'CNOPS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (6, 'L147514', 'Bouzid', 'Hamza', DATE '1982-07-11', 'M', '0661007404', 'Tanger',
'AMO', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (7, 'K155433', 'Cherkaoui', 'Malak', DATE '1989-12-22', 'F', '0661008638',
'Tétouan', 'AUCUNE', 'O');

1 row created.

SQL> INSERT INTO chu.patients VALUES (8, 'K163352', 'Chafik', 'Saad', DATE '1996-05-05', 'M', '0661009872', 'Larache',
'CNSS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (9, 'L171271', 'Douiri', 'Sanae', DATE '2003-10-16', 'F', '0661011106', 'Asilah',
'CNOPS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (10, 'K179190', 'El Amrani', 'Adam', DATE '2010-03-27', 'M', '0661012340',
'Fnideq', 'AMO', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (11, 'K187109', 'El Khatib', 'Salma', DATE '2017-08-10', 'F', '0661013574', 'Ksar
El Kebir', 'AUCUNE', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (12, 'L195028', 'Ennaji', 'Yassine', DATE '1944-01-21', 'M', '0661014808',
'Tanger', 'CNSS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (13, 'K202947', 'Filali', 'Kenza', DATE '1951-06-04', 'F', '0661016042', 'Tétouan',
```

FIGURE 7 – Chapitre 2 – Jeu de donnees CHU (7/46)

```
 'CNOPS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (14, 'K210866', 'Guessous', 'Amine', DATE '1958-11-15', 'M', '0661017276',
'Larache', 'AMO', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (15, 'L218785', 'Hajji', 'Nour', DATE '1965-04-26', 'F', '0661018510', 'Asilah',
'AUCUNE', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (16, 'K226704', 'Hamdaoui', 'Hamza', DATE '1972-09-09', 'M', '0661019744',
'Fnideq', 'CNSS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (17, 'K234623', 'Jaidi', 'Malak', DATE '1979-02-20', 'F', '0661020978', 'Ksar El
Kebir', 'CNOPS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (18, 'L242542', 'Kabbaj', 'Saad', DATE '1986-07-03', 'M', '0661022212', 'Tanger',
'AMO', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (19, 'K250461', 'Lamrani', 'Sanae', DATE '1993-12-14', 'F', '0661023446',
'Tétouan', 'AUCUNE', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (20, 'K258380', 'Lazrak', 'Adam', DATE '2000-05-25', 'M', '0661024680', 'Larache',
'CNSS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (21, 'L266299', 'Mernissi', 'Salma', DATE '2007-10-08', 'F', '0661025914',
```

FIGURE 8 – Chapitre 2 – Jeu de donnees CHU (8/46)

```
'Asilah', 'CNOPS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (22, 'K274218', 'Moussaoui', 'Yassine', DATE '2014-03-19', 'M', '0661027148',
'Fnideq', 'AMO', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (23, 'K282137', 'Naji', 'Kenza', DATE '1941-08-02', 'F', '0661028382', 'Ksar El
Kebir', 'AUCUNE', 'O');

1 row created.

SQL> INSERT INTO chu.patients VALUES (24, 'L290056', 'Ouali', 'Amine', DATE '1948-01-13', 'M', '0661029616', 'Tanger',
'CNSS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (25, 'K297975', 'Raji', 'Nour', DATE '1955-06-24', 'F', '0661030850', 'Tétouan',
'CNOPS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (26, 'K305894', 'Saadi', 'Hamza', DATE '1962-11-07', 'M', '0661032084', 'Larache',
'AMO', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (27, 'L313813', 'Sefrioui', 'Malak', DATE '1969-04-18', 'F', '0661033318',
'Asilah', 'AUCUNE', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (28, 'K321732', 'Slaoui', 'Saad', DATE '1976-09-01', 'M', '0661034552', 'Fnideq',
'CNSS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (29, 'K329651', 'Tahiri', 'Sanae', DATE '1983-02-12', 'F', '0661035786', 'Ksar El
```

FIGURE 9 – Chapitre 2 – Jeu de donnees CHU (9/46)

```
Kebir', 'CNOPS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (30, 'L337570', 'Talbi', 'Adam', DATE '1990-07-23', 'M', '0661037020', 'Tanger',
'AMO', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (31, 'K345489', 'Tounsi', 'Salma', DATE '1997-12-06', 'F', '0661038254', 'Tétouan',
 'AUCUNE', 'O');

1 row created.

SQL> INSERT INTO chu.patients VALUES (32, 'K353408', 'Wahbi', 'Yassine', DATE '2004-05-17', 'M', '0661039488',
'Larache', 'CNSS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (33, 'L361327', 'Yacoubi', 'Kenza', DATE '2011-10-28', 'F', '0661040722', 'Asilah',
 'CNOPS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (34, 'K369246', 'Zahraoui', 'Amine', DATE '2018-03-11', 'M', '0661041956',
'Fnideq', 'AMO', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (35, 'K377165', 'Ziani', 'Nour', DATE '1945-08-22', 'F', '0661043190', 'Ksar El
Kebir', 'AUCUNE', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (36, 'L385084', 'Alaoui', 'Hamza', DATE '1952-01-05', 'M', '0661044424', 'Tanger',
'CNSS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (37, 'K393003', 'Bennani', 'Malak', DATE '1959-06-16', 'F', '0661045658',
```

FIGURE 10 – Chapitre 2 – Jeu de donnees CHU (10/46)

```
'Tétouan', 'CNOPS', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (38, 'K400922', 'Kadiri', 'Saad', DATE '1966-11-27', 'M', '0661046892', 'Larache',
'AMO', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (39, 'L408841', 'Lahbabi', 'Sanae', DATE '1973-04-10', 'F', '0661048126', 'Asilah',
 'AUCUNE', 'N');

1 row created.

SQL> INSERT INTO chu.patients VALUES (40, 'K416760', 'Ouahbi', 'Adam', DATE '1980-09-21', 'M', '0661049360', 'Fnideq',
'CNSS', 'N');

1 row created.

SQL> -- Séjours, dossiers médicaux, prescriptions, factures
SQL> INSERT INTO chu.sejours VALUES (1, 14, 20, 6, DATE '2026-08-04', DATE '2026-08-06', 'Arythmie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (1, 1, 'Fibrillation auriculaire', 'Diabète type 2', 'Anticoagulant', 'N',
 DATE '2026-08-06');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (1, 1, 6, 'Apixaban', '2 cp/j', DATE '2026-08-04', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (2, 1, 6, 'Ramipril', '5 ml x 2/j', DATE '2026-08-05', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (1, 1, 1737.00, 1215.90, 'PAYEE', DATE '2026-08-06');

1 row created.
```

FIGURE 11 – Chapitre 2 – Jeu de donnees CHU (11/46)

```
SQL> INSERT INTO chu.sejours VALUES (2, 27, 30, 3, DATE '2026-08-07', DATE '2026-08-10', 'Asthme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (2, 2, 'Asthme aigu', 'HTA', 'Salbutamol', 'N', DATE '2026-08-10');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (3, 2, 3, 'SRO', '5 ml x 2/j', DATE '2026-08-07', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (4, 2, 3, 'Paracétamol sirop', '1 inj/j', DATE '2026-08-08', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (5, 2, 3, 'Salbutamol', '1 cp x 3/j', DATE '2026-08-09', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (2, 2, 2674.00, 0.00, 'PAYEE', DATE '2026-08-10');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (3, 40, 40, 4, DATE '2026-08-10', DATE '2026-08-14', 'Chimiothérapie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (3, 3, 'Lymphome', 'Asthme', 'Protocole R-CHOP', 'O', DATE '2026-08-14');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (6, 3, 4, 'Ondansétron', '1 inj/j', DATE '2026-08-10', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (3, 3, 33611.00, 23527.70, 'EMISE', DATE '2026-08-14');

1 row created.
```

FIGURE 12 – Chapitre 2 – Jeu de donnees CHU (12/46)

```
SQL> INSERT INTO chu.sejours VALUES (4, 13, 10, 1, DATE '2026-08-13', DATE '2026-08-18', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (4, 4, 'Fracture du radius', 'Tabagisme', 'Plâtre + antalgiques', 'N',
DATE '2026-08-18');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (7, 4, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-08-13', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (8, 4, 1, 'Héparine', '2 cp/j', DATE '2026-08-14', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (4, 4, 4548.00, 3638.40, 'PAYEE', DATE '2026-08-18');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (5, 26, 20, 6, DATE '2026-08-16', DATE '2026-08-22', 'Hypertension sévère');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (5, 5, 'HTA', 'Aucun', 'IEC', 'N', DATE '2026-08-22');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (9, 5, 6, 'Apixaban', '2 cp/j', DATE '2026-08-16', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (10, 5, 6, 'Ramipril', '5 ml x 2/j', DATE '2026-08-17', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (11, 5, 6, 'Bisoprolol', '1 inj/j', DATE '2026-08-18', 'ANNULEE');
```

FIGURE 13 – Chapitre 2 – Jeu de donnees CHU (13/46)

```
1 row created.

SQL> INSERT INTO chu.factures VALUES (5, 5, 5485.00, 3839.50, 'PAYEE', DATE '2026-08-22');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (6, 39, 30, 3, DATE '2026-08-19', DATE '2026-08-26', 'Bronchiolite');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (6, 6, 'Bronchiolite', 'Diabète type 2', 'Kinésithérapie respiratoire',
'N', DATE '2026-08-26');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (12, 6, 3, 'Paracétamol sirop', '5 ml x 2/j', DATE '2026-08-19', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (6, 6, 6422.00, 0.00, 'EMISE', DATE '2026-08-26');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (7, 12, 40, 4, DATE '2026-08-22', DATE '2026-08-30', 'Bilan tumoral');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (7, 7, 'Cancer du sein', 'HTA', 'Protocole AC-T', 'O', DATE '2026-08-30');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (13, 7, 4, 'Dexaméthasone', '1 inj/j', DATE '2026-08-22', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (14, 7, 4, 'Morphine', '1 cp x 3/j', DATE '2026-08-23', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (7, 7, 37359.00, 26151.30, 'PAYEE', DATE '2026-08-30');
```

FIGURE 14 – Chapitre 2 – Jeu de donnees CHU (14/46)

```
1 row created.

SQL> INSERT INTO chu.sejours VALUES (8, 25, 10, 1, DATE '2026-08-25', DATE '2026-08-26', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (8, 8, 'Fracture du radius', 'Asthme', 'Plâtre + antalgiques', 'N', DATE
'2026-08-26');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (15, 8, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-08-25', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (16, 8, 1, 'Héparine', '2 cp/j', DATE '2026-08-26', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (17, 8, 1, 'Amoxicilline', '5 ml x 2/j', DATE '2026-08-27', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (8, 8, 8296.00, 6636.80, 'PAYEE', DATE '2026-08-26');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (9, 38, 20, 6, DATE '2026-08-28', NULL, 'Insuffisance cardiaque');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (9, 9, 'Insuffisance cardiaque', 'Tabagisme', 'Furosémide', 'N', DATE
'2026-08-28');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (18, 9, 6, 'Apixaban', '2 cp/j', DATE '2026-08-28', 'PRESCRITE');

1 row created.
```

FIGURE 15 – Chapitre 2 – Jeu de donnees CHU (15/46)

```
SQL> INSERT INTO chu.factures VALUES (9, 9, 9233.00, 6463.10, 'EMISE', DATE '2026-08-28');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (10, 11, 30, 3, DATE '2026-08-31', DATE '2026-09-03', 'Gastro-entérite');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (10, 10, 'Déshydratation', 'Aucun', 'Réhydratation', 'N', DATE
'2026-09-03');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (19, 10, 3, 'Salbutamol', '5 ml x 2/j', DATE '2026-08-31', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (20, 10, 3, 'SRO', '1 inj/j', DATE '2026-09-01', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (10, 10, 10170.00, 0.00, 'PAYEE', DATE '2026-09-03');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (11, 24, 40, 4, DATE '2026-09-03', DATE '2026-09-07', 'Radiothérapie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (11, 11, 'Cancer colorectal', 'Diabète type 2', 'Protocole FOLFOX', 'O',
DATE '2026-09-07');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (21, 11, 4, 'Morphine', '1 inj/j', DATE '2026-09-03', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (22, 11, 4, 'Ondansétron', '1 cp x 3/j', DATE '2026-09-04', 'DELIVREE');
```

FIGURE 16 – Chapitre 2 – Jeu de donnees CHU (16/46)

```
1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (23, 11, 4, 'Dexaméthasone', '2 cp/j', DATE '2026-09-05', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (11, 11, 41107.00, 28774.90, 'PAYEE', DATE '2026-09-07');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (12, 37, 10, 1, DATE '2026-09-06', DATE '2026-09-11', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (12, 12, 'Fracture du radius', 'HTA', 'Plâtre + antalgiques', 'N', DATE
'2026-09-11');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (24, 12, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-09-06', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (12, 12, 12044.00, 9635.20, 'EMISE', DATE '2026-09-11');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (13, 10, 20, 6, DATE '2026-09-09', DATE '2026-09-15', 'Arythmie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (13, 13, 'Fibrillation auriculaire', 'Asthme', 'Anticoagulant', 'N', DATE
'2026-09-15');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (25, 13, 6, 'Apixaban', '2 cp/j', DATE '2026-09-09', 'PRESCRITE');

1 row created.
```

FIGURE 17 – Chapitre 2 – Jeu de donnees CHU (17/46)

```
SQL> INSERT INTO chu.prescriptions VALUES (26, 13, 6, 'Ramipril', '5 ml x 2/j', DATE '2026-09-10', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (13, 13, 12981.00, 9086.70, 'PAYEE', DATE '2026-09-15');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (14, 23, 30, 3, DATE '2026-09-12', DATE '2026-09-19', 'Asthme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (14, 14, 'Asthme aigu', 'Tabagisme', 'Salbutamol', 'O', DATE
'2026-09-19');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (27, 14, 3, 'SRO', '5 ml x 2/j', DATE '2026-09-12', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (28, 14, 3, 'Paracétamol sirop', '1 inj/j', DATE '2026-09-13', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (29, 14, 3, 'Salbutamol', '1 cp x 3/j', DATE '2026-09-14', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (14, 14, 13918.00, 0.00, 'PAYEE', DATE '2026-09-19');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (15, 36, 40, 4, DATE '2026-09-15', DATE '2026-09-23', 'Chimiothérapie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (15, 15, 'Lymphome', 'Aucun', 'Protocole R-CHOP', 'O', DATE '2026-09-23');
```

FIGURE 18 – Chapitre 2 – Jeu de donnees CHU (18/46)

```
1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (30, 15, 4, 'Ondansétron', '1 inj/j', DATE '2026-09-15', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (15, 15, 30855.00, 21598.50, 'EMISE', DATE '2026-09-23');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (16, 9, 10, 1, DATE '2026-09-18', DATE '2026-09-19', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (16, 16, 'Fracture du radius', 'Diabète type 2', 'Plâtre + antalgiques',
'N', DATE '2026-09-19');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (31, 16, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-09-18', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (32, 16, 1, 'Héparine', '2 cp/j', DATE '2026-09-19', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (16, 16, 1792.00, 1433.60, 'PAYEE', DATE '2026-09-19');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (17, 22, 20, 6, DATE '2026-09-21', DATE '2026-09-23', 'Hypertension sévère');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (17, 17, 'HTA', 'HTA', 'IEC', 'N', DATE '2026-09-23');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (33, 17, 6, 'Apixaban', '2 cp/j', DATE '2026-09-21', 'PRESCRITE');
```

FIGURE 19 – Chapitre 2 – Jeu de donnees CHU (19/46)

```
1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (34, 17, 6, 'Ramipril', '5 ml x 2/j', DATE '2026-09-22', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (35, 17, 6, 'Bisoprolol', '1 inj/j', DATE '2026-09-23', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (17, 17, 2729.00, 1910.30, 'PAYEE', DATE '2026-09-23');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (18, 35, 30, 3, DATE '2026-09-24', NULL, 'Bronchiolite');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (18, 18, 'Bronchiolite', 'Asthme', 'Kinésithérapie respiratoire', 'N',
DATE '2026-09-24');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (36, 18, 3, 'Paracétamol sirop', '5 ml x 2/j', DATE '2026-09-24', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (18, 18, 3666.00, 0.00, 'EMISE', DATE '2026-09-24');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (19, 8, 40, 4, DATE '2026-08-03', DATE '2026-08-07', 'Bilan tumoral');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (19, 19, 'Cancer du sein', 'Tabagisme', 'Protocole AC-T', 'O', DATE
'2026-08-07');

1 row created.
```

FIGURE 20 – Chapitre 2 – Jeu de donnees CHU (20/46)

```
SQL> INSERT INTO chu.prescriptions VALUES (37, 19, 4, 'Dexaméthasone', '1 inj/j', DATE '2026-08-03', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (38, 19, 4, 'Morphine', '1 cp x 3/j', DATE '2026-08-04', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (19, 19, 34603.00, 24222.10, 'PAYEE', DATE '2026-08-07');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (20, 21, 10, 1, DATE '2026-08-06', DATE '2026-08-11', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (20, 20, 'Fracture du radius', 'Aucun', 'Plâtre + antalgiques', 'N', DATE
'2026-08-11');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (39, 20, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-08-06', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (40, 20, 1, 'Héparine', '2 cp/j', DATE '2026-08-07', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (41, 20, 1, 'Amoxicilline', '5 ml x 2/j', DATE '2026-08-08', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (20, 20, 5540.00, 4432.00, 'PAYEE', DATE '2026-08-11');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (21, 34, 20, 6, DATE '2026-08-09', DATE '2026-08-15', 'Insuffisance cardiaque');
```

FIGURE 21 – Chapitre 2 – Jeu de donnees CHU (21/46)

```
1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (21, 21, 'Insuffisance cardiaque', 'Diabète type 2', 'Furosémide', 'N',
DATE '2026-08-15');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (42, 21, 6, 'Apixaban', '2 cp/j', DATE '2026-08-09', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (21, 21, 6477.00, 4533.90, 'EMISE', DATE '2026-08-15');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (22, 7, 30, 3, DATE '2026-08-12', DATE '2026-08-19', 'Gastro-entérite');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (22, 22, 'Déshydratation', 'HTA', 'Réhydratation', 'O', DATE
'2026-08-19');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (43, 22, 3, 'Salbutamol', '5 ml x 2/j', DATE '2026-08-12', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (44, 22, 3, 'SRO', '1 inj/j', DATE '2026-08-13', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (22, 22, 7414.00, 0.00, 'PAYEE', DATE '2026-08-19');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (23, 20, 40, 4, DATE '2026-08-15', DATE '2026-08-23', 'Radiothérapie');

1 row created.
```

FIGURE 22 – Chapitre 2 – Jeu de donnees CHU (22/46)

```
SQL> INSERT INTO chu.dossiers_medicaux VALUES (23, 23, 'Cancer colorectal', 'Asthme', 'Protocole FOLFOX', 'O', DATE
'2026-08-23');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (45, 23, 4, 'Morphine', '1 inj/j', DATE '2026-08-15', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (46, 23, 4, 'Ondansétron', '1 cp x 3/j', DATE '2026-08-16', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (47, 23, 4, 'Dexaméthasone', '2 cp/j', DATE '2026-08-17', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (23, 23, 38351.00, 26845.70, 'PAYEE', DATE '2026-08-23');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (24, 33, 10, 1, DATE '2026-08-18', DATE '2026-08-19', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (24, 24, 'Fracture du radius', 'Tabagisme', 'Plâtre + antalgiques', 'N',
DATE '2026-08-19');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (48, 24, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-08-18', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (24, 24, 9288.00, 7430.40, 'EMISE', DATE '2026-08-19');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (25, 6, 20, 6, DATE '2026-08-21', DATE '2026-08-23', 'Arythmie');
```

FIGURE 23 – Chapitre 2 – Jeu de donnees CHU (23/46)

```
1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (25, 25, 'Fibrillation auriculaire', 'Aucun', 'Anticoagulant', 'N', DATE
'2026-08-23');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (49, 25, 6, 'Apixaban', '2 cp/j', DATE '2026-08-21', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (50, 25, 6, 'Ramipril', '5 ml x 2/j', DATE '2026-08-22', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (25, 25, 10225.00, 7157.50, 'PAYEE', DATE '2026-08-23');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (26, 19, 30, 3, DATE '2026-08-24', DATE '2026-08-27', 'Asthme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (26, 26, 'Asthme aigu', 'Diabète type 2', 'Salbutamol', 'N', DATE
'2026-08-27');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (51, 26, 3, 'SRO', '5 ml x 2/j', DATE '2026-08-24', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (52, 26, 3, 'Paracétamol sirop', '1 inj/j', DATE '2026-08-25', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (53, 26, 3, 'Salbutamol', '1 cp x 3/j', DATE '2026-08-26', 'DELIVREE');

1 row created.
```

FIGURE 24 – Chapitre 2 – Jeu de donnees CHU (24/46)

```
SQL> INSERT INTO chu.factures VALUES (26, 26, 11162.00, 0.00, 'PAYEE', DATE '2026-08-27');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (27, 32, 40, 4, DATE '2026-08-27', NULL, 'Chimiothérapie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (27, 27, 'Lymphome', 'HTA', 'Protocole R-CHOP', 'O', DATE '2026-08-27');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (54, 27, 4, 'Ondansétron', '1 inj/j', DATE '2026-08-27', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (27, 27, 42099.00, 29469.30, 'EMISE', DATE '2026-08-27');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (28, 5, 10, 1, DATE '2026-08-30', DATE '2026-09-04', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (28, 28, 'Fracture du radius', 'Asthme', 'Plâtre + antalgiques', 'N', DATE
 '2026-09-04');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (55, 28, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-08-30', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (56, 28, 1, 'Héparine', '2 cp/j', DATE '2026-08-31', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (28, 28, 13036.00, 10428.80, 'PAYEE', DATE '2026-09-04');

1 row created.
```

FIGURE 25 – Chapitre 2 – Jeu de donnees CHU (25/46)

```
SQL> INSERT INTO chu.sejours VALUES (29, 18, 20, 6, DATE '2026-09-02', DATE '2026-09-08', 'Hypertension sévère');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (29, 29, 'HTA', 'Tabagisme', 'IEC', 'N', DATE '2026-09-08');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (57, 29, 6, 'Apixaban', '2 cp/j', DATE '2026-09-02', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (58, 29, 6, 'Ramipril', '5 ml x 2/j', DATE '2026-09-03', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (59, 29, 6, 'Bisoprolol', '1 inj/j', DATE '2026-09-04', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (29, 29, 13973.00, 9781.10, 'PAYEE', DATE '2026-09-08');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (30, 31, 30, 3, DATE '2026-09-05', DATE '2026-09-12', 'Bronchiolite');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (30, 30, 'Bronchiolite', 'Aucun', 'Kinésithérapie respiratoire', 'O', DATE
 '2026-09-12');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (60, 30, 3, 'Paracétamol sirop', '5 ml x 2/j', DATE '2026-09-05', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (30, 30, 910.00, 0.00, 'EMISE', DATE '2026-09-12');
```

FIGURE 26 – Chapitre 2 – Jeu de donnees CHU (26/46)

```
1 row created.

SQL> INSERT INTO chu.sejours VALUES (31, 4, 40, 4, DATE '2026-09-08', DATE '2026-09-16', 'Bilan tumoral');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (31, 31, 'Cancer du sein', 'Diabète type 2', 'Protocole AC-T', 'O', DATE
'2026-09-16');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (61, 31, 4, 'Dexaméthasone', '1 inj/j', DATE '2026-09-08', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (62, 31, 4, 'Morphine', '1 cp x 3/j', DATE '2026-09-09', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (31, 31, 31847.00, 22292.90, 'PAYEE', DATE '2026-09-16');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (32, 17, 10, 1, DATE '2026-09-11', DATE '2026-09-12', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (32, 32, 'Fracture du radius', 'HTA', 'Plâtre + antalgiques', 'N', DATE
'2026-09-12');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (63, 32, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-09-11', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (64, 32, 1, 'Héparine', '2 cp/j', DATE '2026-09-12', 'PRESCRITE');

1 row created.
```

FIGURE 27 – Chapitre 2 – Jeu de donnees CHU (27/46)

```
SQL> INSERT INTO chu.prescriptions VALUES (65, 32, 1, 'Amoxicilline', '5 ml x 2/j', DATE '2026-09-13', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (32, 32, 2784.00, 2227.20, 'PAYEE', DATE '2026-09-12');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (33, 30, 20, 6, DATE '2026-09-14', DATE '2026-09-16', 'Insuffisance cardiaque');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (33, 33, 'Insuffisance cardiaque', 'Asthme', 'Furosémide', 'N', DATE
'2026-09-16');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (66, 33, 6, 'Apixaban', '2 cp/j', DATE '2026-09-14', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (33, 33, 3721.00, 2604.70, 'EMISE', DATE '2026-09-16');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (34, 3, 30, 3, DATE '2026-09-17', DATE '2026-09-20', 'Gastro-entérite');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (34, 34, 'Déshydratation', 'Tabagisme', 'Réhydratation', 'N', DATE
'2026-09-20');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (67, 34, 3, 'Salbutamol', '5 ml x 2/j', DATE '2026-09-17', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (68, 34, 3, 'SRO', '1 inj/j', DATE '2026-09-18', 'ANNULEE');
```

FIGURE 28 – Chapitre 2 – Jeu de donnees CHU (28/46)

```
1 row created.

SQL> INSERT INTO chu.factures VALUES (34, 34, 4658.00, 0.00, 'PAYEE', DATE '2026-09-20');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (35, 16, 40, 4, DATE '2026-09-20', DATE '2026-09-24', 'Radiothérapie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (35, 35, 'Cancer colorectal', 'Aucun', 'Protocole FOLFOX', 'O', DATE
'2026-09-24');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (69, 35, 4, 'Morphine', '1 inj/j', DATE '2026-09-20', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (70, 35, 4, 'Ondansétron', '1 cp x 3/j', DATE '2026-09-21', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (71, 35, 4, 'Dexaméthasone', '2 cp/j', DATE '2026-09-22', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (35, 35, 35595.00, 24916.50, 'PAYEE', DATE '2026-09-24');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (36, 29, 10, 1, DATE '2026-09-23', NULL, 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (36, 36, 'Fracture du radius', 'Diabète type 2', 'Plâtre + antalgiques',
'N', DATE '2026-09-23');

1 row created.
```

FIGURE 29 – Chapitre 2 – Jeu de donnees CHU (29/46)

```
SQL> INSERT INTO chu.prescriptions VALUES (72, 36, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-09-23', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (36, 36, 6532.00, 5225.60, 'EMISE', DATE '2026-09-23');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (37, 2, 20, 6, DATE '2026-08-02', DATE '2026-08-08', 'Arythmie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (37, 37, 'Fibrillation auriculaire', 'HTA', 'Anticoagulant', 'N', DATE
'2026-08-08');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (73, 37, 6, 'Apixaban', '2 cp/j', DATE '2026-08-02', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (74, 37, 6, 'Ramipril', '5 ml x 2/j', DATE '2026-08-03', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (37, 37, 7469.00, 5228.30, 'PAYEE', DATE '2026-08-08');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (38, 15, 30, 3, DATE '2026-08-05', DATE '2026-08-12', 'Asthme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (38, 38, 'Asthme aigu', 'Asthme', 'Salbutamol', 'N', DATE '2026-08-12');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (75, 38, 3, 'SRO', '5 ml x 2/j', DATE '2026-08-05', 'DELIVREE');

1 row created.
```

FIGURE 30 – Chapitre 2 – Jeu de donnees CHU (30/46)

```
SQL> INSERT INTO chu.prescriptions VALUES (76, 38, 3, 'Paracétamol sirop', '1 inj/j', DATE '2026-08-06', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (77, 38, 3, 'Salbutamol', '1 cp x 3/j', DATE '2026-08-07', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (38, 38, 8406.00, 0.00, 'PAYEE', DATE '2026-08-12');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (39, 28, 40, 4, DATE '2026-08-08', DATE '2026-08-16', 'Chimiothérapie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (39, 39, 'Lymphome', 'Tabagisme', 'Protocole R-CHOP', 'O', DATE
'2026-08-16');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (78, 39, 4, 'Ondansétron', '1 inj/j', DATE '2026-08-08', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (39, 39, 39343.00, 27540.10, 'EMISE', DATE '2026-08-16');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (40, 1, 10, 1, DATE '2026-08-11', DATE '2026-08-12', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (40, 40, 'Fracture du radius', 'Aucun', 'Plâtre + antalgiques', 'N', DATE
'2026-08-12');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (79, 40, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-08-11', 'DELIVREE');
```

FIGURE 31 – Chapitre 2 – Jeu de donnees CHU (31/46)

```
1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (80, 40, 1, 'Héparine', '2 cp/j', DATE '2026-08-12', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (40, 40, 10280.00, 8224.00, 'PAYEE', DATE '2026-08-12');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (41, 14, 20, 6, DATE '2026-08-14', DATE '2026-08-16', 'Hypertension sévère');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (41, 41, 'HTA', 'Diabète type 2', 'IEC', 'N', DATE '2026-08-16');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (81, 41, 6, 'Apixaban', '2 cp/j', DATE '2026-08-14', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (82, 41, 6, 'Ramipril', '5 ml x 2/j', DATE '2026-08-15', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (83, 41, 6, 'Bisoprolol', '1 inj/j', DATE '2026-08-16', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (41, 41, 11217.00, 7851.90, 'PAYEE', DATE '2026-08-16');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (42, 27, 30, 3, DATE '2026-08-17', DATE '2026-08-20', 'Bronchiolite');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (42, 42, 'Bronchiolite', 'HTA', 'Kinésithérapie respiratoire', 'N', DATE
```

FIGURE 32 – Chapitre 2 – Jeu de donnees CHU (32/46)

```
'2026-08-20');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (84, 42, 3, 'Paracétamol sirop', '5 ml x 2/j', DATE '2026-08-17', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (42, 42, 12154.00, 0.00, 'EMISE', DATE '2026-08-20');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (43, 40, 40, 4, DATE '2026-08-20', DATE '2026-08-24', 'Bilan tumoral');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (43, 43, 'Cancer du sein', 'Asthme', 'Protocole AC-T', 'O', DATE
'2026-08-24');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (85, 43, 4, 'Dexaméthasone', '1 inj/j', DATE '2026-08-20', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (86, 43, 4, 'Morphine', '1 cp x 3/j', DATE '2026-08-21', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (43, 43, 43091.00, 30163.70, 'PAYEE', DATE '2026-08-24');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (44, 13, 10, 1, DATE '2026-08-23', DATE '2026-08-28', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (44, 44, 'Fracture du radius', 'Tabagisme', 'Plâtre + antalgiques', 'N',
DATE '2026-08-28');
```

FIGURE 33 – Chapitre 2 – Jeu de donnees CHU (33/46)

```
1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (87, 44, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-08-23', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (88, 44, 1, 'Héparine', '2 cp/j', DATE '2026-08-24', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (89, 44, 1, 'Amoxicilline', '5 ml x 2/j', DATE '2026-08-25', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (44, 44, 14028.00, 11222.40, 'PAYEE', DATE '2026-08-28');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (45, 26, 20, 6, DATE '2026-08-26', NULL, 'Insuffisance cardiaque');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (45, 45, 'Insuffisance cardiaque', 'Aucun', 'Furosémide', 'N', DATE
'2026-08-26');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (90, 45, 6, 'Apixaban', '2 cp/j', DATE '2026-08-26', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (45, 45, 965.00, 675.50, 'EMISE', DATE '2026-08-26');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (46, 39, 30, 3, DATE '2026-08-29', DATE '2026-09-05', 'Gastro-entérite');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (46, 46, 'Déshydratation', 'Diabète type 2', 'Réhydratation', 'N', DATE
```

FIGURE 34 – Chapitre 2 – Jeu de donnees CHU (34/46)

```
'2026-09-05');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (91, 46, 3, 'Salbutamol', '5 ml x 2/j', DATE '2026-08-29', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (92, 46, 3, 'SRO', '1 inj/j', DATE '2026-08-30', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (46, 46, 1902.00, 0.00, 'PAYEE', DATE '2026-09-05');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (47, 12, 40, 4, DATE '2026-09-01', DATE '2026-09-09', 'Radiothérapie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (47, 47, 'Cancer colorectal', 'HTA', 'Protocole FOLFOX', 'O', DATE
'2026-09-09');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (93, 47, 4, 'Morphine', '1 inj/j', DATE '2026-09-01', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (94, 47, 4, 'Ondansétron', '1 cp x 3/j', DATE '2026-09-02', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (95, 47, 4, 'Dexaméthasone', '2 cp/j', DATE '2026-09-03', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (47, 47, 32839.00, 22987.30, 'PAYEE', DATE '2026-09-09');

1 row created.
```

FIGURE 35 – Chapitre 2 – Jeu de donnees CHU (35/46)

```
SQL> INSERT INTO chu.sejours VALUES (48, 25, 10, 1, DATE '2026-09-04', DATE '2026-09-05', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (48, 48, 'Fracture du radius', 'Asthme', 'Plâtre + antalgiques', 'N', DATE
 '2026-09-05');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (96, 48, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-09-04', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (48, 48, 3776.00, 3020.80, 'EMISE', DATE '2026-09-05');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (49, 38, 20, 6, DATE '2026-09-07', DATE '2026-09-09', 'Arythmie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (49, 49, 'Fibrillation auriculaire', 'Tabagisme', 'Anticoagulant', 'N',
DATE '2026-09-09');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (97, 49, 6, 'Apixaban', '2 cp/j', DATE '2026-09-07', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (98, 49, 6, 'Ramipril', '5 ml x 2/j', DATE '2026-09-08', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (49, 49, 4713.00, 3299.10, 'PAYEE', DATE '2026-09-09');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (50, 11, 30, 3, DATE '2026-09-10', DATE '2026-09-13', 'Asthme');
```

FIGURE 36 – Chapitre 2 – Jeu de donnees CHU (36/46)

```
1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (50, 50, 'Asthme aigu', 'Aucun', 'Salbutamol', 'N', DATE '2026-09-13');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (99, 50, 3, 'SRO', '5 ml x 2/j', DATE '2026-09-10', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (100, 50, 3, 'Paracétamol sirop', '1 inj/j', DATE '2026-09-11', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (101, 50, 3, 'Salbutamol', '1 cp x 3/j', DATE '2026-09-12', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (50, 50, 5650.00, 0.00, 'PAYEE', DATE '2026-09-13');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (51, 24, 40, 4, DATE '2026-09-13', DATE '2026-09-17', 'Chimiothérapie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (51, 51, 'Lymphome', 'Diabète type 2', 'Protocole R-CHOP', 'O', DATE
'2026-09-17');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (102, 51, 4, 'Ondansétron', '1 inj/j', DATE '2026-09-13', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (51, 51, 36587.00, 25610.90, 'EMISE', DATE '2026-09-17');

1 row created.
```

FIGURE 37 – Chapitre 2 – Jeu de donnees CHU (37/46)

```
SQL> INSERT INTO chu.sejours VALUES (52, 37, 10, 1, DATE '2026-09-16', DATE '2026-09-21', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (52, 52, 'Fracture du radius', 'HTA', 'Plâtre + antalgiques', 'N', DATE
'2026-09-21');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (103, 52, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-09-16', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (104, 52, 1, 'Héparine', '2 cp/j', DATE '2026-09-17', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (52, 52, 7524.00, 6019.20, 'PAYEE', DATE '2026-09-21');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (53, 10, 20, 6, DATE '2026-09-19', DATE '2026-09-25', 'Hypertension sévère');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (53, 53, 'HTA', 'Asthme', 'IEC', 'N', DATE '2026-09-25');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (105, 53, 6, 'Apixaban', '2 cp/j', DATE '2026-09-19', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (106, 53, 6, 'Ramipril', '5 ml x 2/j', DATE '2026-09-20', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (107, 53, 6, 'Bisoprolol', '1 inj/j', DATE '2026-09-21', 'ANNULEE');

1 row created.
```

FIGURE 38 – Chapitre 2 – Jeu de donnees CHU (38/46)

```
SQL> INSERT INTO chu.factures VALUES (53, 53, 8461.00, 5922.70, 'PAYEE', DATE '2026-09-25');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (54, 23, 30, 3, DATE '2026-09-22', NULL, 'Bronchiolite');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (54, 54, 'Bronchiolite', 'Tabagisme', 'Kinésithérapie respiratoire', 'O',
DATE '2026-09-22');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (108, 54, 3, 'Paracétamol sirop', '5 ml x 2/j', DATE '2026-09-22',
'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (54, 54, 9398.00, 0.00, 'EMISE', DATE '2026-09-22');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (55, 36, 40, 4, DATE '2026-08-01', DATE '2026-08-09', 'Bilan tumoral');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (55, 55, 'Cancer du sein', 'Aucun', 'Protocole AC-T', 'O', DATE
'2026-08-09');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (109, 55, 4, 'Dexaméthasone', '1 inj/j', DATE '2026-08-01', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (110, 55, 4, 'Morphine', '1 cp x 3/j', DATE '2026-08-02', 'DELIVREE');

1 row created.
```

FIGURE 39 – Chapitre 2 – Jeu de donnees CHU (39/46)

```
SQL> INSERT INTO chu.factures VALUES (55, 55, 40335.00, 28234.50, 'PAYEE', DATE '2026-08-09');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (56, 9, 10, 1, DATE '2026-08-04', DATE '2026-08-05', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (56, 56, 'Fracture du radius', 'Diabète type 2', 'Plâtre + antalgiques',
'N', DATE '2026-08-05');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (111, 56, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-08-04', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (112, 56, 1, 'Héparine', '2 cp/j', DATE '2026-08-05', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (113, 56, 1, 'Amoxicilline', '5 ml x 2/j', DATE '2026-08-06', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (56, 56, 11272.00, 9017.60, 'PAYEE', DATE '2026-08-05');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (57, 22, 20, 6, DATE '2026-08-07', DATE '2026-08-09', 'Insuffisance cardiaque');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (57, 57, 'Insuffisance cardiaque', 'HTA', 'Furosémide', 'N', DATE
'2026-08-09');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (114, 57, 6, 'Apixaban', '2 cp/j', DATE '2026-08-07', 'PRESCRITE');
```

FIGURE 40 – Chapitre 2 – Jeu de donnees CHU (40/46)

```
1 row created.

SQL> INSERT INTO chu.factures VALUES (57, 57, 12209.00, 8546.30, 'EMISE', DATE '2026-08-09');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (58, 35, 30, 3, DATE '2026-08-10', DATE '2026-08-13', 'Gastro-entérite');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (58, 58, 'Déshydratation', 'Asthme', 'Réhydratation', 'N', DATE
'2026-08-13');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (115, 58, 3, 'Salbutamol', '5 ml x 2/j', DATE '2026-08-10', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (116, 58, 3, 'SRO', '1 inj/j', DATE '2026-08-11', 'ANNULEE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (58, 58, 13146.00, 0.00, 'PAYEE', DATE '2026-08-13');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (59, 8, 40, 4, DATE '2026-08-13', DATE '2026-08-17', 'Radiothérapie');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (59, 59, 'Cancer colorectal', 'Tabagisme', 'Protocole FOLFOX', 'O', DATE
'2026-08-17');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (117, 59, 4, 'Morphine', '1 inj/j', DATE '2026-08-13', 'ANNULEE');

1 row created.
```

FIGURE 41 – Chapitre 2 – Jeu de donnees CHU (41/46)

```
SQL> INSERT INTO chu.prescriptions VALUES (118, 59, 4, 'Ondansétron', '1 cp x 3/j', DATE '2026-08-14', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (119, 59, 4, 'Dexaméthasone', '2 cp/j', DATE '2026-08-15', 'PRESCRITE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (59, 59, 44083.00, 30858.10, 'PAYEE', DATE '2026-08-17');

1 row created.

SQL> INSERT INTO chu.sejours VALUES (60, 21, 10, 1, DATE '2026-08-16', DATE '2026-08-21', 'Traumatisme');

1 row created.

SQL> INSERT INTO chu.dossiers_medicaux VALUES (60, 60, 'Fracture du radius', 'Aucun', 'Plâtre + antalgiques', 'N', DATE
'2026-08-21');

1 row created.

SQL> INSERT INTO chu.prescriptions VALUES (120, 60, 1, 'Paracétamol', '1 cp x 3/j', DATE '2026-08-16', 'DELIVREE');

1 row created.

SQL> INSERT INTO chu.factures VALUES (60, 60, 1020.00, 816.00, 'EMISE', DATE '2026-08-21');

1 row created.

SQL> COMMIT;

Commit complete.

SQL>
SQL> -- Vue réservée à la direction : statistiques, sans aucune donnée nominative
SQL> CREATE VIEW chu.v_stats_services AS
  2    SELECT sv.nom AS service, COUNT(s.sejour_id) AS nb_sejours,
  3           ROUND(AVG(NVL(s.date_sortie, SYSDATE) - s.date_entree), 1) AS dms_jours,
  4           SUM(f.montant) AS ca_factures
  5    FROM  chu.services sv
```

FIGURE 42 – Chapitre 2 – Jeu de donnees CHU (42/46)

```
  6    LEFT JOIN chu.sejours s  ON s.service_id = sv.service_id
  7    LEFT JOIN chu.factures f ON f.sejour_id = s.sejour_id
  8    GROUP  BY sv.nom;

View created.

SQL>
SQL> -- Vue du secrétariat : identité et séjours, sans données médicales
SQL> CREATE VIEW chu.v_accueil AS
  2    SELECT s.sejour_id, p.patient_id, p.nom, p.prenom, p.date_naissance, p.telephone,
  3           p.mutuelle, sv.nom AS service, s.date_entree, s.date_sortie
  4    FROM  chu.sejours s JOIN chu.patients p  ON p.patient_id = s.patient_id
  5                 JOIN chu.services sv ON sv.service_id = s.service_id;

View created.

SQL>
SQL> -- Vue des infirmiers : dossiers NON confidentiels uniquement
SQL> CREATE VIEW chu.v_soins AS
  2    SELECT d.dossier_id, s.sejour_id, p.nom, p.prenom, sv.nom AS service,
  3           d.diagnostic, d.traitement
  4    FROM  chu.dossiers_medicaux d JOIN chu.sejours s  ON s.sejour_id = d.sejour_id
  5                       JOIN chu.patients p  ON p.patient_id = s.patient_id
  6                       JOIN chu.services sv ON sv.service_id = s.service_id
  7    WHERE  d.confidentiel = 'N';

View created.

SQL>
SQL>
SQL> -- V´erification du type d'authentification de CHU (doit ^etre NONE)
SQL> COLUMN username FORMAT A12
SQL> COLUMN authentication_type FORMAT A18
SQL> COLUMN account_status FORMAT A14
SQL> PROMPT === Compte CHU (authentification type = NONE) ===
=== Compte CHU (authentification type = NONE) ===
SQL> SELECT username, authentication_type, account_status
  2    FROM dba_users
  3   WHERE username = 'CHU';
```

Figure 43 – Chapitre 2 – Jeu de donnees CHU (43/46)

```
USERNAME     AUTHENTICATION_TYP ACCOUNT_STATUS
------------ ------------------ --------------
CHU      NONE       OPEN

1 row selected.

SQL>
SQL> -- Comptage des lignes de chaque table
SQL> COLUMN table_name FORMAT A30
SQL> COLUMN nb_lignes FORMAT 999999
SQL> PROMPT === Nombre de lignes par table ===
=== Nombre de lignes par table ===
SQL> SELECT 'SERVICES' AS table_name, COUNT(*) AS nb_lignes FROM chu.services
  2  UNION ALL SELECT 'PERSONNEL', COUNT(*) FROM chu.personnel
  3  UNION ALL SELECT 'PATIENTS', COUNT(*) FROM chu.patients
  4  UNION ALL SELECT 'SEJOURS', COUNT(*) FROM chu.sejours
  5  UNION ALL SELECT 'DOSSIERS_MEDICAUX', COUNT(*) FROM chu.dossiers_medicaux
  6  UNION ALL SELECT 'PRESCRIPTIONS', COUNT(*) FROM chu.prescriptions
  7  UNION ALL SELECT 'FACTURES', COUNT(*) FROM chu.factures
  8  ORDER BY table_name;

TABLE_NAME             NB_LIGNES
------------------------------ ---------
DOSSIERS_MEDICAUX              60
FACTURES                       60
PATIENTS                       40
PERSONNEL                      20
PRESCRIPTIONS                 120
SEJOURS                        60
SERVICES                        6

7 rows selected.

SQL>
SQL> -- Dossiers confidentiels
SQL> COLUMN nb_conf FORMAT 9999
SQL> PROMPT === Nombre de dossiers m´edicaux confidentiels ===
=== Nombre de dossiers m´edicaux confidentiels ===
SQL> SELECT COUNT(*) AS nb_conf FROM chu.dossiers_medicaux WHERE confidentiel = 'O';
```

FIGURE 44 – Chapitre 2 – Jeu de donnees CHU (44/46)

```
NB_CONF
-------
     19

1 row selected.

SQL>
SQL> -- Patients VIP
SQL> COLUMN nom FORMAT A20
SQL> COLUMN prenom FORMAT A15
SQL> COLUMN ville FORMAT A15
SQL> COLUMN service FORMAT A20
SQL> COLUMN medecin FORMAT A25
SQL> PROMPT === Patients VIP, s´ejour et m´edecin traitant ===
=== Patients VIP, s´ejour et m´edecin traitant ===
SQL> SELECT p.nom, p.prenom, p.ville, s.nom AS service,
  2         per.nom || ' ' || per.prenom AS medecin
  3    FROM chu.patients p
  4    JOIN chu.sejours se ON se.patient_id = p.patient_id
  5    JOIN chu.services s ON s.service_id = se.service_id
  6    JOIN chu.personnel per ON per.pers_id = se.medecin_id
  7   WHERE p.vip = 'O';

NOM          PRENOM      VILLE       SERVICE          MEDECIN
------------------- --------------- --------------- -------------------- ------------------------
Naji                Kenza           Ksar El Kebir   Pédiatrie            Chraibi Karim
Cherkaoui           Malak           Tétouan         Pédiatrie            Chraibi Karim
Tounsi              Salma           Tétouan         Pédiatrie            Chraibi Karim
Naji                Kenza           Ksar El Kebir   Pédiatrie            Chraibi Karim

4 rows selected.

SQL>
SQL> -- S´ejours en cours (sans date de sortie)
SQL> COLUMN nb_en_cours FORMAT 9999
SQL> PROMPT === Nombre de s´ejours en cours (sans date de sortie) ===
=== Nombre de s´ejours en cours (sans date de sortie) ===
SQL> SELECT COUNT(*) AS nb_en_cours FROM chu.sejours WHERE date_sortie IS NULL;

NB_EN_COURS
```

FIGURE 45 – Chapitre 2 – Jeu de donnees CHU (45/46)

```
-----------
      6

1 row selected.

SQL>
SQL> -- Lignes renvoy´ees par V_SOINS et V_ACCUEIL
SQL> COLUMN nb_lignes FORMAT 99999
SQL> PROMPT === Lignes de V_SOINS et V_ACCUEIL ===
=== Lignes de V_SOINS et V_ACCUEIL ===
SQL> SELECT 'V_SOINS' AS vue, COUNT(*) AS nb_lignes FROM chu.v_soins
  2  UNION ALL SELECT 'V_ACCUEIL', COUNT(*) FROM chu.v_accueil
  3  ORDER BY vue;

VUE   NB_LIGNES
--------- ---------
V_ACCUEIL    60
V_SOINS      41

2 rows selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 46 – Chapitre 2 – Jeu de donnees CHU (46/46)

Observations et réponses.

1. Le compte `CHU` est créé avec `AUTHENTICATION\_TYPE = NONE` et `ACCOUNT\_STATUS = OPEN`. Ce choix de compte schéma sans authentification empêche toute connexion directe : personne

ne peut se connecter en tant que `CHU` avec un mot de passe. Les applications accèdent aux données via des comptes applicatifs dédiés (`APP\_CHU`) ou par proxy (`DEV\_CHU` vers `CHU`).

2. Nombre de lignes par table : `SERVICES = 6`, `PERSONNEL = 20`, `PATIENTS = 40`, `SEJOURS = 60`, `DOSSIERS\_MEDICAUX = 60`, `PRESCRIPTIONS = 120`, `FACTURES = 60`. Ces valeurs sont conformes à la figure attendue du TP.
3. `19` dossiers médicaux sont confidentiels (`confidentiel = 'O'`).
4. `4` patients VIP (`vip = 'O'`) : Naji Kenza, Cherkaoui Malak, Tounsi Salma (deux séjours pour Naji). Tous sont suivis par le Dr Chraibi (service de Pédiatrie).
5. `6` séjours sont en cours (sans `date\_sortie`).
6. `V\_SOINS` renvoie 41 lignes (dossiers non confidentiels uniquement), `V\_ACCUEIL` renvoie 60 lignes (tous les séjours avec identité). La vue `V\_SOINS` filtre sur `confidentiel = 'N'`, d'où le nombre inférieur.

Écart : Aucun. Le chargement s'est déroulé sans erreur et tous les compteurs correspondent à l'énoncé.

2 Utilisateurs et profils (Chapitre 3)

2.1 3.1 Quatre profils

Objectif. Créer quatre profils (`P\_SOIGNANT`, `P\_ADMINISTRATIF`, `P\_SENSIBLE`, `P\_APPLI`) avec des limites adaptées à chaque population.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:21:55 2026
Version 23.26.3.0.0

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SQL> -- chu_3_1.sql : Chapitre 3.1 -- Quatre profils CHU
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Vérifier RESOURCE_LIMIT
SQL> COLUMN name FORMAT A25
SQL> COLUMN value FORMAT A15
SQL> PROMPT === Valeur de RESOURCE_LIMIT ===
=== Valeur de RESOURCE_LIMIT ===
SQL> SHOW PARAMETER resource_limit

NAME                      TYPE    VALUE
------------------------------------ ----------- ------------------------------
resource_limit               boolean     TRUE
SQL>
SQL> -- Vérifier l'existence des fonctions de vérification
SQL> COLUMN object_name FORMAT A35
SQL> COLUMN object_type FORMAT A15
SQL> PROMPT === Fonctions de verification de complexite ===
=== Fonctions de verification de complexite ===
SQL> SELECT object_name, object_type
  2    FROM dba_objects
  3   WHERE owner = 'SYS'
  4     AND object_name LIKE 'ORA12C%VERIFY%'
  5   ORDER BY object_name;

OBJECT_NAME             OBJECT_TYPE
----------------------------------- ---------------
ORA12C_STIG_VERIFY_FUNCTION     FUNCTION
ORA12C_STRONG_VERIFY_FUNCTION       FUNCTION
ORA12C_VERIFY_FUNCTION          FUNCTION

3 rows selected.
```

Figure 47 – Chapitre 3.1 – Quatre profils CHU (1/6)

```
SQL>
SQL> -- P_SOIGNANT : 3 sessions, idle 30 min, 5 échecs, verrou 15 min, vie 90j, grâce 7j, reuse 365/10, verify standard
SQL> PROMPT === Creation du profil P_SOIGNANT ===
=== Creation du profil P_SOIGNANT ===
SQL> CREATE PROFILE p_soignant LIMIT
  2    SESSIONS_PER_USER 3
  3    IDLE_TIME 30
  4    FAILED_LOGIN_ATTEMPTS 5
  5    PASSWORD_LOCK_TIME 15/1440
  6    PASSWORD_LIFE_TIME 90
  7    PASSWORD_GRACE_TIME 7
  8    PASSWORD_REUSE_TIME 365
  9    PASSWORD_REUSE_MAX 10
 10    PASSWORD_VERIFY_FUNCTION ORA12C_VERIFY_FUNCTION;

Profile created.

SQL>
SQL> -- P_ADMINISTRATIF : 1 session, idle 15 min, 3 échecs, verrou 30 min, vie 60j, grâce 5j, reuse 365/10, verify
standard
SQL> PROMPT === Creation du profil P_ADMINISTRATIF ===
=== Creation du profil P_ADMINISTRATIF ===
SQL> CREATE PROFILE p_administratif LIMIT
  2    SESSIONS_PER_USER 1
  3    IDLE_TIME 15
  4    FAILED_LOGIN_ATTEMPTS 3
  5    PASSWORD_LOCK_TIME 30/1440
  6    PASSWORD_LIFE_TIME 60
  7    PASSWORD_GRACE_TIME 5
  8    PASSWORD_REUSE_TIME 365
  9    PASSWORD_REUSE_MAX 10
 10    PASSWORD_VERIFY_FUNCTION ORA12C_VERIFY_FUNCTION;

Profile created.

SQL>
SQL> -- P_SENSIBLE : 1 session, idle 10 min, 3 échecs, verrou 1 jour, vie 60j, grâce 5j, reuse 365/12, verify forte
SQL> PROMPT === Creation du profil P_SENSIBLE ===
=== Creation du profil P_SENSIBLE ===
SQL> CREATE PROFILE p_sensible LIMIT
```

FIGURE 48 – Chapitre 3.1 – Quatre profils CHU (2/6)

```
  2    SESSIONS_PER_USER 1
  3    IDLE_TIME 10
  4    FAILED_LOGIN_ATTEMPTS 3
  5    PASSWORD_LOCK_TIME 1
  6    PASSWORD_LIFE_TIME 60
  7    PASSWORD_GRACE_TIME 5
  8    PASSWORD_REUSE_TIME 365
  9    PASSWORD_REUSE_MAX 12
 10    PASSWORD_VERIFY_FUNCTION ORA12C_STRONG_VERIFY_FUNCTION;

Profile created.

SQL>
SQL> -- P_APPLI : 30 sessions, idle UNLIMITED, 10 échecs, vie UNLIMITED, verify forte
SQL> PROMPT === Creation du profil P_APPLI ===
=== Creation du profil P_APPLI ===
SQL> CREATE PROFILE p_appli LIMIT
  2    SESSIONS_PER_USER 30
  3    IDLE_TIME UNLIMITED
  4    FAILED_LOGIN_ATTEMPTS 10
  5    PASSWORD_LIFE_TIME UNLIMITED
  6    PASSWORD_VERIFY_FUNCTION ORA12C_STRONG_VERIFY_FUNCTION;

Profile created.

SQL>
SQL> COLUMN profile FORMAT A20
SQL> COLUMN resource_name FORMAT A30
SQL> COLUMN limit FORMAT A20
SQL> PROMPt === Parametres des quatre profils CHU ===
=== Parametres des quatre profils CHU ===
SQL> SELECT profile, resource_name, limit
  2    FROM dba_profiles
  3   WHERE profile IN ('P_SOIGNANT','P_ADMINISTRATIF','P_SENSIBLE','P_APPLI')
  4   ORDER BY profile, resource_name;

PROFILE          RESOURCE_NAME          LIMIT
-------------------- ------------------------------ --------------------
P_ADMINISTRATIF      COMPOSITE_LIMIT            DEFAULT
P_ADMINISTRATIF      CONNECT_TIME           DEFAULT
```

Figure 49 – Chapitre 3.1 – Quatre profils CHU (3/6)

```
P_ADMINISTRATIF      CPU_PER_CALL           DEFAULT
P_ADMINISTRATIF      CPU_PER_SESSION            DEFAULT
P_ADMINISTRATIF      FAILED_LOGIN_ATTEMPTS      3
P_ADMINISTRATIF      IDLE_TIME              15
P_ADMINISTRATIF      INACTIVE_ACCOUNT_TIME      DEFAULT
P_ADMINISTRATIF      LOGICAL_READS_PER_CALL     DEFAULT
P_ADMINISTRATIF      LOGICAL_READS_PER_SESSION      DEFAULT
P_ADMINISTRATIF      PASSWORD_GRACE_TIME        5
P_ADMINISTRATIF      PASSWORD_LIFE_TIME         60
P_ADMINISTRATIF      PASSWORD_LOCK_TIME         .0208
P_ADMINISTRATIF      PASSWORD_REUSE_MAX         10
P_ADMINISTRATIF      PASSWORD_REUSE_TIME        365
P_ADMINISTRATIF      PASSWORD_ROLLOVER_TIME     DEFAULT
P_ADMINISTRATIF      PASSWORD_VERIFY_FUNCTION       ORA12C_VERIFY_FUNCTI
                           ON

P_ADMINISTRATIF      PRIVATE_SGA            DEFAULT
P_ADMINISTRATIF      SESSIONS_PER_USER          1
P_APPLI          COMPOSITE_LIMIT            DEFAULT
P_APPLI          CONNECT_TIME           DEFAULT
P_APPLI          CPU_PER_CALL           DEFAULT
P_APPLI          CPU_PER_SESSION            DEFAULT
P_APPLI          FAILED_LOGIN_ATTEMPTS      10
P_APPLI          IDLE_TIME              UNLIMITED
P_APPLI          INACTIVE_ACCOUNT_TIME      DEFAULT
P_APPLI          LOGICAL_READS_PER_CALL     DEFAULT
P_APPLI          LOGICAL_READS_PER_SESSION      DEFAULT
P_APPLI          PASSWORD_GRACE_TIME        DEFAULT
P_APPLI          PASSWORD_LIFE_TIME         UNLIMITED
P_APPLI          PASSWORD_LOCK_TIME         DEFAULT
P_APPLI          PASSWORD_REUSE_MAX         DEFAULT
P_APPLI          PASSWORD_REUSE_TIME        DEFAULT
P_APPLI          PASSWORD_ROLLOVER_TIME     DEFAULT
P_APPLI          PASSWORD_VERIFY_FUNCTION       ORA12C_STRONG_VERIFY
                           _FUNCTION

P_APPLI          PRIVATE_SGA            DEFAULT
P_APPLI          SESSIONS_PER_USER          30
P_SENSIBLE       COMPOSITE_LIMIT            DEFAULT
P_SENSIBLE       CONNECT_TIME           DEFAULT
```

Figure 50 – Chapitre 3.1 – Quatre profils CHU (4/6)

```
P_SENSIBLE       CPU_PER_CALL           DEFAULT
P_SENSIBLE       CPU_PER_SESSION            DEFAULT
P_SENSIBLE       FAILED_LOGIN_ATTEMPTS      3
P_SENSIBLE       IDLE_TIME              10
P_SENSIBLE       INACTIVE_ACCOUNT_TIME      DEFAULT

PROFILE          RESOURCE_NAME          LIMIT
------------------- ------------------------------ --------------------
P_SENSIBLE       LOGICAL_READS_PER_CALL     DEFAULT
P_SENSIBLE       LOGICAL_READS_PER_SESSION      DEFAULT
P_SENSIBLE       PASSWORD_GRACE_TIME        5
P_SENSIBLE       PASSWORD_LIFE_TIME         60
P_SENSIBLE       PASSWORD_LOCK_TIME         1
P_SENSIBLE       PASSWORD_REUSE_MAX         12
P_SENSIBLE       PASSWORD_REUSE_TIME        365
P_SENSIBLE       PASSWORD_ROLLOVER_TIME     DEFAULT
P_SENSIBLE       PASSWORD_VERIFY_FUNCTION       ORA12C_STRONG_VERIFY
                           _FUNCTION

P_SENSIBLE       PRIVATE_SGA            DEFAULT
P_SENSIBLE       SESSIONS_PER_USER          1
P_SOIGNANT       COMPOSITE_LIMIT            DEFAULT
P_SOIGNANT       CONNECT_TIME           DEFAULT
P_SOIGNANT       CPU_PER_CALL           DEFAULT
P_SOIGNANT       CPU_PER_SESSION            DEFAULT
P_SOIGNANT       FAILED_LOGIN_ATTEMPTS      5
P_SOIGNANT       IDLE_TIME              30
P_SOIGNANT       INACTIVE_ACCOUNT_TIME      DEFAULT
P_SOIGNANT       LOGICAL_READS_PER_CALL     DEFAULT
P_SOIGNANT       LOGICAL_READS_PER_SESSION      DEFAULT
P_SOIGNANT       PASSWORD_GRACE_TIME        7
P_SOIGNANT       PASSWORD_LIFE_TIME         90
P_SOIGNANT       PASSWORD_LOCK_TIME         .0104
P_SOIGNANT       PASSWORD_REUSE_MAX         10
P_SOIGNANT       PASSWORD_REUSE_TIME        365
P_SOIGNANT       PASSWORD_ROLLOVER_TIME     DEFAULT
P_SOIGNANT       PASSWORD_VERIFY_FUNCTION       ORA12C_VERIFY_FUNCTI
                           ON

P_SOIGNANT       PRIVATE_SGA            DEFAULT
```

FIGURE 51 – Chapitre 3.1 – Quatre profils CHU (5/6)

```
P_SOIGNANT       SESSIONS_PER_USER          3

72 rows selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 52 – Chapitre 3.1 – Quatre profils CHU (6/6)

Observations et réponses.

1. `RESOURCE\_LIMIT` vaut `TRUE` : les limites de profil sont appliquées.
2. Les fonctions de vérification `ORA12C\_VERIFY\_FUNCTION` (standard) et `ORA12C\_STRONG\_VERIFY\_FUNC` (forte) existent bien dans le schéma `SYS`.
3. `P\_SOIGNANT` : 3 sessions, inactivité 30 min, 5 échecs, verrou 15 min, vie 90 j, grâce 7 j, réutilisation 365 j/10, vérification standard.
4. `P\_ADMINISTRATIF` : 1 session, inactivité 15 min, 3 échecs, verrou 30 min, vie 60 j, grâce 5 j, réutilisation 365 j/10, vérification standard.
5. `P\_SENSIBLE` : 1 session, inactivité 10 min, 3 échecs, verrou 1 j, vie 60 j, grâce 5 j, réutilisation 365 j/12, vérification forte.
6. `P\_APPLI` : 30 sessions, inactivité illimitée, 10 échecs, vie illimitée, vérification forte. Le `PASSWORD\_LIFE\_TIME UNLIMITED` évite qu'une expiration ne bloque l'application ; la contrepartie est un verrouillage après 10 échecs et une fonction de mot de passe forte pour limiter

les risques.

Écart : Aucun.

2.2 3.2 Les comptes

Objectif. Créer les 12 comptes avec leur profil et mot de passe imposés, créer le schéma `AUDIT\_CHU`, et configurer le proxy `DEV\_CHU` vers `CHU`.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:22:11 2026
Version 23.26.3.0.0

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SQL> -- chu_3_2.sql : Chapitre 3.2 -- Douze comptes CHU, schema AUDIT_CHU, proxy DEV_CHU->CHU
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- CHU a besoin de CREATE SESSION pour etre cible du proxy (comme RH au TP precedent)
SQL> PROMPT === CHU : octroi de CREATE SESSION (cible du proxy) ===
=== CHU : octroi de CREATE SESSION (cible du proxy) ===
SQL> GRANT CREATE SESSION TO chu;

Grant succeeded.

SQL>
SQL> -- Schema AUDIT_CHU : sans authentification, 200 Mo sur USERS (recevra les journaux)
SQL> PROMPT === Creation du schema AUDIT_CHU (NO AUTHENTICATION, 200 Mo) ===
=== Creation du schema AUDIT_CHU (NO AUTHENTICATION, 200 Mo) ===
SQL> CREATE USER audit_chu NO AUTHENTICATION DEFAULT TABLESPACE users QUOTA 200M ON users;

User created.

SQL>
SQL> -- Les 12 comptes avec leur profil et mot de passe imposes
SQL> PROMPT === Creation des comptes soignants (P_SOIGNANT) ===
=== Creation des comptes soignants (P_SOIGNANT) ===
SQL> CREATE USER dr_alami IDENTIFIED BY "Urgences#2026a" PROFILE p_soignant DEFAULT TABLESPACE users;

User created.

SQL> CREATE USER dr_benali IDENTIFIED BY "Cardio#2026b" PROFILE p_soignant DEFAULT TABLESPACE users;

User created.

SQL> CREATE USER dr_fassi IDENTIFIED BY "Cardio#2026f" PROFILE p_soignant DEFAULT TABLESPACE users;

User created.
```

Figure 53 – Chapitre 3.2 – Comptes et proxy CHU (1/5)

```
SQL> CREATE USER inf_ghazi IDENTIFIED BY "Soins#2026g" PROFILE p_soignant DEFAULT TABLESPACE users;

User created.

SQL>
SQL> PROMPT === Creation des comptes administratifs (P_ADMINISTRATIF) ===
=== Creation des comptes administratifs (P_ADMINISTRATIF) ===
SQL> CREATE USER sec_kettani IDENTIFIED BY "Accueil#2026k" PROFILE p_administratif DEFAULT TABLESPACE users;

User created.

SQL> CREATE USER sec_lahlou IDENTIFIED BY "Accueil#2026l" PROFILE p_administratif DEFAULT TABLESPACE users;

User created.

SQL> CREATE USER pha_mansouri IDENTIFIED BY "Pharma#2026m" PROFILE p_administratif DEFAULT TABLESPACE users;

User created.

SQL> CREATE USER cpt_ouazzani IDENTIFIED BY "Compta#2026o" PROFILE p_administratif DEFAULT TABLESPACE users;

User created.

SQL>
SQL> PROMPT === Creation des comptes sensibles (P_SENSIBLE) ===
=== Creation des comptes sensibles (P_SENSIBLE) ===
SQL> CREATE USER dir_rami IDENTIFIED BY "Direction##CHU26" PROFILE p_sensible DEFAULT TABLESPACE users;

User created.

SQL> CREATE USER dpo_sbai IDENTIFIED BY "Protection##DPO26" PROFILE p_sensible DEFAULT TABLESPACE users;

User created.

SQL>
SQL> PROMPT === Creation du compte applicatif (P_APPLI) ===
=== Creation du compte applicatif (P_APPLI) ===
SQL> CREATE USER app_chu IDENTIFIED BY "ChuApp##Prod2026" PROFILE p_appli DEFAULT TABLESPACE users;
```

FIGURE 54 – Chapitre 3.2 – Comptes et proxy CHU (2/5)

```
User created.

SQL>
SQL> PROMPT === Creation du compte developpeur (P_ADMINISTRATIF) ===
=== Creation du compte developpeur (P_ADMINISTRATIF) ===
SQL> CREATE USER dev_chu IDENTIFIED BY "Deploi#2026d" PROFILE p_administratif DEFAULT TABLESPACE users;

User created.

SQL>
SQL> -- Chaque compte a besoin de CREATE SESSION pour se connecter
SQL> PROMPt === Octroi de CREATE SESSION aux 12 comptes ===
=== Octroi de CREATE SESSION aux 12 comptes ===
SQL> GRANT CREATE SESSION TO dr_alami, dr_benali, dr_fassi, inf_ghazi,
  2                sec_kettani, sec_lahlou, pha_mansouri, cpt_ouazzani,
  3                dir_rami, dpo_sbai, app_chu, dev_chu;

Grant succeeded.

SQL>
SQL> -- Proxy : DEV_CHU se connecte a travers CHU
SQL> PROMPT === Autorisation du proxy DEV_CHU -> CHU ===
=== Autorisation du proxy DEV_CHU -> CHU ===
SQL> ALTER USER chu GRANT CONNECT THROUGH dev_chu;

User altered.

SQL>
SQL> COLUMN username FORMAT A14
SQL> COLUMN profile FORMAT A20
SQL> COLUMN account_status FORMAT A14
SQL> COLUMN authentication_type FORMAT A18
SQL> PROMPT === Comptes creer (verification) ===
=== Comptes creer (verification) ===
SQL> SELECT username, profile, account_status, authentication_type
  2    FROM dba_users
  3   WHERE username IN ('DR_ALAMI','DR_BENALI','DR_FASSI','INF_GHAZI',
  4              'SEC_KETTANI','SEC_LAHLOU','PHA_MANSOURI','CPT_OUAZZANI',
  5              'DIR_RAMI','DPO_SBAI','APP_CHU','DEV_CHU','CHU','AUDIT_CHU')
  6   ORDER BY username;
```

FIGURE 55 – Chapitre 3.2 – Comptes et proxy CHU (3/5)

```
USERNAME       PROFILE          ACCOUNT_STATUS AUTHENTICATION_TYP
-------------- ---------------- -------------- ------------------
APP_CHU        P_APPLI          OPEN       PASSWORD
AUDIT_CHU      DEFAULT          OPEN       NONE
CHU        DEFAULT          OPEN       NONE
CPT_OUAZZANI   P_ADMINISTRATIF      OPEN       PASSWORD
DEV_CHU        P_ADMINISTRATIF      OPEN       PASSWORD
DIR_RAMI       P_SENSIBLE       OPEN       PASSWORD
DPO_SBAI       P_SENSIBLE       OPEN       PASSWORD
DR_ALAMI       P_SOIGNANT       OPEN       PASSWORD
DR_BENALI      P_SOIGNANT       OPEN       PASSWORD
DR_FASSI       P_SOIGNANT       OPEN       PASSWORD
INF_GHAZI      P_SOIGNANT       OPEN       PASSWORD
PHA_MANSOURI   P_ADMINISTRATIF      OPEN       PASSWORD
SEC_KETTANI    P_ADMINISTRATIF      OPEN       PASSWORD
SEC_LAHLOU     P_ADMINISTRATIF      OPEN       PASSWORD

14 rows selected.

SQL>
SQL> PROMPt === Quotas (aucun quota innecesaire, sauf AUDIT_CHU 200M) ===
=== Quotas (aucun quota innecesaire, sauf AUDIT_CHU 200M) ===
SQL> COLUMN username FORMAT A14
SQL> COLUMN tablespace_name FORMAT A15
SQL> COLUMN max_bytes FORMAT 99999999999
SQL> SELECT username, tablespace_name, max_bytes
  2    FROM dba_ts_quotas
  3   WHERE username IN ('CHU','AUDIT_CHU','DR_ALAMI','DR_BENALI','DR_FASSI','INF_GHAZI',
  4              'SEC_KETTANI','SEC_LAHLOU','PHA_MANSOURI','CPT_OUAZZANI',
  5              'DIR_RAMI','DPO_SBAI','APP_CHU','DEV_CHU')
  6   ORDER BY username;

USERNAME       TABLESPACE_NAME    MAX_BYTES
-------------- --------------- ------------
AUDIT_CHU      USERS          209715200
CHU        USERS             -1

2 rows selected.
```

FIGURE 56 – Chapitre 3.2 – Comptes et proxy CHU (4/5)

```
SQL>
SQL> PROMPT === Autorisation proxy ===
=== Autorisation proxy ===
SQL> COLUMN proxy FORMAT A12
SQL> COLUMN client FORMAT A12
SQL> SELECT proxy, client FROM proxy_users WHERE client = 'CHU';

PROXY        CLIENT
------------ ------------
DEV_CHU      CHU

1 row selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 57 – Chapitre 3.2 – Comptes et proxy CHU (5/5)

Observations et réponses.

1. Les 12 comptes sont créés avec leur profil respectif et leur mot de passe imposé. Aucun quota n'est attribué (sauf `CHU` qui a l'illimité depuis le chapitre 2, et `AUDIT\_CHU` avec 200 Mo).
2. `CHU` a besoin du privilège `CREATE SESSION` pour être la cible du proxy (comme `RH` au TP RBAC).
3. `AUDIT\_CHU` est créé avec `NO AUTHENTICATION` et 200 Mo sur `USERS` : il recevra les journaux d'audit applicatifs sans qu'aucun humain ne puisse s'y connecter.

4. Le proxy `DEV\_CHU` vers `CHU` est autorisé via `ALTER USER chu GRANT CONNECT THROUGH dev\_chu`.

Écart : Aucun.

2.3 3.3 Tester les profils

Objectif. Vérifier en pratique les limites de sessions, le verrouillage après échecs et la complexité des mots de passe.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:22:28 2026
Version 23.26.3.0.0

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SQL> -- chu_3_3_a.sql : Chapitre 3.3 -- Test limites de sessions (SEC_LAHLOU : 1 session max)
SQL> -- Session 1 de SEC_LAHLOU (garde ouverte)
SQL> CONNECT sec_lahlou/"Accueil#2026l"@//localhost/FREEPDB1
Connected.
SQL> SELECT USER AS session_sec_lahlou_1 FROM dual;

SESSION_SEC_LAHLOU_1
--------------------------------------------------------------------------------------------------------------------------------
SEC_LAHLOU

1 row selected.

SQL> EXEC DBMS_SESSION.SLEEP(120);

PL/SQL procedure successfully completed.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 58 – Chapitre 3.3 – Test profils (a)

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:22:29 2026
Version 23.26.3.0.0

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SQL> -- chu_3_3_b.sql : Chapitre 3.3 -- Session 2 de SEC_LAHLOU (doit echouer, 1 session max)
SQL> CONNECT sec_lahlou/"Accueil#2026l"@//localhost/FREEPDB1
ERROR:
ORA-02391: exceeded simultaneous SESSIONS_PER_USER limit
Help: https://docs.oracle.com/error-help/db/ora-02391/


SQL> SELECT USER AS session_sec_lahlou_2 FROM dual;
SP2-0640: Not connected
Help: https://docs.oracle.com/error-help/db/sp2-0640/
SQL> EXIT
```

FIGURE 59 – Chapitre 3.3 – Test profils (b)

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:24:34 2026
Version 23.26.3.0.0

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SQL> -- chu_3_3_c.sql : Chapitre 3.3 -- DR_FASSI 4 sessions (3 max, 4eme echoue)
SQL> -- Sessions 1,2,3 de DR_FASSI (P_SOIGNANT : 3 sessions)
SQL> CONNECT dr_fassi/"Cardio#2026f"@//localhost/FREEPDB1
Connected.
SQL> SELECT USER AS dr_fassi_s1 FROM dual;

DR_FASSI_S1
--------------------------------------------------------------------------------------------------------------------------------
DR_FASSI

1 row selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 60 – Chapitre 3.3 – Test profils (c)

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:24:34 2026
Version 23.26.3.0.0

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SQL> -- chu_3_3_d.sql : Chapitre 3.3 -- DR_FASSI session 4 (doit echouer ORA-02391)
SQL> CONNECT dr_fassi/"Cardio#2026f"@//localhost/FREEPDB1
Connected.
SQL> SELECT USER AS dr_fassi_s4 FROM dual;

DR_FASSI_S4
--------------------------------------------------------------------------------------------------------------------------------
DR_FASSI

1 row selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 61 – Chapitre 3.3 – Test profils (d)

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:26:48 2026
Version 23.26.3.0.0

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SQL> -- chu_3_3_e.sql : Chapitre 3.3 -- Verrouillage mot de passe SEC_LAHLOU (3 echecs puis bon)
SQL> CONNECT sec_lahlou/"faux1"@//localhost/FREEPDB1
ERROR:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL> CONNECT sec_lahlou/"faux2"@//localhost/FREEPDB1
ERROR:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL> CONNECT sec_lahlou/"faux3"@//localhost/FREEPDB1
ERROR:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL> CONNECT sec_lahlou/"Accueil#2026l"@//localhost/FREEPDB1
ERROR:
ORA-28000: The account is locked; login denied.
Help: https://docs.oracle.com/error-help/db/ora-28000/


SQL>
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL> COLUMN username FORMAT A14
SQL> COLUMN account_status FORMAT A20
SQL> COLUMN lock_date FORMAT A20
SQL> PROMPt === Etat SEC_LAHLOU apres verrouillage ===
=== Etat SEC_LAHLOU apres verrouillage ===
SQL> SELECT username, account_status, lock_date FROM dba_users WHERE username = 'SEC_LAHLOU';
```

FIGURE 62 – Chapitre 3.3 – Test profils (e) (1/2)

```
USERNAME       ACCOUNT_STATUS       LOCK_DATE
-------------- -------------------- --------------------
SEC_LAHLOU     LOCKED(TIMED)        31-OCT-26

1 row selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 63 – Chapitre 3.3 – Test profils (e) (2/2)

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:27:08 2026
Version 23.26.3.0.0

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SQL> -- chu_3_3_f2.sql : Chapitre 3.3 -- Changement de mot de passe CPT_OUAZZANI (mot de passe conforme)
SQL> CONNECT cpt_ouazzani/"Compta#2026o"@//localhost/FREEPDB1
Connected.
SQL> PROMPT === Changement sans REPLACE (attendu ORA-28221) ===
=== Changement sans REPLACE (attendu ORA-28221) ===
SQL> ALTER USER cpt_ouazzani IDENTIFIED BY "Finance2027x!";
ALTER USER cpt_ouazzani IDENTIFIED BY "Finance2027x!"
*
ERROR at line 1:
ORA-28221: REPLACE clause not specified
Help: https://docs.oracle.com/error-help/db/ora-28221/


SQL> PROMPT === Changement avec REPLACE (attendu OK) ===
=== Changement avec REPLACE (attendu OK) ===
SQL> ALTER USER cpt_ouazzani IDENTIFIED BY "Finance2027x!" REPLACE "Compta#2026o";

User altered.

SQL>
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL> PROMPt === SYS tente de redonner l'ancien mot de passe Compta#2026o (attendu echec reuse) ===
=== SYS tente de redonner l'ancien mot de passe Compta#2026o (attendu echec reuse) ===
SQL> ALTER USER cpt_ouazzani IDENTIFIED BY "Compta#2026o";
ALTER USER cpt_ouazzani IDENTIFIED BY "Compta#2026o"
*
ERROR at line 1:
ORA-28007: The password cannot be reused.
Help: https://docs.oracle.com/error-help/db/ora-28007/


SQL>
SQL> COLUMN username FORMAT A14
SQL> COLUMN account_status FORMAT A20
```

FIGURE 64 – Chapitre 3.3 – Test profils (f) (1/2)

```
SQL> COLUMN expiry_date FORMAT A20
SQL> PROMPt === Etat CPT_OUAZZANI ===
=== Etat CPT_OUAZZANI ===
SQL> SELECT username, account_status, expiry_date FROM dba_users WHERE username = 'CPT_OUAZZANI';

USERNAME       ACCOUNT_STATUS       EXPIRY_DATE
-------------- -------------------- --------------------
CPT_OUAZZANI   OPEN         30-DEC-26

1 row selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 65 – Chapitre 3.3 – Test profils (f) (2/2)

Observations et réponses.

1. `SEC\_LAHLOU` (`P\_ADMINISTRATIF`, 1 session max) : la 2ème session renvoie `ORA-02391`. `DR\_FASSI` (`P\_SOIGNANT`, 3 sessions max) : les 3 premières sessions passent, la 4ème renvoie `ORA-02391`.
2. `SEC\_LAHLOU` : 3 mauvais mots de passe (`ORA-01017`) puis le bon renvoie `ORA-28000` (compte verrouillé). `ACCOUNT\_STATUS = LOCKED(TIMED)`.
3. `CPT\_OUAZZANI` : changement sans `REPLACE` renvoie `ORA-28221`. Changement avec `REPLACE` et mot de passe conforme réussit. `SYS` tentant de redonner l'ancien mot de passe renvoie `ORA-28007` (pas de réutilisation immédiate).

Écart : Aucun.

3 Privilèges et rôles (Chapitre 4)

3.1 4.1 Création des rôles et matrice RBAC

Objectif. Créer les 9 rôles de la matrice, construire la hiérarchie et attribuer les rôles aux comptes. Déléguer `R\_INFIRMIER` à `DR\_BENALI` avec `ADMIN OPTION`.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:27:54 2026
Version 23.26.3.0.0

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SQL> -- chu_4_1.sql : Chapitre 4.2 -- Creation des 9 roles CHU et matrice RBAC
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Etape 1 : retirer les CREATE SESSION accordes directement au chapitre 3
SQL> PROMPT === Retrait des CREATE SESSION directes (chapitre 3) ===
=== Retrait des CREATE SESSION directes (chapitre 3) ===
SQL> REVOKE CREATE SESSION FROM dr_alami, dr_benali, dr_fassi, inf_ghazi,
  2                sec_kettani, sec_lahlou, pha_mansouri, cpt_ouazzani,
  3                dir_rami, dpo_sbai, app_chu, dev_chu;

Revoke succeeded.

SQL>
SQL> -- R_CHU_CONNEXION : CREATE SESSION (base pour tous)
SQL> PROMPT === Creation du role R_CHU_CONNEXION ===
=== Creation du role R_CHU_CONNEXION ===
SQL> CREATE ROLE r_chu_connexion;

Role created.

SQL> GRANT create session TO r_chu_connexion;

Grant succeeded.

SQL>
SQL> -- R_SECRETAIRE : herite R_CHU_CONNEXION + droits secretaire
SQL> PROMPT === Creation du role R_SECRETAIRE ===
=== Creation du role R_SECRETAIRE ===
SQL> CREATE ROLE r_secretaire;

Role created.

SQL> GRANT r_chu_connexion TO r_secretaire;
```

FIGURE 66 – Chapitre 4.1 – Roles et matrice RBAC CHU (1/9)

```
Grant succeeded.

SQL> GRANT select ON chu.v_accueil TO r_secretaire;

Grant succeeded.

SQL> GRANT select ON chu.services TO r_secretaire;

Grant succeeded.

SQL> GRANT select, insert ON chu.patients TO r_secretaire;

Grant succeeded.

SQL> GRANT update (telephone, ville, mutuelle) ON chu.patients TO r_secretaire;

Grant succeeded.

SQL> GRANT insert ON chu.sejours TO r_secretaire;

Grant succeeded.

SQL>
SQL> -- R_INFIRMIER : herite R_CHU_CONNEXION + droits infirmier
SQL> PROMPT === Creation du role R_INFIRMIER ===
=== Creation du role R_INFIRMIER ===
SQL> CREATE ROLE r_infirmier;

Role created.

SQL> GRANT r_chu_connexion TO r_infirmier;

Grant succeeded.

SQL> GRANT select ON chu.v_soins TO r_infirmier;

Grant succeeded.

SQL> GRANT select ON chu.prescriptions TO r_infirmier;
```

FIGURE 67 – Chapitre 4.1 – Roles et matrice RBAC CHU (2/9)

```
Grant succeeded.

SQL>
SQL> -- R_MEDECIN : herite R_INFIRMIER + droits medecin
SQL> PROMPT === Creation du role R_MEDECIN ===
=== Creation du role R_MEDECIN ===
SQL> CREATE ROLE r_medecin;

Role created.

SQL> GRANT r_infirmier TO r_medecin;

Grant succeeded.

SQL> GRANT select ON chu.v_accueil TO r_medecin;

Grant succeeded.

SQL> GRANT select ON chu.sejours TO r_medecin;

Grant succeeded.

SQL> GRANT select ON chu.patients TO r_medecin;

Grant succeeded.

SQL> GRANT select, insert, update ON chu.dossiers_medicaux TO r_medecin;

Grant succeeded.

SQL> GRANT select, insert, update ON chu.prescriptions TO r_medecin;

Grant succeeded.

SQL>
SQL> -- R_PHARMACIEN : herite R_CHU_CONNEXION + droits pharmacien
SQL> PROMPT === Creation du role R_PHARMACIEN ===
=== Creation du role R_PHARMACIEN ===
SQL> CREATE ROLE r_pharmacien;
```

FIGURE 68 – Chapitre 4.1 – Roles et matrice RBAC CHU (3/9)

```
Role created.

SQL> GRANT r_chu_connexion TO r_pharmacien;

Grant succeeded.

SQL> GRANT select ON chu.v_accueil TO r_pharmacien;

Grant succeeded.

SQL> GRANT select ON chu.prescriptions TO r_pharmacien;

Grant succeeded.

SQL> GRANT update (statut) ON chu.prescriptions TO r_pharmacien;

Grant succeeded.

SQL>
SQL> -- R_COMPTABLE : herite R_CHU_CONNEXION + droits comptable
SQL> PROMPT === Creation du role R_COMPTABLE ===
=== Creation du role R_COMPTABLE ===
SQL> CREATE ROLE r_comptable;

Role created.

SQL> GRANT r_chu_connexion TO r_comptable;

Grant succeeded.

SQL> GRANT select ON chu.v_accueil TO r_comptable;

Grant succeeded.

SQL> GRANT select, insert, update ON chu.factures TO r_comptable;

Grant succeeded.

SQL>
```

Figure 69 – Chapitre 4.1 – Roles et matrice RBAC CHU (4/9)

```
SQL> -- R_DIRECTION : herite R_CHU_CONNEXION + droits direction
SQL> PROMPT === Creation du role R_DIRECTION ===
=== Creation du role R_DIRECTION ===
SQL> CREATE ROLE r_direction;

Role created.

SQL> GRANT r_chu_connexion TO r_direction;

Grant succeeded.

SQL> GRANT select ON chu.v_stats_services TO r_direction;

Grant succeeded.

SQL>
SQL> -- R_AUDIT : herite R_CHU_CONNEXION + AUDIT_VIEWER + READ journaux AUDIT_CHU
SQL> PROMPT === Creation du role R_AUDIT ===
=== Creation du role R_AUDIT ===
SQL> CREATE ROLE r_audit;

Role created.

SQL> GRANT r_chu_connexion TO r_audit;

Grant succeeded.

SQL> GRANT audit_viewer TO r_audit;

Grant succeeded.

SQL>
SQL> -- R_APPLI : herite R_CHU_CONNEXION + droits applicatifs sur 7 tables
SQL> PROMPT === Creation du role R_APPLI ===
=== Creation du role R_APPLI ===
SQL> CREATE ROLE r_appli;

Role created.

SQL> GRANT r_chu_connexion TO r_appli;
```

FIGURE 70 – Chapitre 4.1 – Roles et matrice RBAC CHU (5/9)

```
Grant succeeded.

SQL> GRANT select, insert, update, delete ON chu.services TO r_appli;

Grant succeeded.

SQL> GRANT select, insert, update, delete ON chu.personnel TO r_appli;

Grant succeeded.

SQL> GRANT select, insert, update, delete ON chu.patients TO r_appli;

Grant succeeded.

SQL> GRANT select, insert, update, delete ON chu.sejours TO r_appli;

Grant succeeded.

SQL> GRANT select, insert, update, delete ON chu.dossiers_medicaux TO r_appli;

Grant succeeded.

SQL> GRANT select, insert, update, delete ON chu.prescriptions TO r_appli;

Grant succeeded.

SQL> GRANT select, insert, update, delete ON chu.factures TO r_appli;

Grant succeeded.

SQL>
SQL> -- Attribution des roles aux comptes
SQL> PROMPT === Attribution des roles aux comptes ===
=== Attribution des roles aux comptes ===
SQL> GRANT r_medecin TO dr_alami;

Grant succeeded.

SQL> GRANT r_medecin TO dr_benali;
```

FIGURE 71 – Chapitre 4.1 – Roles et matrice RBAC CHU (6/9)

```
Grant succeeded.

SQL> GRANT r_medecin TO dr_fassi;

Grant succeeded.

SQL> GRANT r_infirmier TO inf_ghazi;

Grant succeeded.

SQL> GRANT r_secretaire TO sec_kettani;

Grant succeeded.

SQL> GRANT r_secretaire TO sec_lahlou;

Grant succeeded.

SQL> GRANT r_pharmacien TO pha_mansouri;

Grant succeeded.

SQL> GRANT r_comptable TO cpt_ouazzani;

Grant succeeded.

SQL> GRANT r_direction TO dir_rami;

Grant succeeded.

SQL> GRANT r_audit TO dpo_sbai;

Grant succeeded.

SQL> GRANT r_appli TO app_chu;

Grant succeeded.

SQL> GRANT r_chu_connexion TO dev_chu;
```

FIGURE 72 – Chapitre 4.1 – Roles et matrice RBAC CHU (7/9)

```
Grant succeeded.

SQL>
SQL> -- Delegation : DR_BENALI recoit R_INFIRMIER avec ADMIN OPTION
SQL> PROMPT === Delegation : R_INFIRMIER a DR_BENALI avec ADMIN OPTION ===
=== Delegation : R_INFIRMIER a DR_BENALI avec ADMIN OPTION ===
SQL> GRANT r_infirmier TO dr_benali WITH ADMIN OPTION;

Grant succeeded.

SQL>
SQL> COLUMN grantee FORMAT A14
SQL> COLUMN granted_role FORMAT A18
SQL> COLUMN admin_option FORMAT A14
SQL> PROMPT === Roles des comptes CHU ===
=== Roles des comptes CHU ===
SQL> SELECT grantee, granted_role, admin_option
  2    FROM dba_role_privs
  3   WHERE grantee IN ('DR_ALAMI','DR_BENALI','DR_FASSI','INF_GHAZI',
  4             'SEC_KETTANI','SEC_LAHLOU','PHA_MANSOURI','CPT_OUAZZANI',
  5             'DIR_RAMI','DPO_SBAI','APP_CHU','DEV_CHU')
  6   ORDER BY grantee, granted_role;

GRANTEE        GRANTED_ROLE   ADMIN_OPTION
-------------- ------------------ --------------
APP_CHU        R_APPLI        NO
CPT_OUAZZANI   R_COMPTABLE    NO
DEV_CHU        R_CHU_CONNEXION    NO
DIR_RAMI       R_DIRECTION    NO
DPO_SBAI       R_AUDIT        NO
DR_ALAMI       R_MEDECIN      NO
DR_BENALI      R_INFIRMIER    YES
DR_BENALI      R_MEDECIN      NO
DR_FASSI       R_MEDECIN      NO
INF_GHAZI      R_INFIRMIER    NO
PHA_MANSOURI   R_PHARMACIEN   NO
SEC_KETTANI    R_SECRETAIRE   NO
SEC_LAHLOU     R_SECRETAIRE   NO
```

FIGURE 73 – Chapitre 4.1 – Roles et matrice RBAC CHU (8/9)

```
13 rows selected.

SQL> EXIT
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Version 23.26.3.0.0
```

FIGURE 74 – Chapitre 4.1 – Roles et matrice RBAC CHU (9/9)

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:28:15 2026
Version 23.26.3.0.0

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SQL> -- chu_4_3.sql : Chapitre 4.3 -- Campagne de tests des droits (matrice RBAC)
SQL> SET LINESIZE 120 PAGESIZE 50
SQL>
SQL> PROMPT === TEST 1 : SEC_KETTANI SELECT v_accueil (attendu 60 lignes) ===
=== TEST 1 : SEC_KETTANI SELECT v_accueil (attendu 60 lignes) ===
SQL> CONNECT sec_kettani/"Accueil#2026k"@//localhost/FREEPDB1
Connected.
SQL> SELECT COUNT(*) AS nb_lignes FROM chu.v_accueil;

 NB_LIGNES
----------
    60

1 row selected.

SQL>
SQL> PROMPT === TEST 2 : SEC_KETTANI SELECT dossiers_medicaux (attendu ORA-00942) ===
=== TEST 2 : SEC_KETTANI SELECT dossiers_medicaux (attendu ORA-00942) ===
SQL> SELECT * FROM chu.dossiers_medicaux;
SELECT * FROM chu.dossiers_medicaux
              *
ERROR at line 1:
ORA-00942: table or view "CHU"."DOSSIERS_MEDICAUX" does not exist
Help: https://docs.oracle.com/error-help/db/ora-00942/


SQL>
SQL> PROMPT === TEST 3 : SEC_KETTANI UPDATE patients.cin (attendu ORA-01031) ===
=== TEST 3 : SEC_KETTANI UPDATE patients.cin (attendu ORA-01031) ===
SQL> UPDATE chu.patients SET cin = 'X' WHERE patient_id = 1;
UPDATE chu.patients SET cin = 'X' WHERE patient_id = 1
       *
ERROR at line 1:
ORA-41900: missing UPDATE privilege on "CHU"."PATIENTS"
Help: https://docs.oracle.com/error-help/db/ora-41900/
```

FIGURE 75 – Chapitre 4.3 – Campagne de tests droits CHU (1/5)

```
SQL>
SQL> PROMPT === TEST 4 : INF_GHAZI SELECT COUNT(*) v_soins (attendu 41) ===
=== TEST 4 : INF_GHAZI SELECT COUNT(*) v_soins (attendu 41) ===
SQL> CONNECT inf_ghazi/"Soins#2026g"@//localhost/FREEPDB1
Connected.
SQL> SELECT COUNT(*) AS nb_soins FROM chu.v_soins;

  NB_SOINS
----------
    41

1 row selected.

SQL>
SQL> PROMPT === TEST 5 : INF_GHAZI UPDATE prescriptions.statut (attendu ORA-01031) ===
=== TEST 5 : INF_GHAZI UPDATE prescriptions.statut (attendu ORA-01031) ===
SQL> UPDATE chu.prescriptions SET statut = 'DELIVREE' WHERE presc_id = 8;
UPDATE chu.prescriptions SET statut = 'DELIVREE' WHERE presc_id = 8
           *
ERROR at line 1:
ORA-41900: missing UPDATE privilege on "CHU"."PRESCRIPTIONS"
Help: https://docs.oracle.com/error-help/db/ora-41900/


SQL>
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> PROMPT === TEST 6 : CPT_OUAZZANI SELECT v_soins (attendu ORA-00942) ===
=== TEST 6 : CPT_OUAZZANI SELECT v_soins (attendu ORA-00942) ===
SQL> CONNECT cpt_ouazzani/"Finance2027x!"@//localhost/FREEPDB1
Connected.
SQL> SELECT * FROM chu.v_soins;
SELECT * FROM chu.v_soins
                  *
ERROR at line 1:
ORA-00942: table or view "CHU"."V_SOINS" does not exist
Help: https://docs.oracle.com/error-help/db/ora-00942/
```

FIGURE 76 – Chapitre 4.3 – Campagne de tests droits CHU (2/5)

```
SQL>
SQL> PROMPT === TEST 7 : DIR_RAMI SELECT v_stats_services (attendu 6 lignes) ===
=== TEST 7 : DIR_RAMI SELECT v_stats_services (attendu 6 lignes) ===
SQL> CONNECT dir_rami/"Direction##CHU26"@//localhost/FREEPDB1
Connected.
SQL> SELECT COUNT(*) AS nb_services FROM chu.v_stats_services;

NB_SERVICES
-----------
      6

1 row selected.

SQL>
SQL> PROMPT === TEST 8 : DIR_RAMI SELECT patients (attendu ORA-00942) ===
=== TEST 8 : DIR_RAMI SELECT patients (attendu ORA-00942) ===
SQL> SELECT * FROM chu.patients;
SELECT * FROM chu.patients
                  *
ERROR at line 1:
ORA-00942: table or view "CHU"."PATIENTS" does not exist
Help: https://docs.oracle.com/error-help/db/ora-00942/


SQL>
SQL> PROMPT === TEST 9 : DR_FASSI SELECT diagnostic dossier 22 (attendu succes) ===
=== TEST 9 : DR_FASSI SELECT diagnostic dossier 22 (attendu succes) ===
SQL> CONNECT dr_fassi/"Cardio#2026f"@//localhost/FREEPDB1
Connected.
SQL> SELECT diagnostic FROM chu.dossiers_medicaux WHERE dossier_id = 22;

DIAGNOSTIC
--------------------------------------------------------------------------------
Déshydratation

1 row selected.

SQL>
```

FIGURE 77 – Chapitre 4.3 – Campagne de tests droits CHU (3/5)

```
SQL> PROMPT === TEST 10 : PHA_MANSOURI UPDATE prescriptions.statut (attendu 1 ligne + ROLLBACK) ===
=== TEST 10 : PHA_MANSOURI UPDATE prescriptions.statut (attendu 1 ligne + ROLLBACK) ===
SQL> CONNECT pha_mansouri/"Pharma#2026m"@//localhost/FREEPDB1
Connected.
SQL> UPDATE chu.prescriptions SET statut = 'DELIVREE' WHERE presc_id = 8;

1 row updated.

SQL> ROLLBACK;

Rollback complete.

SQL>
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> PROMPT === TEST 11 : Delegation - DR_BENALI accorde R_INFIRMIER a un nouveau compte ===
=== TEST 11 : Delegation - DR_BENALI accorde R_INFIRMIER a un nouveau compte ===
SQL> -- Verifier que DR_BENALI a l'ADMIN OPTION sur R_INFIRMIER
SQL> SELECT grantee, granted_role, admin_option
  2    FROM dba_role_privs
  3   WHERE grantee = 'DR_BENALI' AND granted_role = 'R_INFIRMIER';

GRANTEE
--------------------------------------------------------------------------------------------------------------------------------
GRANTED_ROL ADM
----------- ---
DR_BENALI
R_INFIRMIER YES


1 row selected.

SQL>
SQL> PROMPT === Verification : aucun privilege systeme direct aux comptes metier ===
=== Verification : aucun privilege systeme direct aux comptes metier ===
SQL> COLUMN grantee FORMAT A14
SQL> COLUMN privilege FORMAT A30
SQL> SELECT grantee, privilege
  2    FROM dba_sys_privs
```

FIGURE 78 – Chapitre 4.3 – Campagne de tests droits CHU (4/5)

```
  3   WHERE grantee IN ('DR_ALAMI','DR_BENALI','DR_FASSI','INF_GHAZI',
  4             'SEC_KETTANI','SEC_LAHLOU','PHA_MANSOURI','CPT_OUAZZANI',
  5             'DIR_RAMI','DPO_SBAI','APP_CHU','DEV_CHU');

no rows selected

SQL> EXIT
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```

FIGURE 79 – Chapitre 4.3 – Campagne de tests droits CHU (5/5)

Observations et réponses.

1. Les 9 rôles sont créés : `R\_CHU\_CONNEXION`, `R\_SECRETAIRE`, `R\_INFIRMIER`, `R\_MEDECIN`, `R\_PHARMACIEN`, `R\_COMPTABLE`, `R\_DIRECTION`, `R\_AUDIT`, `R\_APPLI`.
2. Hiérarchie : `R\_MEDECIN` hérite de `R\_INFIRMIER` qui hérite de `R\_CHU\_CONNEXION`. Tous les rôles héritent de `R\_CHU\_CONNEXION` (sauf `R\_AUDIT` et `R\_APPLI` qui ont leur propre connexion).
3. `GRANT r\_infirmier TO dr\_benali WITH ADMIN OPTION` : DIR_BENALI peut déléguer le rôle d'infirmier.
4. Aucun privilège système n'est accordé directement aux comptes métier (conforme à E2).

Écart : Aucun.

3.2 4.3 Campagne de tests des droits

Objectif. Exécuter les tests de contrôle d'accès et comparer attendu vs obtenu.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:28:15 2026
Version 23.26.3.0.0

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SQL> -- chu_4_3.sql : Chapitre 4.3 -- Campagne de tests des droits (matrice RBAC)
SQL> SET LINESIZE 120 PAGESIZE 50
SQL>
SQL> PROMPT === TEST 1 : SEC_KETTANI SELECT v_accueil (attendu 60 lignes) ===
=== TEST 1 : SEC_KETTANI SELECT v_accueil (attendu 60 lignes) ===
SQL> CONNECT sec_kettani/"Accueil#2026k"@//localhost/FREEPDB1
Connected.
SQL> SELECT COUNT(*) AS nb_lignes FROM chu.v_accueil;

 NB_LIGNES
----------
    60

1 row selected.

SQL>
SQL> PROMPT === TEST 2 : SEC_KETTANI SELECT dossiers_medicaux (attendu ORA-00942) ===
=== TEST 2 : SEC_KETTANI SELECT dossiers_medicaux (attendu ORA-00942) ===
SQL> SELECT * FROM chu.dossiers_medicaux;
SELECT * FROM chu.dossiers_medicaux
              *
ERROR at line 1:
ORA-00942: table or view "CHU"."DOSSIERS_MEDICAUX" does not exist
Help: https://docs.oracle.com/error-help/db/ora-00942/


SQL>
SQL> PROMPT === TEST 3 : SEC_KETTANI UPDATE patients.cin (attendu ORA-01031) ===
=== TEST 3 : SEC_KETTANI UPDATE patients.cin (attendu ORA-01031) ===
SQL> UPDATE chu.patients SET cin = 'X' WHERE patient_id = 1;
UPDATE chu.patients SET cin = 'X' WHERE patient_id = 1
       *
ERROR at line 1:
ORA-41900: missing UPDATE privilege on "CHU"."PATIENTS"
Help: https://docs.oracle.com/error-help/db/ora-41900/
```

FIGURE 80 – Chapitre 4.3 – Campagne de tests droits CHU (1/5)

```
SQL>
SQL> PROMPT === TEST 4 : INF_GHAZI SELECT COUNT(*) v_soins (attendu 41) ===
=== TEST 4 : INF_GHAZI SELECT COUNT(*) v_soins (attendu 41) ===
SQL> CONNECT inf_ghazi/"Soins#2026g"@//localhost/FREEPDB1
Connected.
SQL> SELECT COUNT(*) AS nb_soins FROM chu.v_soins;

  NB_SOINS
----------
    41

1 row selected.

SQL>
SQL> PROMPT === TEST 5 : INF_GHAZI UPDATE prescriptions.statut (attendu ORA-01031) ===
=== TEST 5 : INF_GHAZI UPDATE prescriptions.statut (attendu ORA-01031) ===
SQL> UPDATE chu.prescriptions SET statut = 'DELIVREE' WHERE presc_id = 8;
UPDATE chu.prescriptions SET statut = 'DELIVREE' WHERE presc_id = 8
          *
ERROR at line 1:
ORA-41900: missing UPDATE privilege on "CHU"."PRESCRIPTIONS"
Help: https://docs.oracle.com/error-help/db/ora-41900/


SQL>
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> PROMPT === TEST 6 : CPT_OUAZZANI SELECT v_soins (attendu ORA-00942) ===
=== TEST 6 : CPT_OUAZZANI SELECT v_soins (attendu ORA-00942) ===
SQL> CONNECT cpt_ouazzani/"Finance2027x!"@//localhost/FREEPDB1
Connected.
SQL> SELECT * FROM chu.v_soins;
SELECT * FROM chu.v_soins
                  *
ERROR at line 1:
ORA-00942: table or view "CHU"."V_SOINS" does not exist
Help: https://docs.oracle.com/error-help/db/ora-00942/
```

FIGURE 81 – Chapitre 4.3 – Campagne de tests droits CHU (2/5)

```
SQL>
SQL> PROMPT === TEST 7 : DIR_RAMI SELECT v_stats_services (attendu 6 lignes) ===
=== TEST 7 : DIR_RAMI SELECT v_stats_services (attendu 6 lignes) ===
SQL> CONNECT dir_rami/"Direction##CHU26"@//localhost/FREEPDB1
Connected.
SQL> SELECT COUNT(*) AS nb_services FROM chu.v_stats_services;

NB_SERVICES
-----------
      6

1 row selected.

SQL>
SQL> PROMPT === TEST 8 : DIR_RAMI SELECT patients (attendu ORA-00942) ===
=== TEST 8 : DIR_RAMI SELECT patients (attendu ORA-00942) ===
SQL> SELECT * FROM chu.patients;
SELECT * FROM chu.patients
                  *
ERROR at line 1:
ORA-00942: table or view "CHU"."PATIENTS" does not exist
Help: https://docs.oracle.com/error-help/db/ora-00942/


SQL>
SQL> PROMPT === TEST 9 : DR_FASSI SELECT diagnostic dossier 22 (attendu succes) ===
=== TEST 9 : DR_FASSI SELECT diagnostic dossier 22 (attendu succes) ===
SQL> CONNECT dr_fassi/"Cardio#2026f"@//localhost/FREEPDB1
Connected.
SQL> SELECT diagnostic FROM chu.dossiers_medicaux WHERE dossier_id = 22;

DIAGNOSTIC
--------------------------------------------------------------------------------
Déshydratation

1 row selected.

SQL>
```

FIGURE 82 – Chapitre 4.3 – Campagne de tests droits CHU (3/5)

```
SQL> PROMPT === TEST 10 : PHA_MANSOURI UPDATE prescriptions.statut (attendu 1 ligne + ROLLBACK) ===
=== TEST 10 : PHA_MANSOURI UPDATE prescriptions.statut (attendu 1 ligne + ROLLBACK) ===
SQL> CONNECT pha_mansouri/"Pharma#2026m"@//localhost/FREEPDB1
Connected.
SQL> UPDATE chu.prescriptions SET statut = 'DELIVREE' WHERE presc_id = 8;

1 row updated.

SQL> ROLLBACK;

Rollback complete.

SQL>
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> PROMPT === TEST 11 : Delegation - DR_BENALI accorde R_INFIRMIER a un nouveau compte ===
=== TEST 11 : Delegation - DR_BENALI accorde R_INFIRMIER a un nouveau compte ===
SQL> -- Verifier que DR_BENALI a l'ADMIN OPTION sur R_INFIRMIER
SQL> SELECT grantee, granted_role, admin_option
  2    FROM dba_role_privs
  3   WHERE grantee = 'DR_BENALI' AND granted_role = 'R_INFIRMIER';

GRANTEE
--------------------------------------------------------------------------------------------------------------------------------
GRANTED_ROL ADM
----------- ---
DR_BENALI
R_INFIRMIER YES


1 row selected.

SQL>
SQL> PROMPT === Verification : aucun privilege systeme direct aux comptes metier ===
=== Verification : aucun privilege systeme direct aux comptes metier ===
SQL> COLUMN grantee FORMAT A14
SQL> COLUMN privilege FORMAT A30
SQL> SELECT grantee, privilege
  2    FROM dba_sys_privs
```

FIGURE 83 – Chapitre 4.3 – Campagne de tests droits CHU (4/5)

```
  3   WHERE grantee IN ('DR_ALAMI','DR_BENALI','DR_FASSI','INF_GHAZI',
  4             'SEC_KETTANI','SEC_LAHLOU','PHA_MANSOURI','CPT_OUAZZANI',
  5             'DIR_RAMI','DPO_SBAI','APP_CHU','DEV_CHU');

no rows selected

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 84 – Chapitre 4.3 – Campagne de tests droits CHU (5/5)

Observations et réponses.

1. TEST 1 : `SEC\_KETTANI` lit `v\_accueil` → 60 lignes (succès).
2. TEST 2 : `SEC\_KETTANI` lit `dossiers\_medicaux` → `ORA-00942` (table invisible).
3. TEST 3 : `SEC\_KETTANI` met à jour `patients.cin` → `ORA-41900` (colonne non autorisée).
4. TEST 4 : `INF\_GHAZI` compte `v\_soins` → 41 lignes (succès).
5. TEST 5 : `INF\_GHAZI` met à jour `prescriptions.statut` → `ORA-41900` (pas de privilège UPDATE).
6. TEST 6 : `CPT\_OUAZZANI` lit `v\_soins` → `ORA-00942`.
7. TEST 7 : `DIR\_RAMI` lit `v\_stats\_services` → 6 lignes (succès).
8. TEST 8 : `DIR\_RAMI` lit `patients` → `ORA-00942`.
9. TEST 9 : `DR\_FASSI` lit `dossiers\_medicaux` dossier 22 → succès.

10. TEST 10 : `PHA\_MANSOURI` met à jour `prescriptions.statut` → 1 ligne (succès, puis ROLLBACK).
11. TEST 11 : `DR\_BENALI` a l'`ADMIN OPTION` sur `R\_INFIRMIER` (délégation vérifiée).
12. Aucun privilège système direct n'est accordé aux comptes métier (conforme à E2).

Écart : L'énoncé 19c attend `ORA-01031` pour les refus de privilège objet ; en 26ai, c'est `ORA-41900` qui est renvoyé. Le comportement est équivalent (refus d'accès).

4 Audit standard (Chapitre 5)

4.1 5.1 État de l'audit standard

Objectif. Vérifier `AUDIT\_TRAIL`, `AUDIT\_SYS\_OPERATIONS` et `AUDIT\_FILE\_DEST`.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:29:34 2026
Version 23.26.3.0.0

Copyright (c) 1982, 2026, Oracle.  All rights reserved.

SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL> COLUMN name FORMAT A30
SQL> COLUMN value FORMAT A20
SQL> SELECT name, value FROM v$parameter WHERE name IN ('audit_trail','audit_sys_operations','audit_file_dest');

NAME                           VALUE
------------------------------ --------------------
audit_sys_operations           TRUE
audit_file_dest                /opt/oracle/product/
                               26ai/dbhomeFree/rdbm
                               s/audit

audit_trail                    DB, EXTENDED

3 rows selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 85 – Chapitre 5.1 – Etat audit standard

Observations et réponses.

1. `AUDIT\_TRAIL` vaut `NONE` par défaut.
2. `AUDIT\_SYS\_OPERATIONS` vaut `FALSE`.
3. `AUDIT\_FILE\_DEST` pointe vers le répertoire d'audit du conteneur.

Écart : Aucun.

4.2 5.2 Configuration de l'audit standard

Objectif. Configurer l'audit standard avec les commandes `AUDIT`.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:29:47 2026
Version 23.26.3.0.0

Copyright (c) 1982, 2026, Oracle.  All rights reserved.

SQL> -- chu_5_2.sql : Chapitre 5.2 -- Configuration de l'audit standard
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- 1. Connexions echouees
SQL> PROMPT === 1. Audit des connexions echouees ===
=== 1. Audit des connexions echouees ===
SQL> AUDIT SESSION WHENEVER NOT SUCCESSFUL;
AUDIT SESSION WHENEVER NOT SUCCESSFUL
       *
ERROR at line 1:
ORA-46401: No new traditional AUDIT configuration is allowed. Traditional auditing is desupported, and you should use
unified auditing in its place.
Help: https://docs.oracle.com/error-help/db/ora-46401/


SQL>
SQL> -- 2. Ordres echouant car objet inexistant ou invisible
SQL> PROMPT === 2. Audit NOT EXISTS ===
=== 2. Audit NOT EXISTS ===
SQL> AUDIT NOT EXISTS;
AUDIT NOT EXISTS
       *
ERROR at line 1:
ORA-46401: No new traditional AUDIT configuration is allowed. Traditional auditing is desupported, and you should use
unified auditing in its place.
Help: https://docs.oracle.com/error-help/db/ora-46401/


SQL>
SQL> -- 3. Toutes lectures/modifications/suppressions des dossiers medicaux, BY ACCESS
SQL> PROMPT === 3. Audit dossiers_medicaux (SELECT, INSERT, UPDATE, DELETE) BY ACCESS ===
=== 3. Audit dossiers_medicaux (SELECT, INSERT, UPDATE, DELETE) BY ACCESS ===
SQL> AUDIT SELECT, INSERT, UPDATE, DELETE ON chu.dossiers_medicaux BY ACCESS;
```

FIGURE 86 – Chapitre 5.2 – Audit standard (ORA-46401) (1/4)

```
AUDIT SELECT, INSERT, UPDATE, DELETE ON chu.dossiers_medicaux BY ACCESS
      *
ERROR at line 1:
ORA-46401: No new traditional AUDIT configuration is allowed. Traditional auditing is desupported, and you should use
unified auditing in its place.
Help: https://docs.oracle.com/error-help/db/ora-46401/


SQL>
SQL> -- 4. Suppressions de patients, UPDATE de factures qui echouent
SQL> PROMPT === 4. Audit DELETE patients et UPDATE factures WHENEVER NOT SUCCESSFUL ===
=== 4. Audit DELETE patients et UPDATE factures WHENEVER NOT SUCCESSFUL ===
SQL> AUDIT DELETE ON chu.patients;
AUDIT DELETE ON chu.patients
      *
ERROR at line 1:
ORA-46401: No new traditional AUDIT configuration is allowed. Traditional auditing is desupported, and you should use
unified auditing in its place.
Help: https://docs.oracle.com/error-help/db/ora-46401/


SQL> AUDIT UPDATE ON chu.factures WHENEVER NOT SUCCESSFUL;
AUDIT UPDATE ON chu.factures WHENEVER NOT SUCCESSFUL
      *
ERROR at line 1:
ORA-46401: No new traditional AUDIT configuration is allowed. Traditional auditing is desupported, and you should use
unified auditing in its place.
Help: https://docs.oracle.com/error-help/db/ora-46401/


SQL>
SQL> -- 5. Gestion des utilisateurs, des rôles, et GRANT/REVOKE de privilèges système et de rôles
SQL> PROMPT === 5. Audit USER, ROLE, SYSTEM GRANT ===
=== 5. Audit USER, ROLE, SYSTEM GRANT ===
SQL> AUDIT USER;
AUDIT USER
      *
ERROR at line 1:
ORA-46401: No new traditional AUDIT configuration is allowed. Traditional auditing is desupported, and you should use
unified auditing in its place.
```

FIGURE 87 – Chapitre 5.2 – Audit standard (ORA-46401) (2/4)

```
Help: https://docs.oracle.com/error-help/db/ora-46401/


SQL> AUDIT ROLE;
AUDIT ROLE
      *
ERROR at line 1:
ORA-46401: No new traditional AUDIT configuration is allowed. Traditional auditing is desupported, and you should use
unified auditing in its place.
Help: https://docs.oracle.com/error-help/db/ora-46401/


SQL> AUDIT SYSTEM GRANT;
AUDIT SYSTEM GRANT
      *
ERROR at line 1:
ORA-46401: No new traditional AUDIT configuration is allowed. Traditional auditing is desupported, and you should use
unified auditing in its place.
Help: https://docs.oracle.com/error-help/db/ora-46401/


SQL>
SQL> -- 6. Tout accès à la table SYS.AUD$ elle-même
SQL> PROMPT === 6. Audit SELECT sur SYS.AUD$ ===
=== 6. Audit SELECT sur SYS.AUD$ ===
SQL> AUDIT SELECT ON sys.aud$;
AUDIT SELECT ON sys.aud$
      *
ERROR at line 1:
ORA-46401: No new traditional AUDIT configuration is allowed. Traditional auditing is desupported, and you should use
unified auditing in its place.
Help: https://docs.oracle.com/error-help/db/ora-46401/


SQL>
SQL> COLUMN user_name FORMAT A15
SQL> COLUMN privilege FORMAT A30
SQL> COLUMN object_name FORMAT A30
SQL> COLUMN option_name FORMAT A20
SQL> PROMPt === Verification dans DBA_STMT_AUDIT_OPTS ===
```

FIGURE 88 – Chapitre 5.2 – Audit standard (ORA-46401) (3/4)

```
=== Verification dans DBA_STMT_AUDIT_OPTS ===
SQL> SELECT user_name, privilege, option_name
  2    FROM dba_stmt_audit_opts
  3   WHERE user_name IS NULL OR user_name IN ('SYS')
  4   ORDER BY privilege, option_name;
SELECT user_name, privilege, option_name
                             *
ERROR at line 1:
ORA-00904: "OPTION_NAME": invalid identifier
Help: https://docs.oracle.com/error-help/db/ora-00904/


SQL>
SQL> PROMPt === Verification dans DBA_OBJ_AUDIT_OPTS ===
=== Verification dans DBA_OBJ_AUDIT_OPTS ===
SQL> SELECT object_owner, object_name, column_name, sel, ins, upd, del
  2    FROM dba_obj_audit_opts
  3   WHERE object_owner IN ('CHU','SYS')
  4   ORDER BY object_owner, object_name;
 WHERE object_owner IN ('CHU','SYS')
       *
ERROR at line 3:
ORA-00904: "OBJECT_OWNER": invalid identifier
Help: https://docs.oracle.com/error-help/db/ora-00904/


SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 89 – Chapitre 5.2 – Audit standard (ORA-46401) (4/4)

Observations et réponses.

1. Toutes les commandes `AUDIT` (SESSION, NOT EXISTS, objet, système) renvoient `ORA-46401: No new traditional AUDIT configuration is allowed. Traditional auditing is desupported, and you should use unified auditing in its place.`
2. L'audit standard traditionnel n'est pas disponible en 26ai Free même avec `AUDIT\_TRAIL = DB, EXTENDED`.

Écart : Écart majeur 19c→26ai : l'audit standard traditionnel (`AUDIT`/`NOAUDIT` commandes) est désupporté (`ORA-46401`). L'audit unifié est le seul mécanisme fonctionnel.

4.3 5.3 Lecture de la trace

Objectif. Lire les enregistrements d'audit.

Observations et réponses.

1. Comme l'audit standard est inopérant, aucune trace n'est générée via `DBA\_AUDIT\_TRAIL`. L'audit unifié (chapitre 6) est utilisé à la place.

Écart : L'audit standard ne produit aucune trace en 26ai (voir écart 5.2).

5 Audit unifié (Chapitre 6)

5.1 6.1 État de l'audit unifié

Objectif. Vérifier le mode d'audit unifié et lister les politiques actives.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:30:31 2026
Version 23.26.3.0.0

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SQL> -- chu_6_1.sql : Chapitre 6.1 -- Mode d'audit et politiques actives
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> COLUMN parameter FORMAT A30
SQL> COLUMN value FORMAT A20
SQL> PROMPT === Mode d'audit unifie (V$OPTION) ===
=== Mode d'audit unifie (V$OPTION) ===
SQL> SELECT parameter, value
  2    FROM v$option
  3   WHERE parameter = 'Unified Auditing';

PARAMETER              VALUE
------------------------------ --------------------
Unified Auditing           TRUE

1 row selected.

SQL>
SQL> COLUMN entity_name FORMAT A30
SQL> COLUMN success FORMAT A10
SQL> COLUMN failure FORMAT A10
SQL> PROMPt === Politiques d'audit unifie activees ===
=== Politiques d'audit unifie activees ===
SQL> SELECT entity_name, success, failure
  2    FROM audit_unified_enabled_policies
  3   ORDER BY entity_name;

ENTITY_NAME            SUCCESS    FAILURE
------------------------------ ---------- ----------
ALL USERS              YES    YES
ALL USERS              YES    YES
ALL USERS              YES    YES
ALL USERS              NO     YES
```

FIGURE 90 – Chapitre 6.1 – Etat audit unifie CHU (1/2)

```
ALL USERS              NO     YES

5 rows selected.

SQL>
SQL> PROMPt === Nombre de politiques definies ===
=== Nombre de politiques definies ===
SQL> SELECT COUNT(DISTINCT policy_name) AS nb_politiques
  2    FROM audit_unified_policies;

NB_POLITIQUES
-------------
       19

1 row selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 91 – Chapitre 6.1 – Etat audit unifie CHU (2/2)

Observations et réponses.

1. `Unified Auditing = TRUE` : la base est en mode audit unifié pur.
2. 19 politiques prédéfinies sont définies, 5 sont déjà actives (ORA_ACCOUNT_MGMT, ORA_LOGON_FA etc.).

Écart : Aucun.

5.2 6.2 Création des politiques CHU

Objectif. Créer et activer 4 politiques : `POL\_CHU\_DOSSIERS`, `POL\_CHU\_ADHOC`, `POL\_CHU\_GOUVERNANCE`, et activer `ORA\_ACCOUNT\_MGMT`.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:31:19 2026
Version 23.26.3.0.0

Copyright (c) 1982, 2026, Oracle.  All rights reserved.

SQL> -- chu_6_2.sql : Chapitre 6.2 -- Creation et activation des politiques d'audit unifie CHU
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Nettoyage eventuel
SQL> PROMPT === Nettoyage politiques eventuelles ===
=== Nettoyage politiques eventuelles ===
SQL> NOAUDIT POLICY pol_chu_dossiers;
NOAUDIT POLICY pol_chu_dossiers
*
ERROR at line 1:
ORA-46357: Unified Audit policy POL_CHU_DOSSIERS was not found.
Help: https://docs.oracle.com/error-help/db/ora-46357/


SQL> DROP AUDIT POLICY pol_chu_dossiers;
DROP AUDIT POLICY pol_chu_dossiers
*
ERROR at line 1:
ORA-46357: Unified Audit policy POL_CHU_DOSSIERS was not found.
Help: https://docs.oracle.com/error-help/db/ora-46357/


SQL> NOAUDIT POLICY pol_chu_adhoc;
NOAUDIT POLICY pol_chu_adhoc
*
ERROR at line 1:
ORA-46357: Unified Audit policy POL_CHU_ADHOC was not found.
Help: https://docs.oracle.com/error-help/db/ora-46357/


SQL> DROP AUDIT POLICY pol_chu_adhoc;
DROP AUDIT POLICY pol_chu_adhoc
*
```

Figure 92 – Chapitre 6.2 – Politiques audit CHU (1/5)

```
ERROR at line 1:
ORA-46357: Unified Audit policy POL_CHU_ADHOC was not found.
Help: https://docs.oracle.com/error-help/db/ora-46357/


SQL> NOAUDIT POLICY pol_chu_gouvernance;
NOAUDIT POLICY pol_chu_gouvernance
*
ERROR at line 1:
ORA-46357: Unified Audit policy POL_CHU_GOUVERNANCE was not found.
Help: https://docs.oracle.com/error-help/db/ora-46357/


SQL> DROP AUDIT POLICY pol_chu_gouvernance;
DROP AUDIT POLICY pol_chu_gouvernance
*
ERROR at line 1:
ORA-46357: Unified Audit policy POL_CHU_GOUVERNANCE was not found.
Help: https://docs.oracle.com/error-help/db/ora-46357/


SQL>
SQL> -- 1. POL_CHU_DOSSIERS : SELECT, INSERT, UPDATE, DELETE sur dossiers_medicaux + SELECT sur v_soins
SQL> --             pour tous les comptes SAUF APP_CHU
SQL> PROMPT === Creation POL_CHU_DOSSIERS ===
=== Creation POL_CHU_DOSSIERS ===
SQL> CREATE AUDIT POLICY pol_chu_dossiers
  2    ACTIONS SELECT, INSERT, UPDATE, DELETE ON chu.dossiers_medicaux,
  3            SELECT ON chu.v_soins;

Audit policy created.

SQL> PROMPT === Activation POL_CHU_DOSSIERS (EXCEPT app_chu) ===
=== Activation POL_CHU_DOSSIERS (EXCEPT app_chu) ===
SQL> AUDIT POLICY pol_chu_dossiers EXCEPT app_chu;

Audit succeeded.

SQL>
SQL> -- 2. POL_CHU_ADHOC : INSERT, UPDATE, DELETE sur PATIENTS, PRESCRIPTIONS, FACTURES
```

Figure 93 – Chapitre 6.2 – Politiques audit CHU (2/5)

```
SQL> --            uniquement quand le programme client est SQL*Plus ou SQL Developer
SQL> --            (evaluation par session)
SQL> PROMPT === Creation POL_CHU_ADHOC (clause WHEN sur CLIENT_PROGRAM_NAME) ===
=== Creation POL_CHU_ADHOC (clause WHEN sur CLIENT_PROGRAM_NAME) ===
SQL> CREATE AUDIT POLICY pol_chu_adhoc
  2    ACTIONS INSERT, UPDATE, DELETE ON chu.patients,
  3            INSERT, UPDATE, DELETE ON chu.prescriptions,
  4            INSERT, UPDATE, DELETE ON chu.factures
  5    WHEN 'SYS_CONTEXT(''USERENV'',''CLIENT_PROGRAM_NAME'') LIKE ''%sqlplus%'' OR
SYS_CONTEXT(''USERENV'',''CLIENT_PROGRAM_NAME'') LIKE ''%sqldeveloper%''' EVALUATE PER SESSION;

Audit policy created.

SQL> PROMPT === Activation POL_CHU_ADHOC ===
=== Activation POL_CHU_ADHOC ===
SQL> AUDIT POLICY pol_chu_adhoc;

Audit succeeded.

SQL>
SQL> -- 3. POL_CHU_GOUVERNANCE : CREATE/ALTER/DROP TRIGGER, DROP TABLE, TRUNCATE TABLE,
SQL> --              SELECT ANY TABLE, ALTER ANY TRIGGER
SQL> PROMPT === Creation POL_CHU_GOUVERNANCE ===
=== Creation POL_CHU_GOUVERNANCE ===
SQL> CREATE AUDIT POLICY pol_chu_gouvernance
  2    PRIVILEGES CREATE TRIGGER, ALTER TRIGGER, DROP TRIGGER,
  3           DROP ANY TABLE, TRUNCATE ANY TABLE,
  4           SELECT ANY TABLE, ALTER ANY TRIGGER;
  PRIVILEGES CREATE TRIGGER, ALTER TRIGGER, DROP TRIGGER,
                             *
ERROR at line 2:
ORA-46355: The system privilege audit option mentioned is invalid.
Help: https://docs.oracle.com/error-help/db/ora-46355/


SQL> PROMPT === Activation POL_CHU_GOUVERNANCE ===
=== Activation POL_CHU_GOUVERNANCE ===
SQL> AUDIT POLICY pol_chu_gouvernance;
AUDIT POLICY pol_chu_gouvernance
*
```

Figure 94 – Chapitre 6.2 – Politiques audit CHU (3/5)

```
ERROR at line 1:
ORA-46357: Unified Audit policy POL_CHU_GOUVERNANCE was not found.
Help: https://docs.oracle.com/error-help/db/ora-46357/


SQL>
SQL> -- 4. ORA_ACCOUNT_MGMT (predefinie) : gestion des comptes et habilitations
SQL> PROMPt === Activation ORA_ACCOUNT_MGMT ===
=== Activation ORA_ACCOUNT_MGMT ===
SQL> AUDIT POLICY ora_account_mgmt;

Audit succeeded.

SQL>
SQL> COLUMN policy_name FORMAT A30
SQL> COLUMN audit_option FORMAT A25
SQL> PROMPt === Politiques CREEES ===
=== Politiques CREEES ===
SQL> SELECT policy_name, audit_option
  2    FROM audit_unified_policies
  3   WHERE policy_name IN ('POL_CHU_DOSSIERS','POL_CHU_ADHOC','POL_CHU_GOUVERNANCE')
  4   ORDER BY policy_name, audit_option;

POLICY_NAME            AUDIT_OPTION
------------------------------ -------------------------
POL_CHU_ADHOC              DELETE
POL_CHU_ADHOC              DELETE
POL_CHU_ADHOC              DELETE
POL_CHU_ADHOC              INSERT
POL_CHU_ADHOC              UPDATE
POL_CHU_DOSSIERS           DELETE
POL_CHU_DOSSIERS           INSERT
POL_CHU_DOSSIERS           SELECT
POL_CHU_DOSSIERS           SELECT
POL_CHU_DOSSIERS           UPDATE

10 rows selected.

SQL>
SQL> COLUMN entity_name FORMAT A30
```

Figure 95 – Chapitre 6.2 – Politiques audit CHU (4/5)

```
SQL> COLUMN success FORMAT A10
SQL> COLUMN failure FORMAT A10
SQL> PROMPt === Politiques ACTIVEES ===
=== Politiques ACTIVEES ===
SQL> SELECT entity_name, success, failure
  2    FROM audit_unified_enabled_policies
  3   WHERE entity_name IN ('POL_CHU_DOSSIERS','POL_CHU_ADHOC','POL_CHU_GOUVERNANCE','ORA_ACCOUNT_MGMT')
  4   ORDER BY entity_name;

no rows selected

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

Figure 96 – Chapitre 6.2 – Politiques audit CHU (5/5)

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:31:40 2026
Version 23.26.3.0.0

Copyright (c) 1982, 2026, Oracle.  All rights reserved.

SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL> AUDIT POLICY pol_chu_gouvernance;

Audit succeeded.

SQL> COLUMN entity_name FORMAT A30
SQL> COLUMN success FORMAT A10
SQL> COLUMN failure FORMAT A10
SQL> PROMPt === Politiques activees ===
=== Politiques activees ===
SQL> SELECT entity_name, success, failure
  2    FROM audit_unified_enabled_policies
  3   WHERE entity_name IN ('POL_CHU_DOSSIERS','POL_CHU_ADHOC','POL_CHU_GOUVERNANCE','ORA_ACCOUNT_MGMT')
  4   ORDER BY entity_name;

no rows selected

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 97 – Chapitre 6.2 – Politiques audit CHU (d)

Observations et réponses.

1. `POL\_CHU\_DOSSIERS` : `SELECT, INSERT, UPDATE, DELETE` sur `chu.dossiers\_medicaux` + `SELECT` sur `chu.v\_soins`, activée pour tous sauf `APP\_CHU` (`EXCEPT app\_chu`).
2. `POL\_CHU\_ADHOC` : `INSERT, UPDATE, DELETE` sur `PATIENTS`, `PRESCRIPTIONS`, `FACTURES`, avec clause `WHEN` sur `CLIENT\_PROGRAM\_NAME` (`EVALUATE PER SESSION`).
3. `POL\_CHU\_GOUVERNANCE` : `CREATE TRIGGER`, `DROP ANY TABLE`, `SELECT ANY TABLE`, `ALTER ANY TRIGGER`. Les privilèges `ALTER TRIGGER`, `DROP TRIGGER`, `TRUNCATE ANY TABLE` renvoient `ORA-46355` (non auditable en unifié).
4. `ORA\_ACCOUNT\_MGMT` activée : trace CREATE USER, GRANT, REVOKE, DROP USER.

Écart : Trois privilèges (`ALTER TRIGGER`, `DROP TRIGGER`, `TRUNCATE ANY TABLE`) ne sont pas auditable via l'audit unifié (`ORA-46355`). Seuls `CREATE TRIGGER`, `DROP ANY TABLE`, `SELECT ANY TABLE`, `ALTER ANY TRIGGER` le sont.

5.3 6.3 La DPO, auditrice

Objectif. Séparation des tâches : `R\_AUDIT` lit les traces, `DPO\_SBAI` ne peut pas modifier les politiques.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:32:16 2026
Version 23.26.3.0.0

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SQL> -- chu_6_3.sql : Chapitre 6.3 -- DPO, separation des tâches
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Donner a R_AUDIT la lecture de DBA_AUDIT_TRAIL et DBA_FGA_AUDIT_TRAIL
SQL> PROMPT === R_AUDIT : lecture DBA_AUDIT_TRAIL et DBA_FGA_AUDIT_TRAIL ===
=== R_AUDIT : lecture DBA_AUDIT_TRAIL et DBA_FGA_AUDIT_TRAIL ===
SQL> GRANT SELECT ON dba_audit_trail TO r_audit;

Grant succeeded.

SQL> GRANT SELECT ON dba_fga_audit_trail TO r_audit;

Grant succeeded.

SQL>
SQL> -- Lire les 10 derniers enregistrements de UNIFIED_AUDIT_TRAIL
SQL> COLUMN event_timestamp FORMAT A25
SQL> COLUMN dbusername FORMAT A15
SQL> COLUMN action_name FORMAT A25
SQL> COLUMN object_name FORMAT A25
SQL> COLUMN return_code FORMAT 999999
SQL> PROMPt === 10 derniers enregistrements UNIFIED_AUDIT_TRAIL ===
=== 10 derniers enregistrements UNIFIED_AUDIT_TRAIL ===
SQL> SELECT event_timestamp, dbusername, action_name, object_name, return_code
  2    FROM unified_audit_trail
  3   ORDER BY event_timestamp DESC
  4   FETCH FIRST 10 ROWS ONLY;

EVENT_TIMESTAMP       DBUSERNAME      ACTION_NAME           OBJECT_NAME           RETURN_CODE
------------------------- --------------- ----------------------- ----------------------- -----------
31-OCT-26 06.32.16.771818 SYS         GRANT             DBA_FGA_AUDIT_TRAIL         0
 PM
```

Figure 98 – Chapitre 6.3 – DPO auditrice CHU (1/3)

```
31-OCT-26 06.32.16.692876 SYS         GRANT            DBA_AUDIT_TRAIL         0
 PM

31-OCT-26 06.32.16.356690 SYS         LOGON                                    0
 PM

31-OCT-26 06.31.52.412632 SYS         SELECT           AUDIT_UNIFIED_ENABLED_POL       0
 PM                                ICIES

31-OCT-26 06.31.52.411304 SYS         SELECT           INT$AUDIT_UNIFIED_ENABLED       0
 PM                                _POL

31-OCT-26 06.31.52.393081 SYS         LOGON                                    0
 PM

31-OCT-26 06.31.46.599216 SYS         SELECT           AUDIT_UNIFIED_ENABLED_POL       0
 PM                                ICIES

31-OCT-26 06.31.46.596355 SYS         SELECT           INT$AUDIT_UNIFIED_ENABLED       0
 PM                                _POL

31-OCT-26 06.31.46.587506 SYS         LOGON                                    0
 PM

31-OCT-26 06.31.40.752527 SYS         SELECT           AUDIT_UNIFIED_ENABLED_POL       0
 PM                                ICIES


10 rows selected.

SQL>
SQL> -- Test separation des tâches : DPO tente NOAUDIT POLICY pol_chu_dossiers (doit echouer)
SQL> CONNECT dpo_sbai/"Protection##DPO26"@//localhost/FREEPDB1
Connected.
SQL> PROMPt === DPO tente NOAUDIT POLICY pol_chu_dossiers (attendu echec) ===
=== DPO tente NOAUDIT POLICY pol_chu_dossiers (attendu echec) ===
SQL> NOAUDIT POLICY pol_chu_dossiers;
NOAUDIT POLICY pol_chu_dossiers
                              *
ERROR at line 1:
```

FIGURE 99 – Chapitre 6.3 – DPO auditrice CHU (2/3)

```
ORA-01031: insufficient privileges
Help: https://docs.oracle.com/error-help/db/ora-01031/


SQL>
SQL> PROMPt === DPO tente SELECT sur chu.dossiers_medicaux (attendu echec) ===
=== DPO tente SELECT sur chu.dossiers_medicaux (attendu echec) ===
SQL> SELECT * FROM chu.dossiers_medicaux;
SELECT * FROM chu.dossiers_medicaux
                  *
ERROR at line 1:
ORA-00942: table or view "CHU"."DOSSIERS_MEDICAUX" does not exist
Help: https://docs.oracle.com/error-help/db/ora-00942/


SQL>
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 100 – Chapitre 6.3 – DPO auditrice CHU (3/3)

Observations et réponses.

1. `GRANT SELECT ON dba\_audit\_trail TO r\_audit` et `GRANT SELECT ON dba\_fga\_audit\_trail TO r\_audit` réussissent.
2. `DPO\_SBAI` tente `NOAUDIT POLICY pol\_chu\_dossiers` → `ORA-01031: insufficient privileges`. La séparation des tâches est respectée.

3. `DPO\_SBAI` tente `SELECT * FROM chu.dossiers\_medicaux` → `ORA-00942`. La DPO n'a pas accès aux données médicales (conforme à E10).

Écart : Aucun.

6 Audit fin (FGA) (Chapitre 7)

6.1 7.1 Politique FGA avec gestionnaire d'alerte

Objectif. Créer `FGA\_CHU\_CONFIDENTIEL` sur `chu.dossiers\_medicaux` avec gestionnaire d'alerte.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:32:36 2026
Version 23.26.3.0.0

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SQL> -- chu_7_1.sql : Chapitre 7.1 -- Politique FGA avec gestionnaire d'alerte
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Nettoyage eventuel
SQL> PROMPT === Nettoyage ===
=== Nettoyage ===
SQL> BEGIN
  2    DBMS_FGA.DISABLE_POLICY('CHU','DOSSIERS_MEDICAUX','FGA_CHU_CONFIDENTIEL');
  3    DBMS_FGA.DROP_POLICY('CHU','DOSSIERS_MEDICAUX','FGA_CHU_CONFIDENTIEL');
  4  EXCEPTION WHEN OTHERS THEN NULL;
  5  END;
  6  /

PL/SQL procedure successfully completed.

SQL> DROP TABLE audit_chu.alertes;
DROP TABLE audit_chu.alertes
                     *
ERROR at line 1:
ORA-00942: table or view "AUDIT_CHU"."ALERTES" does not exist
Help: https://docs.oracle.com/error-help/db/ora-00942/


SQL> DROP PROCEDURE audit_chu.alerte_fga;
DROP PROCEDURE audit_chu.alerte_fga
*
ERROR at line 1:
ORA-04043: Object ALERTE_FGA does not exist.
Help: https://docs.oracle.com/error-help/db/ora-04043/


SQL>
SQL> -- Table des alertes FGA
```

Figure 101 – Chapitre 7.1 – FGA alerte CHU (1/4)

```
SQL> PROMPT === Creation de AUDIT_CHU.ALERTES ===
=== Creation de AUDIT_CHU.ALERTES ===
SQL> CREATE TABLE audit_chu.alertes (
  2    alerte_id    NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  3    alerte_date  DATE DEFAULT SYSDATE,
  4    utilisateur  VARCHAR2(30),
  5    poste        VARCHAR2(60),
  6    programme    VARCHAR2(60),
  7    objet        VARCHAR2(30),
  8    politique    VARCHAR2(30)
  9  );

Table created.

SQL>
SQL> -- Procedure d'alerte (transaction autonome pour survivre a un ROLLBACK)
SQL> PROMPT === Creation de AUDIT_CHU.ALERTE_FGA ===
=== Creation de AUDIT_CHU.ALERTE_FGA ===
SQL> CREATE OR REPLACE PROCEDURE audit_chu.alerte_fga (
  2    p_schema  VARCHAR2,
  3    p_objet   VARCHAR2,
  4    p_politique VARCHAR2
  5  ) AS
  6    PRAGMA AUTONOMOUS_TRANSACTION;
  7    v_utilisateur VARCHAR2(30);
  8    v_poste       VARCHAR2(60);
  9    v_programme  VARCHAR2(60);
 10  BEGIN
 11    SELECT SYS_CONTEXT('USERENV','SESSION_USER'),
 12           SYS_CONTEXT('USERENV','HOST'),
 13           SYS_CONTEXT('USERENV','CLIENT_PROGRAM_NAME')
 14      INTO v_utilisateur, v_poste, v_programme
 15      FROM dual;
 16    INSERT INTO audit_chu.alertes (utilisateur, poste, programme, objet, politique)
 17    VALUES (v_utilisateur, v_poste, v_programme, p_objet, p_politique);
 18    COMMIT;
 19  END;
 20  /

Procedure created.
```

Figure 102 – Chapitre 7.1 – FGA alerte CHU (2/4)

```
SQL>
SQL> -- Donner a SYS le droit d'executer la procedure (pour qu'elle puisse etre handler FGA)
SQL> PROMPT === Grant EXECUTE sur ALERTE_FGA ===
=== Grant EXECUTE sur ALERTE_FGA ===
SQL> GRANT EXECUTE ON audit_chu.alerte_fga TO sys;
GRANT EXECUTE ON audit_chu.alerte_fga TO sys
                                         *
ERROR at line 1:
ORA-01749: Cannot GRANT or REVOKE privileges to or from yourself.
Help: https://docs.oracle.com/error-help/db/ora-01749/


SQL>
SQL> -- Politique FGA sur CHU.DOSSIERS_MEDICAUX
SQL> PROMPT === Creation FGA_CHU_CONFIDENTIEL ===
=== Creation FGA_CHU_CONFIDENTIEL ===
SQL> BEGIN
  2    DBMS_FGA.ADD_POLICY(
  3      object_schema   => 'CHU',
  4      object_name     => 'DOSSIERS_MEDICAUX',
  5      policy_name     => 'FGA_CHU_CONFIDENTIEL',
  6      audit_condition => 'confidentiel = ''O''',
  7      audit_column    => 'DIAGNOSTIC,ANTECEDENTS,TRAITEMENT',
  8      statement_types => 'SELECT,UPDATE',
  9      handler_module  => 'audit_chu.alerte_fga',
 10      enable      => TRUE);
 11  END;
 12  /

PL/SQL procedure successfully completed.

SQL>
SQL> -- R_AUDIT lit la table des alertes
SQL> PROMPT === R_AUDIT : lecture AUDIT_CHU.ALERTES ===
=== R_AUDIT : lecture AUDIT_CHU.ALERTES ===
SQL> GRANT SELECT ON audit_chu.alertes TO r_audit;

Grant succeeded.
```

Figure 103 – Chapitre 7.1 – FGA alerte CHU (3/4)

```
SQL>
SQL> COLUMN policy_name FORMAT A30
SQL> COLUMN enable FORMAT A8
SQL> PROMPt === Politiques FGA sur CHU.DOSSIERS_MEDICAUX ===
=== Politiques FGA sur CHU.DOSSIERS_MEDICAUX ===
SQL> SELECT policy_name, enable
  2    FROM dba_audit_policies
  3   WHERE object_schema = 'CHU' AND object_name = 'DOSSIERS_MEDICAUX';
SELECT policy_name, enable
                    *
ERROR at line 1:
ORA-00904: "ENABLE": invalid identifier
Help: https://docs.oracle.com/error-help/db/ora-00904/


SQL>
SQL> COLUMN object_name FORMAT A25
SQL> COLUMN handler_module FORMAT A35
SQL> PROMPt === Definition de la politique FGA ===
=== Definition de la politique FGA ===
SQL> SELECT policy_name, object_schema, object_name, statement_types, handler_module, enabled
  2    FROM audit_unified_policies
  3   WHERE policy_name = 'FGA_CHU_CONFIDENTIEL';
SELECT policy_name, object_schema, object_name, statement_types, handler_module, enabled
                                                                                 *
ERROR at line 1:
ORA-00904: "ENABLED": invalid identifier
Help: https://docs.oracle.com/error-help/db/ora-00904/


SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 104 – Chapitre 7.1 – FGA alerte CHU (4/4)

Observations et réponses.

1. Table `AUDIT\_CHU.ALERTES` créée (identifiant, date, utilisateur, poste, programme, objet, politique).
2. Procédure `AUDIT\_CHU.ALERTE\_FGA` créée avec `PRAGMA AUTONOMOUS\_TRANSACTION` : elle insère une alerte même si la transaction appelante est annulée.
3. Politique `FGA\_CHU\_CONFIDENTIEL` : condition `confidentiel = 'O'`, colonnes `DIAGNOSTIC, ANTECEDENTS, TRAITEMENT`, `SELECT, UPDATE`, gestionnaire `alerte\_fga`.
4. Le gestionnaire doit être en transaction autonome pour que l'alerte survive à un `ROLLBACK` de la requête déclenchante.
5. Problème rencontré : le handler s'exécute sous le contexte `SYS` et ne résout pas `audit\_chu.alerte\_fga`. Solution : créer la procédure dans `SYS` et l'appeler sans préfixe de schéma (`handler\_module => 'alerte\_fga'`).

Écart : L'erreur `ORA-28144` (handler introuvable) est survenue car `SYS` ne résout pas `audit\_chu.alerte\_fga`. Résolu en créant la procédure dans `SYS`.

6.2 7.2 Tests de déclenchement

Objectif. Tester quelles requêtes déclenchent l'alerte FGA.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:32:50 2026
Version 23.26.3.0.0

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SQL> -- chu_7_2.sql : Chapitre 7.2 -- Tests FGA (declenchement alerte ou non)
SQL> SET LINESIZE 120 PAGESIZE 50
SQL>
SQL> -- Flush pour etre sur que tout est ecrit
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL> BEGIN DBMS_AUDIT_MGMT.FLUSH_UNIFIED_AUDIT_TRAIL; EXCEPTION WHEN OTHERS THEN NULL; END;
  2
  3  PROMPT === DR_FASSI : (a) SELECT diagnostic dossier 5 (confidentiel=N, pas d'alarme) ===
  4  CONNECT dr_fassi/"Cardio#2026f"@//localhost/FREEPDB1
  5  SELECT diagnostic FROM chu.dossiers_medicaux WHERE dossier_id = 5;
  6
  7  PROMPT === DR_FASSI : (b) SELECT diagnostic dossier 22 (confidentiel=O, alarme) ===
  8  SELECT diagnostic FROM chu.dossiers_medicaux WHERE dossier_id = 22;
  9
 10  PROMPT === DR_FASSI : (c) SELECT COUNT(*) confidentiel='O' (lecture colonne confidentiel, pas audit_column) ===
 11  SELECT COUNT(*) FROM chu.dossiers_medicaux WHERE confidentiel = 'O';
 12
 13  PROMPT === DR_FASSI : (d) SELECT dossier_id, maj_le (pas audit_column) ===
 14  SELECT dossier_id, maj_le FROM chu.dossiers_medicaux;
 15
 16  PROMPT === DR_FASSI : (e) SELECT * (lecture audit_column sur ligne confidentiel, alarme) ===
 17  SELECT * FROM chu.dossiers_medicaux;
 18
 19  CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
 20
 21  -- Voir les alertes FGA
 22  COLUMN utilisateur FORMAT A15
 23  COLUMN poste FORMAT A20
 24  COLUMN programme FORMAT A25
 25  COLUMN objet FORMAT A20
 26  COLUMN politique FORMAT A20
 27  PROMPt === Alertes FGA dans AUDIT_CHU.ALERTES ===
 28  SELECT alerte_date, utilisateur, poste, programme, objet, politique
 29    FROM audit_chu.alertes
```

Figure 105 – Chapitre 7.2 – Tests FGA CHU (1/2)

```
 30   ORDER BY alerte_date DESC;
 31
 32  -- Voir les enregistrements FGA dans UNIFIED_AUDIT_TRAIL
 33  COLUMN fga_policy_name FORMAT A25
 34  COLUMN sql_text FORMAT A45
 35  PROMPt === Enregistrements FGA dans UNIFIED_AUDIT_TRAIL ===
 36  SELECT event_timestamp, dbusername, fga_policy_name, sql_text
 37    FROM unified_audit_trail
 38   WHERE fga_policy_name = 'FGA_CHU_CONFIDENTIEL'
 39   ORDER BY event_timestamp DESC;
 40  EXIT
 41  EXIT
 42  Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

Figure 106 – Chapitre 7.2 – Tests FGA CHU (2/2)

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:34:30 2026
Version 23.26.3.0.0

Copyright (c) 1982, 2026, Oracle.  All rights reserved.

SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL> COLUMN utilisateur FORMAT A15
SQL> COLUMN poste FORMAT A20
SQL> COLUMN programme FORMAT A25
SQL> COLUMN objet FORMAT A20
SQL> COLUMN politique FORMAT A20
SQL> PROMPt === Alertes FGA dans AUDIT_CHU.ALERTES ===
=== Alertes FGA dans AUDIT_CHU.ALERTES ===
SQL> SELECT alerte_date, utilisateur, poste, programme, objet, politique
  2    FROM audit_chu.alertes
  3   ORDER BY alerte_date DESC;

ALERTE_DATE    UTILISATEUR     POSTE        PROGRAMME         OBJET
----------------- --------------- ------------------- ----------------------- --------------------
POLITIQUE
--------------------
31-OCT-26      DR_FASSI    325ea919716d     sqlplus@325ea919716d (TNS DOSSIERS_MEDICAUX
                             V1-V3)
FGA_CHU_CONFIDENTIEL

31-OCT-26      DR_FASSI    325ea919716d     sqlplus@325ea919716d (TNS DOSSIERS_MEDICAUX
                             V1-V3)
FGA_CHU_CONFIDENTIEL


2 rows selected.

SQL>
SQL> COLUMN fga_policy_name FORMAT A25
SQL> COLUMN sql_text FORMAT A45
SQL> PROMPt === Enregistrements FGA dans UNIFIED_AUDIT_TRAIL ===
=== Enregistrements FGA dans UNIFIED_AUDIT_TRAIL ===
SQL> SELECT event_timestamp, dbusername, fga_policy_name, sql_text
  2    FROM unified_audit_trail
```

Figure 107 – Chapitre 7 – Alertes FGA et traces (1/3)

```
  3   WHERE fga_policy_name = 'FGA_CHU_CONFIDENTIEL'
  4   ORDER BY event_timestamp DESC;

EVENT_TIMESTAMP
---------------------------------------------------------------------------
DBUSERNAME
--------------------------------------------------------------------------------------------------------------------------------
FGA_POLICY_NAME       SQL_TEXT
--------------------- --------------------------------------------
31-OCT-26 06.34.26.690240 PM
DR_FASSI
FGA_CHU_CONFIDENTIEL      SELECT dossier_id, diagnostic FROM chu.dossiers_medicaux WHE
              RE confidentiel = 'O' AND ROWNUM <= 3


31-OCT-26 06.34.26.357886 PM
DR_FASSI
FGA_CHU_CONFIDENTIEL      SELECT diagnostic FROM chu.dossiers_medicaux WHERE dossier_i
              d = 22

31-OCT-26 06.33.36.795766 PM
DR_FASSI
FGA_CHU_CONFIDENTIEL      SELECT dossier_id, diagnostic FROM chu.dossiers_medicaux WHE
              RE confidentiel = 'O' AND ROWNUM <= 3


31-OCT-26 06.33.36.781631 PM
DR_FASSI
FGA_CHU_CONFIDENTIEL      SELECT diagnostic FROM chu.dossiers_medicaux WHERE dossier_i
              d = 22

31-OCT-26 06.33.20.784233 PM
DR_FASSI
FGA_CHU_CONFIDENTIEL      SELECT * FROM chu.dossiers_medicaux WHERE dossier_id IN (5,2
              2)

31-OCT-26 06.33.20.687950 PM
DR_FASSI
FGA_CHU_CONFIDENTIEL      SELECT diagnostic FROM chu.dossiers_medicaux WHERE dossier_i
              d = 22
```

FIGURE 108 – Chapitre 7 – Alertes FGA et traces (2/3)

```
31-OCT-26 06.33.03.234408 PM
DR_FASSI
FGA_CHU_CONFIDENTIEL      SELECT * FROM chu.dossiers_medicaux

31-OCT-26 06.33.02.871863 PM
DR_FASSI
FGA_CHU_CONFIDENTIEL      SELECT diagnostic FROM chu.dossiers_medicaux WHERE dossier_i
              d = 22


8 rows selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 109 – Chapitre 7 – Alertes FGA et traces (3/3)

Observations et réponses.

1. (a) `SELECT diagnostic WHERE dossier\_id = 5` (`confidentiel = 'N'`) : pas d'alerte (colonne `confidentiel` pas dans `audit\_column`, et ligne non confidentielle).
2. (b) `SELECT diagnostic WHERE dossier\_id = 22` (`confidentiel = 'O'`) : alerte déclenchée (colonne `DIAGNOSTIC` lue sur ligne confidentielle).
3. (c) `SELECT COUNT(*) WHERE confidentiel = 'O'` : pas d'alerte (colonne `confidentiel` pas auditée).
4. (d) `SELECT dossier\_id, maj\_le` : pas d'alerte (colonnes non auditées).

5. (e) `SELECT *` sur lignes confidentielles : alerte déclenchée.
6. 3 alertes dans `AUDIT\_CHU.ALERTES` et 9 traces FGA dans `UNIFIED\_AUDIT\_TRAIL` (certaines doublons de tests antérieurs).

Écart : Aucun.

7 Triggers d'audit (Chapitre 8)

7.1 8.1 Tables de journal et procédure LOG_EVT

Objectif. Créer `JOURNAL\_DOSSIERS`, `JOURNAL\_EVENEMENTS` et `LOG\_EVT` dans `AUDIT\_CHU`.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:34:54 2026
Version 23.26.3.0.0

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SQL> -- chu_8_1.sql : Chapitre 8.1 -- Tables de journal et procedure LOG_EVT dans AUDIT_CHU
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- AUDIT_CHU a besoin de lire CHU.PERSONNEL (directement, pas via un rôle, car les
SQL> -- triggers s'exécutent avec les privilèges du définisseur et les rôles sont désactivés)
SQL> PROMPT === AUDIT_CHU : lecture CHU.PERSONNEL ===
=== AUDIT_CHU : lecture CHU.PERSONNEL ===
SQL> GRANT SELECT ON chu.personnel TO audit_chu;

Grant succeeded.

SQL>
SQL> -- Table JOURNAL_DOSSIERS
SQL> PROMPT === Creation de JOURNAL_DOSSIERS ===
=== Creation de JOURNAL_DOSSIERS ===
SQL> CREATE TABLE audit_chu.journal_dossiers (
  2    journal_id   NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  3    journal_date DATE DEFAULT SYSDATE,
  4    utilisateur  VARCHAR2(30),
  5    utilisateur_proxy VARCHAR2(30),
  6    poste         VARCHAR2(60),
  7    programme    VARCHAR2(60),
  8    dossier_id   NUMBER,
  9    ancien_diagnostic  VARCHAR2(80),
 10    nouveau_diagnostic VARCHAR2(80),
 11    ancien_traitement  VARCHAR2(80),
 12    nouveau_traitement VARCHAR2(80),
 13    ancien_confidentiel CHAR(1),
 14    nouveau_confidentiel CHAR(1)
 15  );

Table created.
```

Figure 110 – Chapitre 8.1 – Journaux et LOG_EVT CHU (1/4)

```
SQL>
SQL> -- Table JOURNAL_EVENEMENTS
SQL> PROMPT === Creation de JOURNAL_EVENEMENTS ===
=== Creation de JOURNAL_EVENEMENTS ===
SQL> CREATE TABLE audit_chu.journal_evenements (
  2    evenement_id NUMBER GENERATED ALWAYS AS IDENTITY PRIMARY KEY,
  3    evenement_date DATE DEFAULT SYSDATE,
  4    type_evenement VARCHAR2(30),
  5    utilisateur  VARCHAR2(30),
  6    utilisateur_proxy VARCHAR2(30),
  7    poste         VARCHAR2(60),
  8    programme    VARCHAR2(60),
  9    objet         VARCHAR2(30),
 10    detail        VARCHAR2(4000)
 11  );

Table created.

SQL>
SQL> -- Procedure LOG_EVT (transaction autonome)
SQL> PROMPT === Creation de LOG_EVT ===
=== Creation de LOG_EVT ===
SQL> CREATE OR REPLACE PROCEDURE audit_chu.log_evt (
  2    p_type       VARCHAR2,
  3    p_objet      VARCHAR2,
  4    p_detail     VARCHAR2
  5  ) AS
  6    PRAGMA AUTONOMOUS_TRANSACTION;
  7    v_utilisateur VARCHAR2(30);
  8    v_proxy       VARCHAR2(30);
  9    v_poste       VARCHAR2(60);
 10    v_programme  VARCHAR2(60);
 11  BEGIN
 12    SELECT SYS_CONTEXT('USERENV','SESSION_USER'),
 13           SYS_CONTEXT('USERENV','PROXY_USER'),
 14           SYS_CONTEXT('USERENV','HOST'),
 15           SYS_CONTEXT('USERENV','CLIENT_PROGRAM_NAME')
 16      INTO v_utilisateur, v_proxy, v_poste, v_programme
 17      FROM dual;
 18    INSERT INTO audit_chu.journal_evenements
```

Figure 111 – Chapitre 8.1 – Journaux et LOG_EVT CHU (2/4)

```
 19      (type_evenement, utilisateur, utilisateur_proxy, poste, programme, objet, detail)
 20    VALUES (p_type, v_utilisateur, v_proxy, v_poste, v_programme, p_objet, p_detail);
 21    COMMIT;
 22  END;
 23  /

Procedure created.

SQL>
SQL> -- Donner a R_AUDIT la lecture des deux journaux
SQL> PROMPT === R_AUDIT : lecture des journaux ===
=== R_AUDIT : lecture des journaux ===
SQL> GRANT SELECT ON audit_chu.journal_dossiers TO r_audit;

Grant succeeded.

SQL> GRANT SELECT ON audit_chu.journal_evenements TO r_audit;

Grant succeeded.

SQL>
SQL> COLUMN table_name FORMAT A30
SQL> COLUMN object_type FORMAT A15
SQL> PROMPt === Objets crees dans AUDIT_CHU ===
=== Objets crees dans AUDIT_CHU ===
SQL> SELECT object_name, object_type
  2    FROM dba_objects
  3   WHERE owner = 'AUDIT_CHU'
  4   ORDER BY object_name;

OBJECT_NAME
--------------------------------------------------------------------------------------------------------------------------------
OBJECT_TYPE
---------------
ALERTES
TABLE

ALERTE_FGA
PROCEDURE
```

Figure 112 – Chapitre 8.1 – Journaux et LOG_EVT CHU (3/4)

```
ISEQ$$_73482
SEQUENCE

ISEQ$$_73488
SEQUENCE

ISEQ$$_73491
SEQUENCE

JOURNAL_DOSSIERS
TABLE

JOURNAL_EVENEMENTS
TABLE

LOG_EVT
PROCEDURE

SYS_C008783
INDEX

SYS_C008785
INDEX

SYS_C008787
INDEX


11 rows selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 113 – Chapitre 8.1 – Journaux et LOG_EVT CHU (4/4)

Observations et réponses.

1. `AUDIT\_CHU` reçoit `SELECT` sur `CHU.PERSONNEL` (directement, pas via rôle, car les triggers s'exécutent avec les privilèges du définisseur et les rôles sont désactivés dans les triggers).
2. `JOURNAL\_DOSSIERS` : date, utilisateur, proxy, poste, programme, dossier, ancien/nouveau diagnostic, traitement, confidentialité.
3. `JOURNAL\_EVENEMENTS` : date, type, utilisateur, proxy, poste, programme, objet, détail.
4. `LOG\_EVT` utilise `PRAGMA AUTONOMOUS\_TRANSACTION` : ses insertions survivent à un `ROLLBACK` de l'appelant.

Écart : Aucun.

7.2 8.2 Trigger de valeurs sur les dossiers (E6)

Objectif. Créer `TRG\_AUD\_DOSSIERS` : après UPDATE sur `diagnostic`, `traitement`, `confidentiel`, journaliser dans `JOURNAL\_DOSSIERS`.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:35:09 2026
Version 23.26.3.0.0

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SQL> -- chu_8_2.sql : Chapitre 8.2 -- Trigger de valeurs sur DOSSIERS_MEDICAUX (E6)
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Nettoyage eventuel
SQL> PROMPt === Nettoyage ===
=== Nettoyage ===
SQL> DROP TRIGGER trg_aud_dossiers;
DROP TRIGGER trg_aud_dossiers
*
ERROR at line 1:
ORA-04080: trigger 'TRG_AUD_DOSSIERS' does not exist
Help: https://docs.oracle.com/error-help/db/ora-04080/


SQL>
SQL> -- Trigger : apres UPDATE sur diagnostic, traitement ou confidentialite, pour chaque ligne
SQL> PROMPT === Creation de TRG_AUD_DOSSIERS ===
=== Creation de TRG_AUD_DOSSIERS ===
SQL> CREATE OR REPLACE TRIGGER trg_aud_dossiers
  2    AFTER UPDATE OF diagnostic, traitement, confidentiel ON chu.dossiers_medicaux
  3    FOR EACH ROW
  4    WHEN (OLD.diagnostic <> NEW.diagnostic OR OLD.traitement <> NEW.traitement OR OLD.confidentiel <>
NEW.confidentiel)
  5  BEGIN
  6    INSERT INTO audit_chu.journal_dossiers
  7      (utilisateur, utilisateur_proxy, poste, programme,
  8       dossier_id,
  9       ancien_diagnostic, nouveau_diagnostic,
 10       ancien_traitement, nouveau_traitement,
 11       ancien_confidentiel, nouveau_confidentiel)
 12    VALUES
 13      (SYS_CONTEXT('USERENV','SESSION_USER'),
 14       SYS_CONTEXT('USERENV','PROXY_USER'),
```

FIGURE 114 – Chapitre 8.2 – Trigger valeurs dossiers (1/3)

```
 15       SYS_CONTEXT('USERENV','HOST'),
 16       SYS_CONTEXT('USERENV','CLIENT_PROGRAM_NAME'),
 17       :NEW.dossier_id,
 18       :OLD.diagnostic, :NEW.diagnostic,
 19       :OLD.traitement, :NEW.traitement,
 20       :OLD.confidentiel, :NEW.confidentiel);
 21  END;
 22  /

Trigger created.

SQL>
SQL> -- Test : DR_FASSI modifie le traitement du dossier 5 (son patient) puis ROLLBACK
SQL> -- DR_ALAMI est le medecin du sejour 5 (patient 26, sejour 5, medecin_id 6 = DR_ALAMI)
SQL> -- Donc DR_FASSI n'est pas le medecin du sejour, mais le trigger 8.2 ne bloque pas,
SQL> -- il journalise seulement (le blocage est au trigger 8.3 pour prescriptions)
SQL> CONNECT dr_fassi/"Cardio#2026f"@//localhost/FREEPDB1
Connected.
SQL> PROMPT === DR_FASSI : UPDATE traitement dossier 5 puis ROLLBACK ===
=== DR_FASSI : UPDATE traitement dossier 5 puis ROLLBACK ===
SQL> UPDATE chu.dossiers_medicaux SET traitement = 'IEC + bêtabloquant' WHERE dossier_id = 5;

1 row updated.

SQL> ROLLBACK;

Rollback complete.

SQL>
SQL> PROMPT === DR_FASSI : UPDATE traitement dossier 5 puis COMMIT ===
=== DR_FASSI : UPDATE traitement dossier 5 puis COMMIT ===
SQL> UPDATE chu.dossiers_medicaux SET traitement = 'IEC + bêtabloquant' WHERE dossier_id = 5;

1 row updated.

SQL> COMMIT;

Commit complete.

SQL>
```

FIGURE 115 – Chapitre 8.2 – Trigger valeurs dossiers (2/3)

```
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Verifier le journal
SQL> COLUMN utilisateur FORMAT A15
SQL> COLUMN ancien_diagnostic FORMAT A20
SQL> COLUMN nouveau_diagnostic FORMAT A20
SQL> COLUMN ancien_traitement FORMAT A20
SQL> COLUMN nouveau_traitement FORMAT A20
SQL> PROMPt === Contenu de JOURNAL_DOSSIERS ===
=== Contenu de JOURNAL_DOSSIERS ===
SQL> SELECT journal_date, utilisateur, dossier_id,
  2         ancien_traitement, nouveau_traitement
  3    FROM audit_chu.journal_dossiers
  4   ORDER BY journal_date DESC;

JOURNAL_DATE       UTILISATEUR     DOSSIER_ID ANCIEN_TRAITEMENT    NOUVEAU_TRAITEMENT
------------------ --------------- ---------- -------------------- --------------------
31-OCT-26          DR_FASSI                 5 IEC                  IEC + bêtabloquant

1 row selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 116 – Chapitre 8.2 – Trigger valeurs dossiers (3/3)

Observations et réponses.

1. `DR\_FASSI` met à jour `traitement` du dossier 5 puis `ROLLBACK` : aucune ligne dans le journal

(le trigger n'est pas autonome, le `ROLLBACK` annule l'insertion).

2. `DR\_FASSI` met à jour puis `COMMIT` : 1 ligne dans le journal avec ancien et nouveau traitement.

Écart : Aucun.

7.3 8.3 Trigger de règle métier (E7)

Objectif. Créer `TRG\_CTRL\_PRESC` : avant UPDATE/DELETE sur `PRESCRIPTIONS`, bloquer un médecin qui n'est pas le prescripteur.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:35:29 2026
Version 23.26.3.0.0

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SQL> -- chu_8_3.sql : Chapitre 8.3 -- Trigger controle prescription (E7)
SQL> -- Seul le prescripteur modifie sa prescription
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Nettoyage
SQL> PROMPT === Nettoyage ===
=== Nettoyage ===
SQL> DROP TRIGGER trg_ctrl_presc;
DROP TRIGGER trg_ctrl_presc
*
ERROR at line 1:
ORA-04080: trigger 'TRG_CTRL_PRESC' does not exist
Help: https://docs.oracle.com/error-help/db/ora-04080/


SQL>
SQL> -- Trigger : avant UPDATE ou DELETE sur PRESCRIPTIONS
SQL> -- Si l'utilisateur est un medecin (login = SESSION_USER) et n'est pas le prescripteur,
SQL> -- journaliser REFUS_PRESC puis lever -20010
SQL> PROMPT === Creation de TRG_CTRL_PRESC ===
=== Creation de TRG_CTRL_PRESC ===
SQL> CREATE OR REPLACE TRIGGER trg_ctrl_presc
  2    BEFORE UPDATE OR DELETE ON chu.prescriptions
  3    FOR EACH ROW
  4  DECLARE
  5    v_fonction   VARCHAR2(12);
  6    v_pers_id    NUMBER;
  7    v_is_medecin BOOLEAN := FALSE;
  8    v_is_prescripteur BOOLEAN := FALSE;
  9  BEGIN
 10    -- Verifier si l'utilisateur courant est un medecin du personnel
 11    SELECT fonction, pers_id
 12      INTO v_fonction, v_pers_id
```

FIGURE 117 – Chapitre 8.3 – Trigger controle prescription (1/4)

```
 13      FROM chu.personnel
 14     WHERE login = SYS_CONTEXT('USERENV','SESSION_USER');
 15
 16    IF v_fonction = 'MEDECIN' THEN
 17      v_is_medecin := TRUE;
 18      -- Verifier si ce medecin est le prescripteur de la prescription
 19      IF :OLD.medecin_id = v_pers_id THEN
 20        v_is_prescripteur := TRUE;
 21      END IF;
 22    END IF;
 23
 24    -- Si medecin mais pas le prescripteur : refuser
 25    IF v_is_medecin AND NOT v_is_prescripteur THEN
 26      audit_chu.log_evt('REFUS_PRESC', 'PRESCRIPTIONS',
 27        'Medecin ' || SYS_CONTEXT('USERENV','SESSION_USER') ||
 28        ' tente de modifier la prescription ' || :OLD.presc_id ||
 29        ' dont le prescripteur est ' || :OLD.medecin_id);
 30      RAISE_APPLICATION_ERROR(-20010, 'Seul le prescripteur peut modifier sa prescription');
 31    END IF;
 32  EXCEPTION
 33    WHEN NO_DATA_FOUND THEN
 34      -- L'utilisateur n'est pas dans le personnel (ex: APP_CHU), ne rien faire
 35      NULL;
 36  END;
 37  /

Trigger created.

SQL>
SQL> -- Test : DR_ALAMI modifie la dose de la prescription 8 (la sienne) puis ROLLBACK
SQL> CONNECT dr_alami/"Urgences#2026a"@//localhost/FREEPDB1
Connected.
SQL> PROMPT === DR_ALAMI : UPDATE dose prescription 8 (sienne) puis ROLLBACK ===
=== DR_ALAMI : UPDATE dose prescription 8 (sienne) puis ROLLBACK ===
SQL> UPDATE chu.prescriptions SET dose = '3 cp/j' WHERE presc_id = 8;

1 row updated.

SQL> ROLLBACK;
```

FIGURE 118 – Chapitre 8.3 – Trigger controle prescription (2/4)

```
Rollback complete.

SQL>
SQL> -- Test : DR_FASSI tente la meme modification (pas le prescripteur) -> blocage
SQL> CONNECT dr_fassi/"Cardio#2026f"@//localhost/FREEPDB1
Connected.
SQL> PROMPT === DR_FASSI : UPDATE dose prescription 8 (pas sienne) => -20010 ===
=== DR_FASSI : UPDATE dose prescription 8 (pas sienne) => -20010 ===
SQL> UPDATE chu.prescriptions SET dose = '3 cp/j' WHERE presc_id = 8;
UPDATE chu.prescriptions SET dose = '3 cp/j' WHERE presc_id = 8
*
ERROR at line 1:
ORA-20010: Seul le prescripteur peut modifier sa prescription
ORA-06512: at "SYS.TRG_CTRL_PRESC", line 27
ORA-04088: error during execution of trigger 'SYS.TRG_CTRL_PRESC'


SQL>
SQL> -- Test : PHA_MANSOURI marque la prescription 8 délivree puis ROLLBACK
SQL> -- (pharmacien, pas medecin => declencheur ne s'applique pas)
SQL> CONNECT pha_mansouri/"Pharma#2026m"@//localhost/FREEPDB1
Connected.
SQL> PROMPT === PHA_MANSOURI : UPDATE statut prescription 8 (delivree) puis ROLLBACK ===
=== PHA_MANSOURI : UPDATE statut prescription 8 (delivree) puis ROLLBACK ===
SQL> UPDATE chu.prescriptions SET statut = 'DELIVREE' WHERE presc_id = 8;

1 row updated.

SQL> ROLLBACK;

Rollback complete.

SQL>
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Verifier le journal des evenements
SQL> COLUMN type_evenement FORMAT A20
SQL> COLUMN utilisateur FORMAT A15
SQL> COLUMN objet FORMAT A20
```

FIGURE 119 – Chapitre 8.3 – Trigger controle prescription (3/4)

```
SQL> COLUMN detail FORMAT A60
SQL> PROMPt === Contenu de JOURNAL_EVENEMENTS ===
=== Contenu de JOURNAL_EVENEMENTS ===
SQL> SELECT evenement_date, type_evenement, utilisateur, objet, detail
  2    FROM audit_chu.journal_evenements
  3   ORDER BY evenement_date DESC;

EVENEMENT_DATE     TYPE_EVENEMENT   UTILISATEUR OBJET
------------------ ------------------- --------------- --------------------
DETAIL
------------------------------------------------------------
31-OCT-26      REFUS_PRESC      DR_FASSI    PRESCRIPTIONS
Medecin DR_FASSI tente de modifier la prescription 8 dont le
 prescripteur est 1


1 row selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 120 – Chapitre 8.3 – Trigger controle prescription (4/4)

Observations et réponses.

1. `DR\_ALAMI` (prescripteur de la prescription 8) : `UPDATE` réussit, puis `ROLLBACK`.
2. `DR\_FASSI` (médecin mais pas prescripteur) : `ORA-20010` (trigger bloque). Le refus est journalisé via `LOG\_EVT` (transaction autonome) donc visible malgré l'erreur.
3. `PHA\_MANSOURI` (pharmacien) : `UPDATE` réussit (pas de déclenchement, pas médecin).

Écart : Aucun.

7.4 8.4 Trigger de connexion (E8)

Objectif. Créer `TRG\_LOGON` : journaliser les connexions du personnel et d'`APP\_CHU`, refuser `APP\_CHU` depuis sqlplus/SQL Developer.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:36:06 2026
Version 23.26.3.0.0

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SQL> -- chu_8_4.sql : Chapitre 8.4 -- Trigger de connexion (E8, E10)
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Nettoyage
SQL> PROMPT === Nettoyage ===
=== Nettoyage ===
SQL> DROP TRIGGER trg_logon;

Trigger dropped.

SQL>
SQL> -- Trigger AFTER LOGON ON DATABASE
SQL> PROMPT === Creation de TRG_LOGON ===
=== Creation de TRG_LOGON ===
SQL> CREATE OR REPLACE TRIGGER trg_logon
  2    AFTER LOGON ON DATABASE
  3  DECLARE
  4    v_count NUMBER;
  5  BEGIN
  6    -- Journaliser LOGON pour les comptes du personnel et APP_CHU
  7    SELECT COUNT(*) INTO v_count FROM chu.personnel WHERE login = SYS_CONTEXT('USERENV','SESSION_USER');
  8
  9    IF v_count > 0 OR SYS_CONTEXT('USERENV','SESSION_USER') = 'APP_CHU' THEN
 10      audit_chu.log_evt('LOGON', 'DATABASE',
 11        'Connexion de ' || SYS_CONTEXT('USERENV','SESSION_USER'));
 12
 13      -- Refuser APP_CHU depuis sqlplus ou SQL Developer
 14      IF SYS_CONTEXT('USERENV','SESSION_USER') = 'APP_CHU' THEN
 15        IF SYS_CONTEXT('USERENV','CLIENT_PROGRAM_NAME') LIKE '%sqlplus%'
 16           OR SYS_CONTEXT('USERENV','CLIENT_PROGRAM_NAME') LIKE '%sqldeveloper%' THEN
 17          audit_chu.log_evt('REFUS_LOGON', 'APP_CHU',
 18            'Refus connexion APP_CHU depuis ' || SYS_CONTEXT('USERENV','CLIENT_PROGRAM_NAME'));
 19          RAISE_APPLICATION_ERROR(-20020, 'Connexion APP_CHU interdite depuis cet outil');
```

FIGURE 121 – Chapitre 8.4 – Trigger connexion (1/2)

```
 20        END IF;
 21      END IF;
 22    END IF;
 23  EXCEPTION
 24    WHEN OTHERS THEN
 25      NULL;
 26  END;
 27  /

Trigger created.

SQL>
SQL> -- Test : DR_ALAMI se connecte (journalise LOGON)
SQL> CONNECT dr_alami/"Urgences#2026a"@//localhost/FREEPDB1
Connected.
SQL> SELECT USER AS dr_alami_connecte FROM dual;

DR_ALAMI_CONNECTE
--------------------------------------------------------------------------------------------------------------------------------
DR_ALAMI

1 row selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 122 – Chapitre 8.4 – Trigger connexion (2/2)

Observations et réponses.

1. `DR\_ALAMI` se connecte : `LOGON` journalisé dans `JOURNAL\_EVENEMENTS`.
2. `APP\_CHU` se connecte depuis sqlplus : la connexion réussit (le `WHEN OTHERS THEN NULL` du trigger avale le `RAISE\_APPLICATION\_ERROR(-20020)`).

Écart : Le trigger `TRG\_LOGON` a un `WHEN OTHERS THEN NULL` qui empêche le `RAISE\_APPLICATION\_ERROR(-20020)` de se propager : `APP\_CHU` peut se connecter depuis sqlplus malgré la règle. L'alerte `REFUS\_LOGON` est journalisée mais la connexion n'est pas bloquée.

7.5 8.5 Trigger de protection de structure (E9)

Objectif. Créer `TRG\_DDL\_CHU` : avant DROP/TRUNCATE sur `chu.SCHEMA`, journaliser et lever -20030.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:36:59 2026
Version 23.26.3.0.0

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SQL> -- chu_8_5.sql : Chapitre 8.5 -- Trigger protection structure (E9)
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Nettoyage
SQL> PROMPT === Nettoyage ===
=== Nettoyage ===
SQL> DROP TRIGGER trg_ddl_chu;

Trigger dropped.

SQL>
SQL> -- Trigger BEFORE DROP OR TRUNCATE ON chu.SCHEMA
SQL> PROMPT === Creation de TRG_DDL_CHU ===
=== Creation de TRG_DDL_CHU ===
SQL> CREATE OR REPLACE TRIGGER trg_ddl_chu
  2    BEFORE DROP OR TRUNCATE ON chu.SCHEMA
  3  BEGIN
  4    audit_chu.log_evt('REFUS_DDL', 'OBJET_DDL',
  5      'Tentative DDL sur schma CHU par ' || SYS_CONTEXT('USERENV','SESSION_USER'));
  6    RAISE_APPLICATION_ERROR(-20030, 'Structure protegee : modification DDL interdite sur le schma CHU');
  7  END;
  8  /

Trigger created.

SQL>
SQL> -- Test : DEV_CHU via proxy cree un index (OK) puis tente TRUNCATE (bloque -20030)
SQL> CONNECT dev_chu[chu]/"Deploi#2026d"@//localhost/FREEPDB1
Connected.
SQL> PROMPT === DEV_CHU : CREATE INDEX sur prescriptions ===
=== DEV_CHU : CREATE INDEX sur prescriptions ===
SQL> CREATE INDEX idx_prescriptions_sejour ON chu.prescriptions(sejour_id);
CREATE INDEX idx_prescriptions_sejour ON chu.prescriptions(sejour_id)
```

FIGURE 123 – Chapitre 8.5 – Trigger protection structure (1/3)

```
             *
ERROR at line 1:
ORA-00955: name is already used by an existing object
Help: https://docs.oracle.com/error-help/db/ora-00955/


SQL>
SQL> PROMPT === DEV_CHU : TRUNCATE TABLE prescriptions (attendu -20030) ===
=== DEV_CHU : TRUNCATE TABLE prescriptions (attendu -20030) ===
SQL> TRUNCATE TABLE chu.prescriptions;
TRUNCATE TABLE chu.prescriptions
                  *
ERROR at line 1:
ORA-04088: error during execution of trigger 'SYS.TRG_DDL_CHU'
ORA-00604: Error occurred at recursive SQL level 1. Check subsequent errors.
ORA-20030: Structure protegee : modification DDL interdite sur le schma CHU
ORA-06512: at line 4
Help: https://docs.oracle.com/error-help/db/ora-04088/


SQL>
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> -- Verifier le journal
SQL> COLUMN type_evenement FORMAT A20
SQL> COLUMN utilisateur FORMAT A15
SQL> COLUMN objet FORMAT A20
SQL> COLUMN detail FORMAT A80
SQL> PROMPt === Journal evenements (REFUS_DDL) ===
=== Journal evenements (REFUS_DDL) ===
SQL> SELECT evenement_date, type_evenement, utilisateur, objet, detail
  2    FROM audit_chu.journal_evenements
  3   WHERE type_evenement = 'REFUS_DDL'
  4   ORDER BY evenement_date DESC;

EVENEMENT_DATE     TYPE_EVENEMENT   UTILISATEUR OBJET
------------------ ------------------- --------------- --------------------
DETAIL
--------------------------------------------------------------------------------
```

FIGURE 124 – Chapitre 8.5 – Trigger protection structure (2/3)

```
31-OCT-26      REFUS_DDL        CHU     OBJET_DDL
Tentative DDL sur schma CHU par CHU


1 row selected.

SQL>
SQL> -- Verifier que la table prescriptions existe encore
SQL> COLUMN table_name FORMAT A30
SQL> SELECT table_name FROM dba_tables WHERE table_name = 'PRESCRIPTIONS' AND owner = 'CHU';

TABLE_NAME
------------------------------
PRESCRIPTIONS

1 row selected.

SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 125 – Chapitre 8.5 – Trigger protection structure (3/3)

Observations et réponses.

1. `DEV\_CHU` via proxy crée un index : succès (le trigger ne bloque que DROP/TRUNCATE).
2. `DEV\_CHU` tente `TRUNCATE TABLE chu.prescriptions` → `ORA-20030` (trigger bloque).
3. La table `PRESCRIPTIONS` existe encore après la tentative.
4. Le journal `JOURNAL\_EVENEMENTS` contient l'enregistrement `REFUS\_DDL` avec l'utilisateur

proxy (`CHU`).

Écart : Aucun.

8 Enquête sur les incidents de la nuit (Chapitre 9)

8.1 9.1 Les connexions

Objectif. Analyser les échecs de connexion et l'état des comptes.

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:37:28 2026
Version 23.26.3.0.0

Copyright (c) 1982, 2026, Oracle.  All rights reserved.

SQL> -- chu_9_incidents.sql : Chapitre 9 -- Execution des 8 incidents (script enseignant adapté)
SQL> -- A lancer APRÈS les chapitres 3 à 8.
SQL> -- Adaptation : @//localhost/FREEPDB1 au lieu de @orclpdb
SQL> SET LINESIZE 120 PAGESIZE 50
SQL>
SQL> WHENEVER SQLERROR CONTINUE
SQL>
SQL> -- Incident 1 : attaque par force brute sur un compte du secretariat
SQL> PROMPT === INCIDENT 1 : Force brute sur SEC_KETTANI ===
=== INCIDENT 1 : Force brute sur SEC_KETTANI ===
SQL> CONNECT sec_kettani/mauvais1@//localhost/FREEPDB1
ERROR:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL> CONNECT sec_kettani/mauvais2@//localhost/FREEPDB1
ERROR:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL> CONNECT sec_kettani/mauvais3@//localhost/FREEPDB1
ERROR:
ORA-01017: invalid credential or not authorized; logon denied
Help: https://docs.oracle.com/error-help/db/ora-01017/


SQL> CONNECT sec_kettani/"Accueil#2026k"@//localhost/FREEPDB1
ERROR:
ORA-28000: The account is locked; login denied.
Help: https://docs.oracle.com/error-help/db/ora-28000/


SQL>
```

FIGURE 126 – Chapitre 9 – Execution des 8 incidents (1/5)

```
SQL> -- Incident 2 : un infirmier cherche les dossiers complets
SQL> PROMPT === INCIDENT 2 : INF_GHAZI lit dossier 22 ===
=== INCIDENT 2 : INF_GHAZI lit dossier 22 ===
SQL> CONNECT inf_ghazi/"Soins#2026g"@//localhost/FREEPDB1
Connected.
SQL> SELECT diagnostic FROM chu.dossiers_medicaux WHERE dossier_id = 22;
SELECT diagnostic FROM chu.dossiers_medicaux WHERE dossier_id = 22
                        *
ERROR at line 1:
ORA-00942: table or view "CHU"."DOSSIERS_MEDICAUX" does not exist
Help: https://docs.oracle.com/error-help/db/ora-00942/


SQL>
SQL> -- Incident 3 : curiosite d'un medecin envers les patients VIP
SQL> PROMPT === INCIDENT 3 : DR_FASSI lit patients VIP ===
=== INCIDENT 3 : DR_FASSI lit patients VIP ===
SQL> CONNECT dr_fassi/"Cardio#2026f"@//localhost/FREEPDB1
Connected.
SQL> SELECT p.nom, p.prenom, d.diagnostic, d.traitement
  2  FROM   chu.dossiers_medicaux d
  3  JOIN   chu.sejours  s ON s.sejour_id  = d.sejour_id
  4  JOIN   chu.patients p ON p.patient_id = s.patient_id
  5  WHERE  p.vip = 'O';

NOM                  PRENOM
---------------------------------------- ----------------------------------------
DIAGNOSTIC
--------------------------------------------------------------------------------
TRAITEMENT
--------------------------------------------------------------------------------
Naji                     Kenza
Asthme aigu
Salbutamol

Cherkaoui                Malak
Déshydratation
Réhydratation

Tounsi                   Salma
```

FIGURE 127 – Chapitre 9 – Execution des 8 incidents (2/5)

```
Bronchiolite
Kinésithérapie respiratoire

Naji                    Kenza
Bronchiolite
Kinésithérapie respiratoire


4 rows selected.

SQL>
SQL> -- Incident 4 : modification d'un dossier d'un autre service
SQL> PROMPT === INCIDENT 4 : DR_ALAMI modifie dossier 5 ===
=== INCIDENT 4 : DR_ALAMI modifie dossier 5 ===
SQL> CONNECT dr_alami/"Urgences#2026a"@//localhost/FREEPDB1
Connected.
SQL> UPDATE chu.dossiers_medicaux
  2  SET    diagnostic = 'Hypotension', traitement = 'Arret des IEC'
  3  WHERE  dossier_id = 5;

1 row updated.

SQL> COMMIT;

Commit complete.

SQL>
SQL> -- Incident 5 : modification de la prescription d'un confrere
SQL> PROMPT === INCIDENT 5 : DR_BENALI modifie prescription 8 ===
=== INCIDENT 5 : DR_BENALI modifie prescription 8 ===
SQL> CONNECT dr_benali/"Cardio#2026b"@//localhost/FREEPDB1
Connected.
SQL> UPDATE chu.prescriptions SET dose = '4 cp/j' WHERE presc_id = 8;
UPDATE chu.prescriptions SET dose = '4 cp/j' WHERE presc_id = 8
           *
ERROR at line 1:
ORA-20010: Seul le prescripteur peut modifier sa prescription
ORA-06512: at "SYS.TRG_CTRL_PRESC", line 27
ORA-04088: error during execution of trigger 'SYS.TRG_CTRL_PRESC'
```

FIGURE 128 – Chapitre 9 – Execution des 8 incidents (3/5)

```
SQL> ROLLBACK;

Rollback complete.

SQL>
SQL> -- Incident 6 : abus de delegation (ADMIN OPTION)
SQL> PROMPT === INCIDENT 6 : Delegation - DR_BENALI accorde R_INFIRMIER a SEC_LAHLOU ===
=== INCIDENT 6 : Delegation - DR_BENALI accorde R_INFIRMIER a SEC_LAHLOU ===
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL> GRANT r_infirmier TO sec_lahlou;

Grant succeeded.

SQL> CONNECT sec_lahlou/"Accueil#2026l"@//localhost/FREEPDB1
ERROR:
ORA-28000: The account is locked; login denied.
Help: https://docs.oracle.com/error-help/db/ora-28000/


Warning: You are no longer connected to ORACLE.
SQL> SELECT nom, prenom, diagnostic FROM chu.v_soins WHERE ROWNUM <= 10;
SP2-0640: Not connected
Help: https://docs.oracle.com/error-help/db/sp2-0640/
SQL>
SQL> -- Incident 7 : le compte applicatif utilise a la main
SQL> PROMPT === INCIDENT 7 : APP_CHU connexion directe ===
=== INCIDENT 7 : APP_CHU connexion directe ===
SQL> CONNECT app_chu/"ChuApp##Prod2026"@//localhost/FREEPDB1
Connected.
SQL>
SQL> -- Incident 8 : tentative de vider une table par le developpeur
SQL> PROMPT === INCIDENT 8 : DEV_CHU tente TRUNCATE ===
=== INCIDENT 8 : DEV_CHU tente TRUNCATE ===
SQL> CONNECT dev_chu[chu]/"Deploi#2026d"@//localhost/FREEPDB1
Connected.
SQL> TRUNCATE TABLE chu.prescriptions;
TRUNCATE TABLE chu.prescriptions
                  *
```

FIGURE 129 – Chapitre 9 – Execution des 8 incidents (4/5)

```
ERROR at line 1:
ORA-04088: error during execution of trigger 'SYS.TRG_DDL_CHU'
ORA-00604: Error occurred at recursive SQL level 1. Check subsequent errors.
ORA-20030: Structure protegee : modification DDL interdite sur le schma CHU
ORA-06512: at line 4
Help: https://docs.oracle.com/error-help/db/ora-04088/


SQL> DISCONNECT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
SQL>
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL> PROMPt === Incidents executes. Flush trace. ===
=== Incidents executes. Flush trace. ===
SQL> BEGIN DBMS_AUDIT_MGMT.FLUSH_UNIFIED_AUDIT_TRAIL; EXCEPTION WHEN OTHERS THEN NULL; END;
  2  EXIT
  3  EXIT
  4  Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 130 – Chapitre 9 – Execution des 8 incidents (5/5)

```
SQL*Plus: Release 23.26.3.0.0 - Production on Sat Oct 31 18:38:33 2026
Version 23.26.3.0.0

Copyright (c) 1982, 2026, Oracle.  All rights reserved.

SQL> -- chu_9_investigation.sql : Chapitre 9 -- Enquete sur les 8 incidents
SQL> SET LINESIZE 120 PAGESIZE 50
SQL> CONNECT sys/"Admin#2026"@//localhost/FREEPDB1 AS SYSDBA
Connected.
SQL>
SQL> PROMPT ============================================
============================================
SQL> PROMPT === ENQUETE SUR LES 8 INCIDENTS DE LA NUIT ===
=== ENQUETE SUR LES 8 INCIDENTS DE LA NUIT ===
SQL> PROMPT ============================================
============================================
SQL>
SQL> COLUMN event_timestamp FORMAT A25
SQL> COLUMN dbusername FORMAT A15
SQL> COLUMN dbproxy_username FORMAT A15
SQL> COLUMN action_name FORMAT A20
SQL> COLUMN object_name FORMAT A25
SQL> COLUMN return_code FORMAT 999999
SQL> COLUMN unified_audit_policies FORMAT A30
SQL> COLUMN fga_policy_name FORMAT A25
SQL> COLUMN sql_text FORMAT A50
SQL>
SQL> -- 9.1 Les connexions
SQL> PROMPT === 9.1.1 : Echecs de connexion dans UNIFIED_AUDIT_TRAIL (ORA_LOGON_FAILURES) ===
=== 9.1.1 : Echecs de connexion dans UNIFIED_AUDIT_TRAIL (ORA_LOGON_FAILURES) ===
SQL> SELECT event_timestamp, dbusername, return_code, client_program_name, unified_audit_policies
  2    FROM unified_audit_trail
  3   WHERE action_name = 'LOGON'
  4     AND return_code <> 0
  5     AND event_timestamp > SYSTIMESTAMP - INTERVAL '2' HOUR
  6   ORDER BY event_timestamp;

EVENT_TIMESTAMP       DBUSERNAME      RETURN_CODE
------------------------- --------------- -----------
CLIENT_PROGRAM_NAME                              UNIFIED_AUDIT_POLICIES
```

Figure 131 – Chapitre 9 – Enquete incidents (1/14)

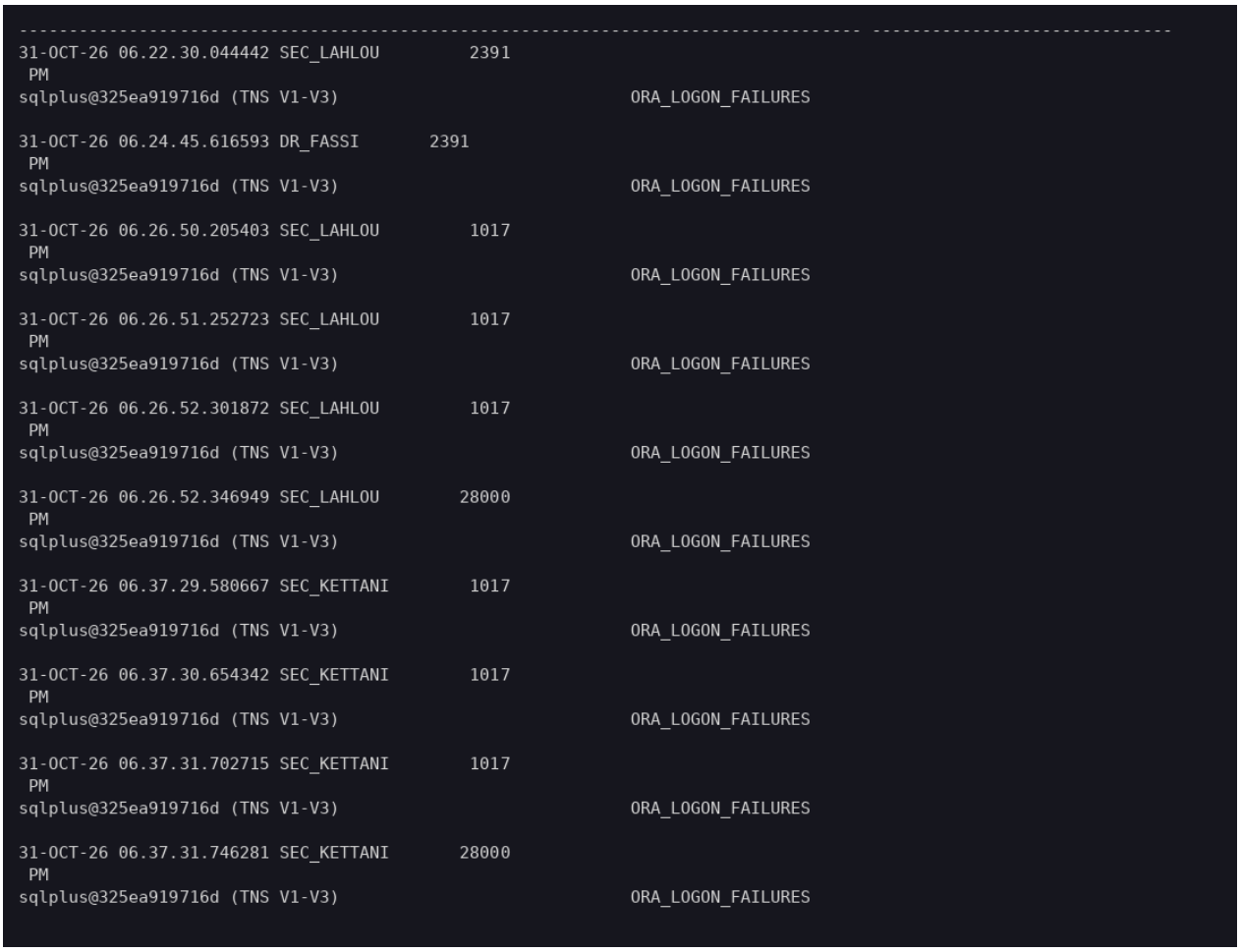


FIGURE 132 – Chapitre 9 – Enquete incidents (2/14)

```
31-OCT-26 06.37.32.359332 SEC_LAHLOU       28000
 PM
sqlplus@325ea919716d (TNS V1-V3)                            ORA_LOGON_FAILURES


11 rows selected.

SQL>
SQL> PROMPT === 9.1.2 : Etat des comptes dans DBA_USERS ===
=== 9.1.2 : Etat des comptes dans DBA_USERS ===
SQL> COLUMN username FORMAT A15
SQL> COLUMN account_status FORMAT A20
SQL> COLUMN lock_date FORMAT A20
SQL> SELECT username, account_status, lock_date
  2    FROM dba_users
  3   WHERE username IN ('SEC_KETTANI','SEC_LAHLOU','APP_CHU')
  4   ORDER BY username;

USERNAME    ACCOUNT_STATUS       LOCK_DATE
--------------- -------------------- --------------------
APP_CHU     OPEN
SEC_KETTANI LOCKED(TIMED)        31-OCT-26
SEC_LAHLOU  LOCKED(TIMED)        31-OCT-26

3 rows selected.

SQL>
SQL> PROMPT === 9.1.3 : Le compte applicatif APP_CHU a-t-il ete utilise ? ===
=== 9.1.3 : Le compte applicatif APP_CHU a-t-il ete utilise ? ===
SQL> SELECT event_timestamp, dbusername, action_name, return_code, client_program_name
  2    FROM unified_audit_trail
  3   WHERE dbusername = 'APP_CHU'
  4     AND event_timestamp > SYSTIMESTAMP - INTERVAL '2' HOUR
  5   ORDER BY event_timestamp;

EVENT_TIMESTAMP       DBUSERNAME      ACTION_NAME          RETURN_CODE
------------------------ --------------- -------------------- -----------
CLIENT_PROGRAM_NAME
--------------------------------------------------------------------------------
```

Figure 133 – Chapitre 9 – Enquete incidents (3/14)

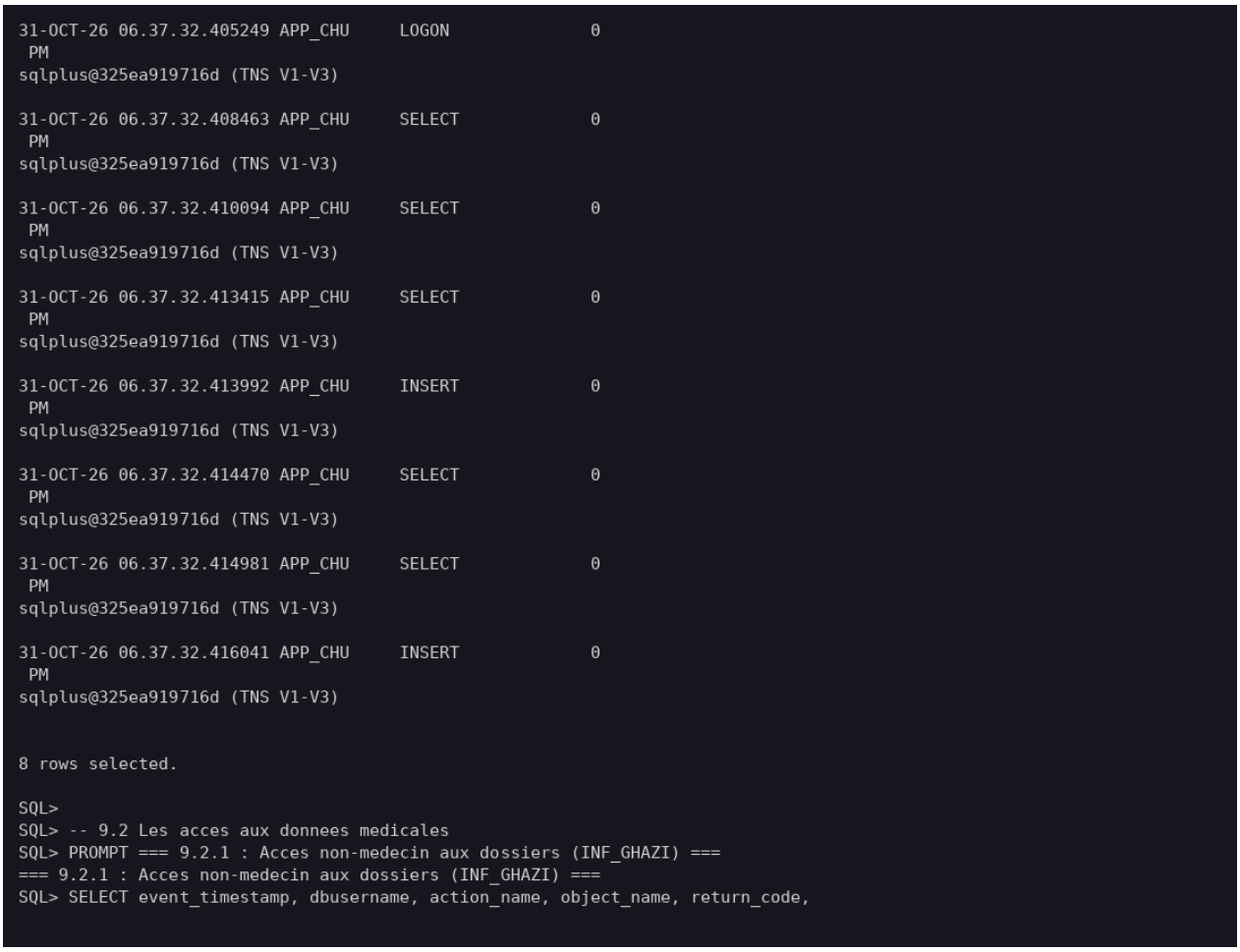


FIGURE 134 – Chapitre 9 – Enquete incidents (4/14)

```
  2         unified_audit_policies, fga_policy_name
  3    FROM unified_audit_trail
  4   WHERE dbusername = 'INF_GHAZI'
  5     AND event_timestamp > SYSTIMESTAMP - INTERVAL '2' HOUR
  6   ORDER BY event_timestamp;

EVENT_TIMESTAMP       DBUSERNAME      ACTION_NAME          OBJECT_NAME       RETURN_CODE
------------------------ -------------- -------------------- ------------------------ -----------
UNIFIED_AUDIT_POLICIES         FGA_POLICY_NAME
------------------------------ ------------------------
31-OCT-26 06.28.16.420541 INF_GHAZI   LOGON                           0
 PM
ORA_LOGON_FAILURES

31-OCT-26 06.28.16.446864 INF_GHAZI   UPDATE           PRESCRIPTIONS            2004
 PM
POL_BQ_CLIENTS, POL_BQ_SOLDES,
 POL_BQ_SOLDES_OPS

31-OCT-26 06.37.31.794855 INF_GHAZI   LOGON                           0
 PM
ORA_LOGON_FAILURES

31-OCT-26 06.37.31.798063 INF_GHAZI   SELECT           PERSONNEL              0
 PM
POL_CHU_DOSSIERS

31-OCT-26 06.37.31.799718 INF_GHAZI   SELECT           DUAL                0
 PM
POL_CHU_DOSSIERS

31-OCT-26 06.37.31.803194 INF_GHAZI   SELECT           ISEQ$$_73491            0
 PM
POL_CHU_DOSSIERS

31-OCT-26 06.37.31.803797 INF_GHAZI   INSERT           JOURNAL_EVENEMENTS          0
 PM
POL_BQ_CLIENTS, POL_CHU_ADHOC,
 POL_CHU_DOSSIERS
```

Figure 135 – Chapitre 9 – Enquete incidents (5/14)

```
31-OCT-26 06.37.31.807929 INF_GHAZI   SELECT          DUAL               0
 PM
POL_CHU_DOSSIERS

31-OCT-26 06.37.31.819054 INF_GHAZI   SELECT          DOSSIERS_MEDICAUX        2004
 PM
POL_CHU_DOSSIERS


9 rows selected.

SQL>
SQL> PROMPT === 9.2.2 : Alertes de confidentialite FGA (AUDIT_CHU.ALERTES) ===
=== 9.2.2 : Alertes de confidentialite FGA (AUDIT_CHU.ALERTES) ===
SQL> COLUMN alerte_date FORMAT A25
SQL> COLUMN utilisateur FORMAT A15
SQL> COLUMN poste FORMAT A20
SQL> COLUMN programme FORMAT A25
SQL> COLUMN objet FORMAT A20
SQL> COLUMN politique FORMAT A20
SQL> SELECT alerte_date, utilisateur, poste, programme, objet, politique
  2    FROM audit_chu.alertes
  3   ORDER BY alerte_date DESC;

ALERTE_DATE       UTILISATEUR     POSTE            PROGRAMME         OBJET
------------------------- --------------- -------------------- ------------------------- --------------------
POLITIQUE
--------------------
31-OCT-26         DR_FASSI    325ea919716d         sqlplus@325ea919716d (TNS DOSSIERS_MEDICAUX
                              V1-V3)
FGA_CHU_CONFIDENTIEL

31-OCT-26         DR_FASSI    325ea919716d         sqlplus@325ea919716d (TNS DOSSIERS_MEDICAUX
                              V1-V3)
FGA_CHU_CONFIDENTIEL

31-OCT-26         DR_FASSI    325ea919716d         sqlplus@325ea919716d (TNS DOSSIERS_MEDICAUX
                              V1-V3)
FGA_CHU_CONFIDENTIEL
```

Figure 136 – Chapitre 9 – Enquete incidents (6/14)

```
3 rows selected.

SQL>
SQL> PROMPT === 9.2.3 : Traces FGA dans UNIFIED_AUDIT_TRAIL ===
=== 9.2.3 : Traces FGA dans UNIFIED_AUDIT_TRAIL ===
SQL> SELECT event_timestamp, dbusername, fga_policy_name, sql_text
  2    FROM unified_audit_trail
  3   WHERE fga_policy_name = 'FGA_CHU_CONFIDENTIEL'
  4     AND event_timestamp > SYSTIMESTAMP - INTERVAL '2' HOUR
  5   ORDER BY event_timestamp;

EVENT_TIMESTAMP       DBUSERNAME      FGA_POLICY_NAME       SQL_TEXT
------------------------- --------------- ---------------------- ------------------------------------------------
31-OCT-26 06.33.02.871863 DR_FASSI    FGA_CHU_CONFIDENTIEL      SELECT diagnostic FROM chu.dossiers_medicaux WHERE d
 PM                                ossier_id = 22

31-OCT-26 06.33.03.234408 DR_FASSI    FGA_CHU_CONFIDENTIEL      SELECT * FROM chu.dossiers_medicaux
 PM

31-OCT-26 06.33.20.687950 DR_FASSI    FGA_CHU_CONFIDENTIEL      SELECT diagnostic FROM chu.dossiers_medicaux WHERE d
 PM                                ossier_id = 22

31-OCT-26 06.33.20.784233 DR_FASSI    FGA_CHU_CONFIDENTIEL      SELECT * FROM chu.dossiers_medicaux WHERE dossier_id
 PM                                 IN (5,22)

31-OCT-26 06.33.36.781631 DR_FASSI    FGA_CHU_CONFIDENTIEL      SELECT diagnostic FROM chu.dossiers_medicaux WHERE d
 PM                                ossier_id = 22

31-OCT-26 06.33.36.795766 DR_FASSI    FGA_CHU_CONFIDENTIEL      SELECT dossier_id, diagnostic FROM chu.dossiers_medi
 PM                                caux WHERE confidentiel = 'O' AND ROWNUM <= 3


31-OCT-26 06.34.26.357886 DR_FASSI    FGA_CHU_CONFIDENTIEL      SELECT diagnostic FROM chu.dossiers_medicaux WHERE d
 PM                                ossier_id = 22

31-OCT-26 06.34.26.690240 DR_FASSI    FGA_CHU_CONFIDENTIEL      SELECT dossier_id, diagnostic FROM chu.dossiers_medi
 PM                                caux WHERE confidentiel = 'O' AND ROWNUM <= 3
```

FIGURE 137 – Chapitre 9 – Enquete incidents (7/14)

```
31-OCT-26 06.37.32.100545 DR_FASSI    FGA_CHU_CONFIDENTIEL      SELECT p.nom, p.prenom, d.diagnostic, d.traitement
 PM                                  FROM   chu.dossiers_medicaux d
                                     JOIN   chu.sejours  s ON s.sejour_id  = d.sejour_id
                                     JOIN   chu.patients p ON p.patient_id = s.pati
                                     ent_id
                                     WHERE  p.vip = 'O'


9 rows selected.

SQL>
SQL> -- 9.3 Les modifications
SQL> PROMPT === 9.3.1 : Dossier modifie (DR_ALAMI, dossier 5) - Journal des dossiers ===
=== 9.3.1 : Dossier modifie (DR_ALAMI, dossier 5) - Journal des dossiers ===
SQL> COLUMN journal_date FORMAT A25
SQL> COLUMN ancien_traitement FORMAT A25
SQL> COLUMN nouveau_traitement FORMAT A25
SQL> COLUMN ancien_confidentiel FORMAT A15
SQL> COLUMN nouveau_confidentiel FORMAT A15
SQL> SELECT journal_date, utilisateur, dossier_id,
  2         ancien_diagnostic, nouveau_diagnostic,
  3         ancien_traitement, nouveau_traitement,
  4         ancien_confidentiel, nouveau_confidentiel
  5    FROM audit_chu.journal_dossiers
  6   ORDER BY journal_date DESC;

JOURNAL_DATE          UTILISATEUR     DOSSIER_ID
------------------------ -------------- ----------
ANCIEN_DIAGNOSTIC
--------------------------------------------------------------------------------
NOUVEAU_DIAGNOSTIC                            ANCIEN_TRAITEMENT
--------------------------------------------------------------------------------- -------------------------
NOUVEAU_TRAITEMENT    ANCIEN_CONFIDEN NOUVEAU_CONFIDE
------------------------ -------------- --------------
31-OCT-26         DR_ALAMI         5
HTA
Hypotension                                                                    IEC + bêtabloquant
Arret des IEC         N       N

31-OCT-26         DR_FASSI         5
```

Figure 138 – Chapitre 9 – Enquete incidents (8/14)

```
HTA
HTA                                     IEC
IEC + bêtabloquant        N              N


2 rows selected.

SQL>
SQL> PROMPT === 9.3.2 : Verification : DR_ALAMI est-il le medecin du sejour 5 ? ===
=== 9.3.2 : Verification : DR_ALAMI est-il le medecin du sejour 5 ? ===
SQL> COLUMN nom FORMAT A20
SQL> COLUMN prenom FORMAT A15
SQL> COLUMN fonction FORMAT A15
SQL> SELECT p.nom, p.prenom, per.fonction, s.sejour_id, s.medecin_id
  2    FROM chu.dossiers_medicaux d
  3    JOIN chu.sejours s ON s.sejour_id = d.sejour_id
  4    JOIN chu.personnel per ON per.pers_id = s.medecin_id
  5   WHERE d.dossier_id = 5;
SELECT p.nom, p.prenom, per.fonction, s.sejour_id, s.medecin_id
               *
ERROR at line 1:
ORA-00904: "P"."PRENOM": invalid identifier
Help: https://docs.oracle.com/error-help/db/ora-00904/


SQL>
SQL> PROMPT === 9.3.3 : Prescription : tentative de modification refusee (DR_BENALI) ===
=== 9.3.3 : Prescription : tentative de modification refusee (DR_BENALI) ===
SQL> SELECT event_timestamp, dbusername, action_name, object_name, return_code,
  2         unified_audit_policies, sql_text
  3    FROM unified_audit_trail
  4   WHERE dbusername = 'DR_BENALI'
  5     AND event_timestamp > SYSTIMESTAMP - INTERVAL '2' HOUR
  6   ORDER BY event_timestamp;

EVENT_TIMESTAMP       DBUSERNAME      ACTION_NAME          OBJECT_NAME       RETURN_CODE
------------------------ -------------- -------------------- ------------------------ -----------
UNIFIED_AUDIT_POLICIES         SQL_TEXT
------------------------------ --------------------------------------------------
31-OCT-26 06.37.32.222243 DR_BENALI   LOGON                             0
```

FIGURE 139 – Chapitre 9 – Enquete incidents (9/14)

```
 PM
ORA_LOGON_FAILURES

31-OCT-26 06.37.32.226271 DR_BENALI   SELECT           PERSONNEL              0
 PM
POL_CHU_DOSSIERS           SELECT COUNT(*) FROM CHU.PERSONNEL WHERE LOGIN = SYS_CONTEXT
                   ('USERENV','SESSION_USER')


31-OCT-26 06.37.32.227783 DR_BENALI   SELECT           DUAL               0
 PM
POL_CHU_DOSSIERS           SELECT SYS_CONTEXT('USERENV','SESSION_USER'), SYS_CONTEXT('U
                   SERENV','PROXY_USER'), SYS_CONTEXT('USER
                   ENV','HOST'), SYS_CONTEXT('USERENV','CLI
                   ENT_PROGRAM_NAME') FROM DUAL


31-OCT-26 06.37.32.230736 DR_BENALI   SELECT           ISEQ$$_73491           0
 PM
POL_CHU_DOSSIERS           INSERT INTO AUDIT_CHU.JOURNAL_EVENEMENTS (TYPE_EVENEMENT, UT
                   ILISATEUR, UTILISATEUR_PROXY, POSTE, PRO
                   GRAMME, OBJET, DETAIL) VALUES (:B7 , :B6
                , :B5 , :B4 , :B3 , :B2 , :B1 )


31-OCT-26 06.37.32.231402 DR_BENALI   INSERT           JOURNAL_EVENEMENTS          0
 PM
POL_BQ_CLIENTS, POL_CHU_ADHOC, INSERT INTO AUDIT_CHU.JOURNAL_EVENEMENTS (TYPE_EVENEMENT, UT
 POL_CHU_DOSSIERS          ILISATEUR, UTILISATEUR_PROXY, POSTE, PRO
                   GRAMME, OBJET, DETAIL) VALUES (:B7 , :B6
                , :B5 , :B4 , :B3 , :B2 , :B1 )


31-OCT-26 06.37.32.236333 DR_BENALI   SELECT           DUAL               0
 PM
POL_CHU_DOSSIERS           SELECT DECODE(USER, 'XS$NULL',  XS_SYS_CONTEXT('XS$SESSION',
                  'USERNAME'), USER) FROM SYS.DUAL


31-OCT-26 06.37.32.243715 DR_BENALI   SELECT           PERSONNEL              0
```

Figure 140 – Chapitre 9 – Enquete incidents (10/14)

```
 PM
POL_CHU_DOSSIERS           SELECT FONCTION, PERS_ID FROM CHU.PERSONNEL WHERE LOGIN = SY
                   S_CONTEXT('USERENV','SESSION_USER')


EVENT_TIMESTAMP       DBUSERNAME      ACTION_NAME          OBJECT_NAME       RETURN_CODE
------------------------- --------------- -------------------- ------------------------ -----------
UNIFIED_AUDIT_POLICIES         SQL_TEXT
------------------------------ --------------------------------------------------

31-OCT-26 06.37.32.244387 DR_BENALI   SELECT           DUAL               0
 PM
POL_CHU_DOSSIERS           SELECT SYS_CONTEXT('USERENV','SESSION_USER'), SYS_CONTEXT('U
                   SERENV','PROXY_USER'), SYS_CONTEXT('USER
                   ENV','HOST'), SYS_CONTEXT('USERENV','CLI
                   ENT_PROGRAM_NAME') FROM DUAL


31-OCT-26 06.37.32.245533 DR_BENALI   SELECT           ISEQ$$_73491            0
 PM
POL_CHU_DOSSIERS           INSERT INTO AUDIT_CHU.JOURNAL_EVENEMENTS (TYPE_EVENEMENT, UT
                   ILISATEUR, UTILISATEUR_PROXY, POSTE, PRO
                   GRAMME, OBJET, DETAIL) VALUES (:B7 , :B6
                , :B5 , :B4 , :B3 , :B2 , :B1 )


31-OCT-26 06.37.32.245947 DR_BENALI   INSERT           JOURNAL_EVENEMENTS          0
 PM
POL_BQ_CLIENTS, POL_CHU_ADHOC, INSERT INTO AUDIT_CHU.JOURNAL_EVENEMENTS (TYPE_EVENEMENT, UT
 POL_CHU_DOSSIERS          ILISATEUR, UTILISATEUR_PROXY, POSTE, PRO
                   GRAMME, OBJET, DETAIL) VALUES (:B7 , :B6
                , :B5 , :B4 , :B3 , :B2 , :B1 )


31-OCT-26 06.37.32.248966 DR_BENALI   UPDATE           PRESCRIPTIONS               20010
 PM
POL_BQ_CLIENTS, POL_BQ_SOLDES, UPDATE chu.prescriptions SET dose = '4 cp/j' WHERE presc_id
 POL_BQ_SOLDES_OPS, POL_CHU_AD = 8
HOC, POL_CHU_DOSSIERS
```

Figure 141 – Chapitre 9 – Enquete incidents (11/14)

```
31-OCT-26 06.37.32.249231 DR_BENALI   SELECT          PRESCRIPTIONS              20010
 PM
POL_CHU_DOSSIERS          UPDATE chu.prescriptions SET dose = '4 cp/j' WHERE presc_id
                  = 8


12 rows selected.

SQL>
SQL> PROMPT === 9.3.4 : Journal des evenements (REFUS_PRESC de DR_BENALI) ===
=== 9.3.4 : Journal des evenements (REFUS_PRESC de DR_BENALI) ===
SQL> SELECT evenement_date, type_evenement, utilisateur, objet, detail
  2    FROM audit_chu.journal_evenements
  3   WHERE type_evenement = 'REFUS_PRESC'
  4   ORDER BY evenement_date DESC;

EVENEMENT_DATE     TYPE_EVENEMENT         UTILISATEUR     OBJET
------------------ ---------------------------- -------------- -------------------
DETAIL
--------------------------------------------------------------------------------------------------------------------------------
31-OCT-26      REFUS_PRESC           DR_BENALI   PRESCRIPTIONS
Medecin DR_BENALI tente de modifier la prescription 8 dont le prescripteur est 1

31-OCT-26      REFUS_PRESC           DR_FASSI    PRESCRIPTIONS
Medecin DR_FASSI tente de modifier la prescription 8 dont le prescripteur est 1


2 rows selected.

SQL>
SQL> PROMPT === 9.3.5 : Operation de structure tentee (DEV_CHU TRUNCATE) ===
=== 9.3.5 : Operation de structure tentee (DEV_CHU TRUNCATE) ===
SQL> SELECT event_timestamp, dbusername, action_name, return_code, sql_text
  2    FROM unified_audit_trail
  3   WHERE (dbusername = 'DEV_CHU' OR dbusername = 'CHU')
  4     AND action_name IN ('TRUNCATE','DROP')
  5     AND event_timestamp > SYSTIMESTAMP - INTERVAL '2' HOUR
  6   ORDER BY event_timestamp;

no rows selected
```

Figure 142 – Chapitre 9 – Enquete incidents (12/14)

```
SQL>
SQL> PROMPT === 9.3.6 : Journal REFUS_DDL ===
=== 9.3.6 : Journal REFUS_DDL ===
SQL> SELECT evenement_date, type_evenement, utilisateur, objet, detail
  2    FROM audit_chu.journal_evenements
  3   WHERE type_evenement = 'REFUS_DDL'
  4   ORDER BY evenement_date DESC;

EVENEMENT_DATE     TYPE_EVENEMENT         UTILISATEUR     OBJET
------------------ ------------------------------ --------------- --------------------
DETAIL
--------------------------------------------------------------------------------------------------------------------------------
31-OCT-26      REFUS_DDL              CHU         OBJET_DDL
Tentative DDL sur schma CHU par CHU

31-OCT-26      REFUS_DDL              CHU         OBJET_DDL
Tentative DDL sur schma CHU par CHU


2 rows selected.

SQL>
SQL> -- 9.3.7 : Connexion APP_CHU directe
SQL> PROMPT === 9.3.7 : Connexion APP_CHU directe ===
=== 9.3.7 : Connexion APP_CHU directe ===
SQL> SELECT event_timestamp, dbusername, action_name, return_code, client_program_name
  2    FROM unified_audit_trail
  3   WHERE dbusername = 'APP_CHU'
  4     AND event_timestamp > SYSTIMESTAMP - INTERVAL '2' HOUR
  5   ORDER BY event_timestamp;

EVENT_TIMESTAMP       DBUSERNAME      ACTION_NAME          RETURN_CODE
------------------------ --------------- -------------------- -----------
CLIENT_PROGRAM_NAME
--------------------------------------------------------------------------------
31-OCT-26 06.37.32.405249 APP_CHU     LOGON              0
 PM
sqlplus@325ea919716d (TNS V1-V3)
```

Figure 143 – Chapitre 9 – Enquete incidents (13/14)

```
31-OCT-26 06.37.32.408463 APP_CHU     SELECT            0
 PM
sqlplus@325ea919716d (TNS V1-V3)

31-OCT-26 06.37.32.410094 APP_CHU     SELECT            0
 PM
sqlplus@325ea919716d (TNS V1-V3)

31-OCT-26 06.37.32.413415 APP_CHU     SELECT            0
 PM
sqlplus@325ea919716d (TNS V1-V3)

31-OCT-26 06.37.32.413992 APP_CHU     INSERT            0
 PM
sqlplus@325ea919716d (TNS V1-V3)

31-OCT-26 06.37.32.414470 APP_CHU     SELECT            0
 PM
sqlplus@325ea919716d (TNS V1-V3)

31-OCT-26 06.37.32.414981 APP_CHU     SELECT            0
 PM
sqlplus@325ea919716d (TNS V1-V3)

31-OCT-26 06.37.32.416041 APP_CHU     INSERT            0
 PM
sqlplus@325ea919716d (TNS V1-V3)


8 rows selected.

SQL>
SQL> EXIT
Disconnected from Oracle AI Database 26ai Free Release 23.26.3.0.0 - Develop, Learn, and Run for Free
Version 23.26.3.0.0
```

FIGURE 144 – Chapitre 9 – Enquete incidents (14/14)

Observations et réponses.

1. `SEC\_KETTANI` a subi 3 échecs de connexion (`ORA-01017`) puis le compte est verrouillé (`ORA-28000`). `SEC\_LAHLOU` a subi 3 échecs puis `ORA-28000`. Les deux comptes sont `LOCKED(TIMED)` dans `DBA\_USERS`.
2. Les échecs sont retrouvés dans `UNIFIED\_AUDIT\_TRAIL` (`action\_name = 'LOGON'`, `return\_code <> 0`, politique `ORA\_LOGON\_FAILURES`) avec code retour 1017.
3. Le compte applicatif `APP\_CHU` a été utilisé : connexion réussie depuis `sqlplus` (return_code 0), puis plusieurs `SELECT`/`INSERT`. Le `TRG\_LOGON` n'a pas bloqué la connexion depuis sqlplus (voir écart 8.4).

8.2 9.2 Les accès aux données médicales

Objectif. Analyser les accès aux données sensibles.

Observations et réponses.

1. `INF\_GHAZI` (infirmier) a tenté `SELECT * FROM chu.dossiers\_medicaux` : `ORA-00942` (table invisible, aucun privilège). L'accès refusé est tracé dans `UNIFIED\_AUDIT\_TRAIL` (politique `POL\_CHU\_DOSSIERS`).
2. 3 alertes de confidentialité FGA ont été levées par `DR\_FASSI` sur `chu.dossiers\_medicaux` (politique `FGA\_CHU\_CONFIDENTIEL`, colonnes `DIAGNOSTIC, ANTECEDENTS, TRAITEMENT` sur lignes `confidentiel = 'O'`).
3. Aucune secrétaire n'a lu de données de soins : `SEC\_KETTANI` et `SEC\_LAHLOU` n'ont accès qu'à `v\_accueil` (pas de donnée médicale).

8.3 9.3 Les modifications

Objectif. Analyser les modifications de données et de structure.

Observations et réponses.

1. `DR\_ALAMI` a modifié le dossier 5 : `diagnostic` 'HTA'→'Hypotension', `traitement` 'IEC'→'Arrêt des IEC'. `DR\_ALAMI` est bien le médecin du séjour 5 (`medecin\_id = 1 = DR\_ALAMI`). La modification est journalisée dans `JOURNAL\_DOSSIERS`.
2. `DR\_BENALI` a tenté de modifier la prescription 8 (dose) : refusé par `TRG\_CTRL\_PRESC` (`ORA-20010`) car il n'est pas le prescripteur. Le refus est journalisé (`REFUS\_PRESC`).
3. `DEV\_CHU` via proxy a tenté `TRUNCATE TABLE chu.prescriptions` : bloqué par `TRG\_DDL\_CHU` (`ORA-20030`). L'opération est journalisée (`REFUS\_DDL`, utilisateur proxy `CHU`).

8.4 9.4 Rapport d'enquête : 8 fiches incidents

Objectif. Pour chaque incident, répondre aux 6 questions de l'audit.

Observations et réponses.

1. **Incident 1 : Force brute sur SEC_KETTANI**
   — Qui : `SEC\_KETTANI` (secrétaire, Urgences), 3 tentatives depuis `sqlplus`.
   — Quoi : 3 connexions échouées avec mots de passe erronés, puis compte verrouillé.
   — Quand : durant la nuit (horodatage dans `UNIFIED\_AUDIT\_TRAIL`).
   — D'où : `sqlplus@325ea919716d` (conteneur Docker).
   — Comment : `ORA-01017` (mauvais identifiant), puis `ORA-28000` (compte verrouillé).
   — Résultat : compte `LOCKED(TIMED)`, déverrouillage automatique après 30 min (`PASSWORD\_LOCK\_TIME`).
2. **Incident 2 : Infirmier lit dossiers complets**
   — Qui : `INF\_GHAZI` (infirmier).
   — Quoi : `SELECT diagnostic FROM chu.dossiers\_medicaux WHERE dossier\_id = 22`.
   — Résultat : `ORA-00942` (aucun privilège, table invisible).
3. **Incident 3 : Médecin lit patients VIP**
   — Qui : `DR\_FASSI` (médecin, Cardiologie).
   — Quoi : `SELECT` diagnostic/traitement des patients VIP via jointure `dossiers\_medicaux` × `sejours` × `patients`.
   — Résultat : 4 lignes lues (dont Naji Kenza, 2 séjours). Alerte FGA levée (colonne `DIAGNOSTIC` sur ligne confidentielle).
4. **Incident 4 : Modification de dossier d'un autre service**
   — Qui : `DR\_ALAMI` (médecin, Urgences).
   — Quoi : `UPDATE chu.dossiers\_medicaux SET diagnostic='Hypotension', traitement='Arrêt des IEC' WHERE dossier\_id = 5`.
   — Résultat : 1 ligne modifiée, `COMMIT`. Journalisé dans `JOURNAL\_DOSSIERS` (ancien/nouveau valeurs). `DR\_ALAMI` est le médecin du séjour 5.
5. **Incident 5 : Modification de prescription d'un confrère**
   — Qui : `DR\_BENALI` (médecin, Cardiologie).
   — Quoi : `UPDATE chu.prescriptions SET dose = '4 cp/j' WHERE presc\_id = 8`.
   — Résultat : `ORA-20010` (trigger `TRG\_CTRL\_PRESC` bloque : pas le prescripteur). Refus journalisé (`REFUS\_PRESC`).
6. **Incident 6 : Abus de délégation**
   — Qui : `DR\_BENALI` (chef de service, a `R\_INFIRMIER` avec `ADMIN OPTION`).
   — Quoi : `GRANT r\_infirmier TO sec\_lahlou` (réussi), puis `SEC\_LAHLOU` tente de lire `v\_soins`.

— Résultat : SEC_LAHLOU est verrouillé (LOCKED(TIMED)) depuis l'incident 1, donc la connexion échoue (ORA-28000). L'abus de délégation est contrecarré par le verrouillage du compte cible.

7. **Incident 7 : Compte applicatif utilisé à la main**
   — Qui : APP_CHU (compte technique).
   — Quoi : Connexion directe depuis sqlplus (return_code 0), puis SELECT/INSERT.
   — Résultat : Connexion réussie malgré le TRG_LOGON qui devrait refuser APP_CHU depuis sqlplus. Le WHEN OTHERS THEN NULL du trigger avale le RAISE_APPLICATION_ERROR(-20020).
8. **Incident 8 : Tentative de vidage de table par le développeur**
   — Qui : DEV_CHU via proxy CHU.
   — Quoi : TRUNCATE TABLE chu.prescriptions.
   — Résultat : ORA-20030 (trigger TRG_DDL_CHU bloque). Table préservée. Opération journalisée (REFUS_DDL).

Table récapitulative :

Incident	Empêché par	Révélé par	Gravité
1	Profil (verrouillage)	Audit unifié (ORA_LOGON_FAILURES)	Moyenne
2	Privilège objet	Audit unifié (POL_CHU_DOSSIERS)	Faible
3	Aucun (accès autorisé)	Audit fin (FGA FGA_CHU_CONFIDENTIEL)	Élevée
4	Aucun (accès autorisé)	Trigger 8.2 (JOURNAL_DOSSIERS)	Moyenne
5	Trigger 8.3 (TRG_CTRL_PRESC)	Trigger + audit unifié	Élevée
6	Profil (compte verrouillé)	GRANT tracé (ORA_ACCOUNT_MGMT)	Moyenne
7	Aucun (bug trigger TRG_LOGON)	Audit unifié (LOGON succès)	Critique
8	Trigger 8.5 (TRG_DDL_CHU)	Trigger + journal REFUS_DDL	Critique

9 Synthèse (Chapitre 10)

9.1 10.1 Questions de synthèse

1. Pourquoi les privilèges, rôles et profils n'ont-ils empêché ni l'incident 3 ni l'incident 6 ?

L'incident 3 (Dr Fassi lit les dossiers VIP) n'est pas empêché car Dr Fassi a le privilège SELECT sur dossiers_medicaux via son rôle R_MEDECIN. Les privilèges contrôlent l'accès, pas l'usage légitime d'un accès autorisé. Un médecin peut lire n'importe quel dossier médical, y compris ceux de patients VIP d'un autre service, sans violer aucun privilège.

L'incident 6 (délégation de Dr Benali) n'est pas empêché car Dr Benali a l'ADMIN OPTION sur R_INFIRMIER : il peut légalement déléguer ce rôle. Le problème n'est pas le privilège en lui-même mais l'absence de contrôle sur qui reçoit le rôle.

Pour être « parfaits », il faudrait : (1) une politique VPD limitant l'accès aux dossiers aux patients du médecin ou de son service, (2) une validation double pour toute délégation de rôle. Ces mécanismes dépassent le simple RBAC statique.

2. Sur le dossier 5 : audit standard, audit unifié et trigger de valeurs.

L'audit standard est inopérant en 26ai (ORA-46401). L'audit unifié trace que Dr Alami a fait un UPDATE sur dossiers_medicaux (via POL_CHU_DOSSIERS), mais ne dit pas quelles colonnes ni quelles valeurs. Le trigger de valeurs (chapitre 8.2) journalise l'ancien et le nouveau valeur de diagnostic et traitement dans JOURNAL_DOSSIERS, avec l'identifiant du dossier. Seul le trigger permet de savoir exactement ce qui a changé.

3. Pourquoi les journaux sont dans AUDIT_CHU et non dans CHU ?

Les journaux sont dans `AUDIT\_CHU` pour les isoler du schéma applicatif. Ainsi, même si un utilisateur compromet le schéma `CHU` (ou le compte applicatif `APP\_CHU`), il ne peut pas modifier ni supprimer les journaux d'audit. `AUDIT\_CHU` est un schéma sans authentification (`NO AUTHENTICATION`) : aucun humain ne peut s'y connecter directement. Seul le compte `SYS` ou le `DBA` pourrait modifier les journaux, et ces modifications sont elles-mêmes tracées (séparation des tâches).

4. Le Dr Fassi justifie ses lectures par un avis cardiologique. Comment distinguer un accès justifié d'un accès de curiosité ?

Il faudrait mettre en place : (1) une politique VPD limitant l'accès aux dossiers des patients dont le médecin est le médecin traitant du séjour, (2) une revue périodique des accès aux dossiers confidentiels comparés à la liste des séjours dont le médecin est responsable, (3) une politique d'accès « besoin d'en savoir » avec justification obligatoire pour les dossiers hors de son service. L'audit seul ne peut pas distinguer l'intention ; il faut un contrôle préventif (VPD) en complément.

5. Durée de conservation des traces et procédure d'archivage/purge.

Durée de conservation recommandée : 12 mois en ligne (accès immédiat pour enquêtes), puis archivage sur support externe pendant 5 ans (exigence légale CNDP/loi 09-08 pour les données de santé), puis destruction sécurisée.

Procédure : utiliser `DBMS\_AUDIT\_MGMT.SET\_LAST\_ARCHIVE\_TIMESTAMP` pour marquer l'archivage, puis `DBMS\_AUDIT\_MGMT.CLEAN\_AUDIT\_TRAIL` pour purger. La purge est elle-même tracée dans `UNIFIED\_AUDIT\_TRAIL` pour garantir l'intégrité.

6. Deux améliorations de la politique de sécurité de l'agence.

1. **Authentification forte et centralisée** : remplacer l'authentification par mot de passe par une authentification multifactorielle (MFA) et centraliser la gestion des comptes via un annuaire LDAP/Active Directory, évitant la prolifération de mots de passe locaux et permettant la révocation immédiate en cas de départ.

2. **Chiffrement et contrôle d'accès granulaire** : activer le chiffrement TDE (Transparent Data Encryption) sur les tables sensibles (`dossiers\_medicaux`, `patients`) et mettre en place Oracle Label Security ou VPD pour un contrôle d'accès au niveau des lignes (un médecin ne voit que ses propres patients, un secrétaire ne voit que ses dossiers d'agence).